Lab #4 TokTickIT Actions Taken, Dashboards, and Final Regression

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Student ID 67070501027, **GitHub username:** N0TAW00D

Answer Part 1: Git Use with Engineering Workflow

Question — How does the Lab 4 work progress from an approved engineering contract to a reviewed release on main?

Lab 4's product release reached main at d07fefe on 13 October 2026: N0TAW00D merged PR #104 into lab4-staging at 1141738, and PR #103 promoted that tree to main. Release-gate tests, builds, lint and migration checks ran on tree 91a8f1a, identical to the product-release tree at d07fefe. PR #105 later merged the documentation-only final-main record at 6d38208 on 17 October. Main at 6d38208 differs from d07fefe only in four Lab 4 documents—ai-use.md, reviewer.md, tests.md and ui-spec.md—so the run is not described as having executed on 6d38208. Figure 1.2 shows all seven Lab 4 Issues individually in Done with their feature PRs linked; the Project API independently confirms those statuses. Figure 1.3 is a recent-commit network view; the PR ledger records the merge sequence and SHAs.

Work progressed from orientation and stakeholder decisions through an approved specification and test plan, Issue decomposition, test-first feature branches, independent audits, peer review, staging integration, and release review. The specification was approved on 3 October before feature implementation, and PR #95 preserved that contract. Later amendments record accepted decisions. PR #97's approval was recorded after merge; the ledger does not present it as pre-merge approval.

Repository, Project and release sources

What	URL / state
Repository	https://github.com/Palapluem/toktickit_cpe334
GitHub Project	https://github.com/users/Palapluem/projects/2
Accepted staging	https://github.com/Palapluem/toktickit_cpe334/commit/876511f31d8eb9b8cf957bb1c6421ed4bab877d5
Release PR #103	PR #103 — MERGED 13 October 2026 by N0TAW00D; release commit d07fefe
Product release commit	https://github.com/Palapluem/toktickit_cpe334/commit/d07fefe23dffafd7dd8257d125c97ffbed4c7250 — Lab 4 release
Current main after PR #105	https://github.com/Palapluem/toktickit_cpe334/commit/6d38208192b999b9210e389c12377889666db8ed — documentation-only merge; product code remains at d07fefe.

Issues

Issue	Work / feature PR	URL
#89	Engineering contract and test plan; PR #95	https://github.com/Palapluem/toktickit_cpe334/issues/89
#90	Actions Taken API and seed; PR #98	https://github.com/Palapluem/toktickit_cpe334/issues/90
#91	Actions Taken UI; PR #101	https://github.com/Palapluem/toktickit_cpe334/issues/91
#92	Ticket workflow and history; PR #100	https://github.com/Palapluem/toktickit_cpe334/issues/92
#93	Role dashboards; PR #99	https://github.com/Palapluem/toktickit_cpe334/issues/93
#94	Hardening and release integration; PR #102	https://github.com/Palapluem/toktickit_cpe334/issues/94
#96	Isolated API test transport; PR #97	https://github.com/Palapluem/toktickit_cpe334/issues/96

Pull Requests

PR / Issue	Branch flow / verified merge evidence	URL
#95 / #89	lab4-staging ← feature/89-lab4-engineering-contract; MERGED f366a47; by N0TAW00D; contract	https://github.com/Palapluem/toktickit_cpe334/pull/95
#97 / #96	lab4-staging ← fix/96-stabilize-server-tests; MERGED 63f7c55; by N0TAW00D; test transport	https://github.com/Palapluem/toktickit_cpe334/pull/97
#98 / #90	lab4-staging ← feature/90-actions-taken-foundation; MERGED ad717ac; by N0TAW00D; Actions Taken foundation	https://github.com/Palapluem/toktickit_cpe334/pull/98

PR / Issue	Branch flow / verified merge evidence	URL
#100 / #92	lab4-staging ← feature/92-ticket-workflow; MERGED 576d93c; by N0TAW00D; workflow and history	https://github.com/Palapluem/toktickit_cpe334/pull/100
#101 / #91	lab4-staging ← feature/91-actions-taken-ui; MERGED 50d542c; by N0TAW00D; Actions Taken UI	https://github.com/Palapluem/toktickit_cpe334/pull/101
#99 / #93	lab4-staging ← feature/93-role-dashboards; MERGED b5d75ed; by N0TAW00D; role dashboards	https://github.com/Palapluem/toktickit_cpe334/pull/99
#102 / #94	lab4-staging ← feature/94-final-hardening-and-release; MERGED 876511f; by N0TAW00D; hardening and integrated evidence	https://github.com/Palapluem/toktickit_cpe334/pull/102
#104	lab4-staging; MERGED 1141738; by N0TAW00D; release-review follow-up, screenshots and docs	https://github.com/Palapluem/toktickit_cpe334/pull/104
#103	main ← lab4-staging; MERGED d07fefe; by N0TAW00D; Lab 4 release	https://github.com/Palapluem/toktickit_cpe334/pull/103
#105	main ← docs/lab-04-final-main-record-20261014; MERGED 6d38208; by N0TAW00D; documentation record	https://github.com/Palapluem/toktickit_cpe334/pull/105

Lab 4 Peer Review Log

Status: Started 3 October 2026. Rows are added from GitHub's actual review, merge and comment records as each Pull Request is reviewed; approval or merge is never inferred from a commit author.

1. Identity and policy

Role	Name	Student ID	GitHub
Author	Wisit Suwannao (วิศิษฐ์ สุวรรณเนาว์)	67070501042	`Palapluem`
Reviewer and merger	Natthawat Primsirikunawut (นัธทวัฒน์ ปริมสิริคุณาวุฒิ)	67070501027	`N0TAW00D`

Repository: `Palapluem/toktickit_cpe334`. Flow: `feature/<issue>-<name>` → `lab4-staging` → one release Pull Request → `main`.

- The reviewer checks each Pull Request against its Issue's acceptance criteria and the Lab 4 contracts, and is the one who merges.
- The author replies to every comment.
- Fixes stay on the same branch and Pull Request.
- No Pull Request exists only to record that another Pull Request merged.

2. Merge ledger

PR	Issue	Base ← head	Reviewed head SHA	Review state	Merged by	Merge commit	Date
#97	#96	`lab4-staging` ← `fix/96-stabilize-server-tests`	`0069884`	COMMENTED 16:29 UTC; APPROVED 16:37 UTC (submitted two minutes after the merge)	`N0TAW00D`	`63f7c55`	2026-10-04 16:35 UTC
#95	#89	`lab4-staging` ← `feature/89-lab4-engineering-contract`	`4e62848` ; fixed since at `4f49df8`	COMMENTED 16:29 UTC; APPROVED 16:37 UTC on `4e62848` ; COMMENTED 16:50 UTC; APPROVED 16:52 UTC on `4f49df8`	`N0TAW00D`	`f366a47`	2026-10-04 16:52 UTC
#98	#90	`lab4-staging` ← `feature/90-actions-taken-foundation`	`948ff56` ; approved at `321e1c2` after the rebase and the fixes	COMMENTED 16:29 UTC; CHANGES_REQUESTED 16:37 UTC (4 Oct); APPROVED 2026-10-11 14:47 UTC. My reply was posted 5 Oct 02:47 UTC	`N0TAW00D`	`ad717ac`	2026-10-11 16:43 UTC
#99	#93	`lab4-staging` ← `feature/93-role-dashboards` (base `feature/90-`	`432fe13` ; approved at `7111d67` , conditionally;	CHANGES_REQUESTED 2026-10-07 16:52 UTC; APPROVED 2026-	`N0TAW00D`	`b5d75ed`	2026-10-11 17:01

Figure 1.1a — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 1/9).

#99	#93	`lab4-staging` ← `feature/93-role-dashboards` (base `feature/90-actions-taken-foundation` until #98 merged)	`432fe13` ; approved at `7111d67` , conditionally; merged at `c7c4d21` after the conditions	CHANGES_REQUESTED 2026-10-07 16:52 UTC; APPROVED 2026-10-11 14:47 UTC	`N0TAW00D`	`b5d75ed`	2026-10-11 17:01 UTC
#102	#94	`lab4-staging` ← `feature/94-final-hardening-and-release`	`a14f5e5` reviewed; final head `ab53f8a` (later changes were documentation-only relative to code head `5685095`)	CHANGES_REQUESTED 2026-10-12 07:51 UTC; APPROVED 2026-10-12 08:46 UTC	`N0TAW00D`	`876511f`	2026-10-12 09:18 UTC
#100	#92	`lab4-staging` ← `feature/92-ticket-workflow` (base `feature/90-actions-taken-foundation` until #98 merged)	`cde6394` ; approved at `22bde51`	CHANGES_REQUESTED 2026-10-07 16:52 UTC; APPROVED 2026-10-11 14:47 UTC	`N0TAW00D`	`576d93c`	2026-10-11 16:43 UTC
#101	#91	`lab4-staging` ← `feature/91-actions-taken-ui` (base `feature/92-ticket-workflow` until #100 merged)	`ab3767f` ; approved at `aa96cca`	CHANGES_REQUESTED 2026-10-07 16:52 UTC; APPROVED 2026-10-11 14:47 UTC	`N0TAW00D`	`50d542c`	2026-10-11 16:44 UTC
#104	none (review-readiness follow-up to #102)	`lab4-staging` ← `fix/lab4-review-readiness`	`b2ca5cd`	APPROVED 2026-10-13 05:40 UTC (re-ran the gate on a fresh PostgreSQL 16; two non-blocking notes)	`N0TAW00D`	`1141738`	2026-10-13 05:40 UTC
#103	release of #89 to #94 and	`main` ← `lab4-`	`1141738` (staging after #102 and	APPROVED 2026-10-13 12:47 UTC (re-ran the gate; four notes	`N0TAW00D`	`d07fefe` (tree `91a8f1a` , identical	2026-10-13 12:47

Figure 1.1b — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 2/9).

	follow-up to #102)	`review-readiness`		PostgreSQL 16; two non-blocking notes)			05:40 UTC
#103	release of #89 to #94 and #96	`main` ← `lab4-staging`	`1141738` (staging after #102 and #104)	APPROVED 2026-10-13 12:47 UTC (re-ran the gate; four notes after the merge)	`N0TAW00D`	`d07fefe` (tree `91a8f1a`, identical to `1141738`)	2026-10-13 12:47 UTC

Times are UTC as GitHub records them (Asia/Bangkok is seven hours ahead).

3. Findings and responses

PR	Reviewer finding (link)	Author response (link)	Fix commit / test	Resolution
#95	Seed count: the specification says nine, #98 seeds eight	Response	`4f49df8` fixes "Nine Actions" to "Eight Actions" (the only occurrence); MIG-04	Fixed; waiting for the reviewer to merge
#95, #98	Merge #95 before #98, so the specification predates the implementation (labsheet Part 2)	Acknowledged in the response above	Merge order #97, #95, #98	Followed
#98	Contract drift: five deviations must be written into specification §11 and `api-spec.md` in this PR	Response	`9c06707` : specification §11.16, `api-spec.md` §1, §2 and §3, `tests.md` (API-04, API-10, API-14, API-18, MIG-04, MIG-05)	Fixed; waiting for the reviewer's second look
#98	Rebase after #97 and #95 merge	Response	Rebased onto `lab4-staging` after #97 and #95 merged; head `321e1c2`	Done; waiting for the reviewer's second look
#98	Confirm the inactive-assignee test and its ID (labsheet Part 6)	Response	API-03 ("an inactive IT Staff user", "deactivated after the list was fetched") and API-04 in `tests.md`	Answered; waiting for the reviewer's second look
#99	Branch flow (§11.1): retarget to `lab4-staging` once #98 merges	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	None yet: this PR was started on #98's head so the dashboards could be built while #98 was in review. #98's commits reach `lab4-staging` through a merge commit, so the base changes without a rebase	Done: GitHub showed `lab4-staging` as the base when the reviewer merged it
#99	Update `reviewer.md` with this PR's review and the response (Part 1)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	This row, and the refresh of the #95, #97 and #98 rows to GitHub's records	Done
			Checked by script on 2026-10-08:	

Figure 1.1c — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 3/9).

#99	Update `reviewer.md` with this PR's review and the response (Part 1)	by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	This row, and the refresh of the #95, #97 and #98 rows to GitHub's records	Done
#99	Confirm every dashboard AC in `tests.md` maps to an existing test file (Part 3)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	Checked by script on 2026-10-08: AC-02, AC-26 to AC-33 and AC-40 each list tests whose files exist and contain the test ID (SEC-04 and SEC-05 sit in the `SEC-03/04` and `SEC-03/05/07` blocks). The check also found one wrong capture name, `requester-dashboard/first-use-empty`, which is `first-use-empty-fixture` on disk; `tests.md` and `ui-spec.md` §10 are corrected	Done
#99	Shorten the PR description: summary list first, long reasoning collapsed (AGENTS.md)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	The description was rewritten: a short list, then the details under `<details>`	Done
#100	Branch flow (§11.1): retarget to `lab4-staging` once #98 merges	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	None yet: this PR needs the Action tables of #98. #98's commits reach `lab4-staging` through a merge commit, so the base changes without a rebase	Done: GitHub showed `lab4-staging` as the base when the reviewer merged it
#100	Add this PR's review and the response to `reviewer.md` (Part 1)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	This row, and the refresh of the #95, #97 and #98 rows to GitHub's records (`d565eb5`)	Done
#100	Confirm §5.2 matches `workflowRules.ts` and the test oracle, role restrictions included (Part 7)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	UNIT-08 reads the §5.2 table as text and compares each of 8 statuses × 3 roles with `transitions.ts`, the gate in `workflowRules.ts` and `transition-oracle.ts` (26 tests, 8 injected faults each caught; `79086a5`). WF-06 sends every cell through the API for the three roles; SEC-02 refuses another Requester's Ticket (404); WF-01 shows a Requester never reaches the gate (403)	Confirmed, 0 differences
		No comment was		

Figure 1.1d — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 4/9).

	included (Part 7)	reviewer's approval of 11 Oct	through the API for the three roles; SEC-02 refuses another Requester's Ticket (404); WF-01 shows a Requester never reaches the gate (403)	
#100	Shorten the PR description: list first, reasoning collapsed (AGENTS.md)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	The description was rewritten: a four-line summary, then the reasoning under `<details>`	Done
#101	Missing evidence (Part 9): `artifacts/lab-04/screenshots/actions-taken/` does not exist	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	28 captures at desktop 1280, tablet 834 and phone 390 px: list, create form, validation, edit panel and the Requester's read-only view at all three widths, plus the other states (`ce61d33`). New test E2E-10 produces them; looking at the images showed the View button and status badge breaking over lines in the table at 1280 px, so a one-line assertion was written first and failed (`1f12430`), then fixed (`01c58de`)	Done
#101	Branch flow (§11.1): retarget to `lab4-staging` as parents merge	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	None yet: this PR builds on #100, which builds on #98. The base changes as each parent merges (merge commits, so no rebase)	Done: GitHub showed `lab4-staging` as the base when the reviewer merged it
#101	`reviewer.md` lists only #95, #97 and #98: add this PR's review and response (Part 1)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	The ledger now has #98, #99, #100 and #101 with GitHub's records, and each finding with its response	Done
#101	Shorten the PR description: list first, reasoning collapsed (AGENTS.md)	No comment was posted; answered by the fix in the next column and confirmed in the reviewer's approval of 11 Oct	The description was rewritten: a short list, then the reasoning under `<details>`	Done
#98	Nit: `reviewer.md` shows the #98 response as pending; make sure it is current once the stack lands	This ledger	#100 updated the response links (`d565eb5`); this version adds the approvals and merge details	Done
	Must fix before merge: the	No comment was	The description now says it writes	

Figure 1.1e — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 5/9).

#98	Nit: `reviewer.md` shows the #98 response as pending; make sure it is current once the stack lands	This ledger	#100 updated the response links (`d565eb5`); this version adds the approvals and merge details	Done
#100	Must fix before merge: the description says the "appears resolved" indication writes an event; the code writes none (BR-26)	No comment was posted; the description was corrected on GitHub on 11 Oct	The description now says it writes no event. The error came from my own shortening of the description on 8 Oct	Done before the merge
#100	Nit: the indication's `updatedAt` can differ from `requesterResolvedAt` when a later write is stored; §11.17 item 12 says "the same instant"	Agreed	`9a338a2` aligns §11.17 item 12 with the code; the WF-17 expectation was already about the later instant	Done in the final pass (#94)
#100	Nit: the cancel cascade sets each Action's `updatedAt` without the never-backwards guard	Agreed; BR-35 covers only the Ticket	Recorded as a known limit (`5a09a97` , `tests.md` §7 and specification §11.19 item 6)	Recorded
#99	Before merge: bring the merged stack into this branch (8 files conflict) and re-run server, client and E2E	The merge was made instead of a rebase, so the reviewed commits kept their hashes; the tree is the same	`ef68e76` (eight files, plus a duplicate `ActionStatusBadge` that git did not flag), `c7c4d21` ; server 758/758, client 325/325, E2E 59/59, counts updated in `9c4792a`	Done before the merge
#99	Before merge: `staffMetricWhere` and `requesterMetricWhere` are used only by a unit test	Removed, with the test cases that exercised them	`a465c3d` (UNIT-07 limited to the shared ACTIVE predicate and the drill-down queries; three injected faults still fail it), `9017167` (the helpers removed)	Done before the merge
#99	Nits: import `OPEN_ACTION_STATUSES` from `workflowRules.ts` ; the five dashboard queries are not one snapshot	Import done; the snapshot is accepted	`9017167` ; specification §11.19 item 5	Done
#102	`audit.spec.ts:53` : the 401 allowlist is wider than its stated reason (it would swallow a 500 on `/api/auth/me` , or a 401 console line from any URL)	Reply	Declarations now name status or failure, method and exact path, and license only the one console line each causes: UNIT-09 (`b62ea6e` red against a stub, `716f5f5` the rules, `aaa39f5` a pinned case; eight injected faults each fail it) and every spec rewritten (`5685095`)	Fixed
#102	`ui-spec.md` §11 row 19 claims "no no-op control", which E2E-06 does not assert	Reply; I narrowed the claim instead of adding a click-everything test	Row 19 now lists what is asserted and says a no-op control is not; `tests.md` §7 and specification §11.19 item 2 say the same; the AC-41 row of the traceability table says it is proved in part	Fixed

Figure 1.1f — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 6/9).

#102	`ui-spec.md` §11 row 19 claims "no no-op control", which E2E-06 does not assert	Reply; I narrowed the claim instead of adding a click-everything test	Row 19 now lists what is asserted and says a no-op control is not; `tests.md` §7 and specification §11.19 item 2 say the same; the AC-41 row of the traceability table says it is proved in part	Fixed
#102	Not verified by the reviewer: server (758) and E2E (64) suites need Postgres; re-run before the release PR	Comment: re-run on the code head of this round (see `tests.md` §6)	server and E2E results in `tests.md` §6	Done
#104	`tests.md` (header and §6.3) and `ui-spec.md` row 15 call the follow-up "local and unmerged"; false once it merges	No comment was posted; answered by the correction in the next column	The statements are rewritten in the final- `main` documentation pull request (this change)	Fixed after the merge
#104	The first-person reflection in `ai-use.md` must be the student's own words	No comment was posted. The reflection text was confirmed by the student on 13 October 2026; the file's status line says it was drafted for review and confirmed	`ai-use.md` status line and "My Reflection" (`b2ca5cd`)	Answered
#103	After the merge: `tests.md` (header, §6.3) and `ui-spec.md` row 15 still say "local and unmerged"	No comment was posted; answered by the correction in the next column	Rewritten in the final- `main` documentation pull request (`tests.md` header and §6, `ui-spec.md` rows 4 and 15, `ai-use.md`)	Fixed after the merge
#103	After the merge: `reviewer.md` §4 says the release pull request is "open and unreviewed"	No comment was posted; answered by the correction in the next column	§2 and §4 now record the review of #103 and #104, the approvals and the merge commits	Fixed after the merge
#103	After the merge: the reflection in `ai-use.md` must be the student's own words	No comment was posted; see the #104 row above	The student confirmed the text on 13 October 2026	Answered
#103	After the merge: close Issue #94 by hand, move its card to Done, make the Development-panel link, then run the final- `main` gate and record it	No comment was posted	Issues #89 to #94 and #96 are closed and the seven cards are Done (read from GitHub on 14 October 2026: all 37 cards of the board are Done); the final- `main` gate was run on `d07fefe` on 14 October 2026 and is recorded in `tests.md` §6.4	Done. The release pull request #103 itself is not linked to #94 in the Development panel (the card links #102)

4. Release

Item	Value

Figure 1.1g — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 7/9).

			tests.md §6.4	links #102)

4. Release

Item	Value
Staging gate evidence (commit, suites, builds, lint)	`lab4-staging` `876511f` ; code verification is recorded against code head `5685095` because PR #102's later changes through `ab53f8a` were documentation-only. Results: server 771/771 in default and shuffled order, client 328/328, both builds pass, lint has 3 known warnings, migration 11/11, Chromium E2E 64/64, cross-browser `e2e/lab-04` 70/72 (the two WebKit A11Y-01 Tab-walk limits). The review-readiness follow-up merged as #104 (`1141738`) is identified in `tests.md` §6.3
Release PR, review and merge	PR #103 (`lab4-staging` → `main`): APPROVED by `N0TAW00D` on 2026-10-13 at 12:47 UTC at `1141738` , merged by him at 12:47:55 UTC; merge commit `d07fefe` , tree `91a8f1a` , identical to `1141738`
Final `main` verification record	Run on `d07fefe` on 2026-10-14: server 771/771 (default and shuffled), client 328/328, builds, lint 0 errors (3 known warnings), migration 11/11, Chromium E2E 64/64, `e2e/lab-04` on Firefox, WebKit and Edge 70/72 (the two WebKit A11Y-01 limits), `npm audit` 0 vulnerabilities; details in `tests.md` §6.4. The reviewer's own run of the same tree is in his approval of #103

5. Reciprocal reviews — partner repository (kept separate)

Reviews performed by `Palapluem` on `N0TAW00D/TokTickIT` . Their tests, commits and screenshots are never evidence for this repository.

Partner PR	Reviewed head SHA	Review state and date	Key finding	Merge (by whom)
#101 `docs(lab-04): Lab 4 specification, API spec, UI spec and test plan`	`64a8e43` ; approved at `065c01c` after six commits that resolved every point	CHANGES_REQUESTED 2026-10-12 08:29 UTC; APPROVED 2026-10-12 09:05 UTC	The resolution gate (at least one `COMPLETED` Action before `RESOLVED` , BR-17 and D-06, test WF-03) contradicts their own rule that earlier tests are edited only for D-12, D-13 and D-18 (§4.2, AC-42): their Lab 3 test `ticket-status.api.test.ts` resolves owned Tickets without Actions and expects `200` . Five non-blocking suggestions in the review; all five were adopted, and I checked each against the partner's code and documents before approving	merged by `Palapluem` on 2026-10-12 09:20 UTC into the partner's `lab4-staging` (`a7c9f92`), after the approval
			Round 1: validation	

Figure 1.1h — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 8/9).

			were adopted, and I checked each against the partner's code and documents before approving	
#105 `feat(actions): Actions Taken API — list, create, update (#94)`	`623b902`, then `7afe760`; approved at `7e733eb`	CHANGES_REQUESTED 2026-10-14 06:03 UTC; CHANGES_REQUESTED 2026-10-15 07:39 UTC; APPROVED 2026-10-15 10:30 UTC	Round 1: validation order (BR-34) of POST and PATCH, the concurrent idempotency test, UUID v4 for `clientRequestId`, out-of-range assignee IDs. Round 2: malformed-JSON order, impossible calendar dates, UUID case for idempotency, `415` for a JSON request with no body, attachment-note text. Round 3 approval: all five fixed (checked by running the 704 tests and by undoing each fix); three non-blocking notes (an untested idempotency hash for `attachmentNotes`, an undocumented `413`, redundant offset checks)	merged by `Palapluem` on 2026-10-15 10:30 UTC into the partner's `lab4-staging` (`d0a797c`), ten seconds after the approval
#106 `feat(dashboard): Requester and IT Staff dashboard API (#97)`	`a0c36c2`	APPROVED 2026-10-15 10:31 UTC	Ran the 640 tests and checked the responses against api-spec §4 to §6; nine injected faults, seven caught; two untested BR-23 cases (a Ticket resolved and then reopened; the latest, not the first, change as `resolvedAt`), a helper used only by its own test, and an unspecified list order, all non-blocking	not merged when this was written: after #105 merged the branch conflicts in `server/tests/lab-04/authorization.api.test.ts`, which both pull requests create

Figure 1.1i — Rendered reviewer.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 9/9).

These nine overlapping segments present the complete reviewer.md GitHub Preview at main 6d38208, recording the final-main release and peer-review ledger. In the partner repository, PR #105 (third round) is merged; PR #106 remains approved but unmerged because of a conflict in `server/tests/lab-04/authorization.api.test.ts`.

GitHub Project board

A read-only check on 13 October confirmed that Issues #89, #90, #91, #92, #93, #94 and #96 are closed. The Project API shows all seven Lab 4 cards in Done and links them to their feature PRs: #89/#95; #90/#98; #91/#101; #92/#100; #93/#99; #94/#102; #96/#97. Figure 1.2 shows the individual cards, their statuses and linked PRs. For legibility, the screenshot uses a temporary label:lab-4 filter; the saved Project view and board records were not changed. The API is an independent status cross-check. Issue #94 was moved to Done after PR #103 merged and release-gate verification was recorded for main's tree.

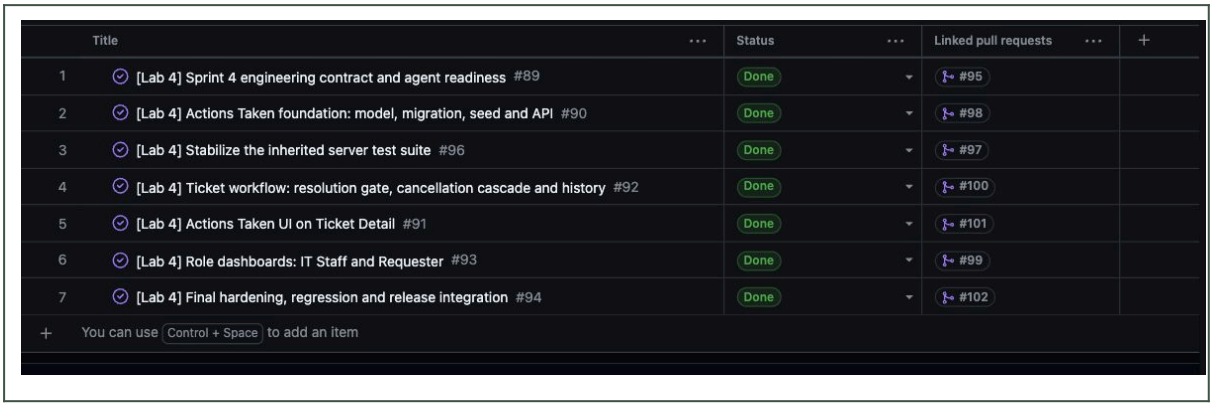


Figure 1.2 — Final GitHub Project board showing all seven Lab 4 Issues in Done and their merged PRs linked (captured 13 October 2026).

Network graph and commit evidence

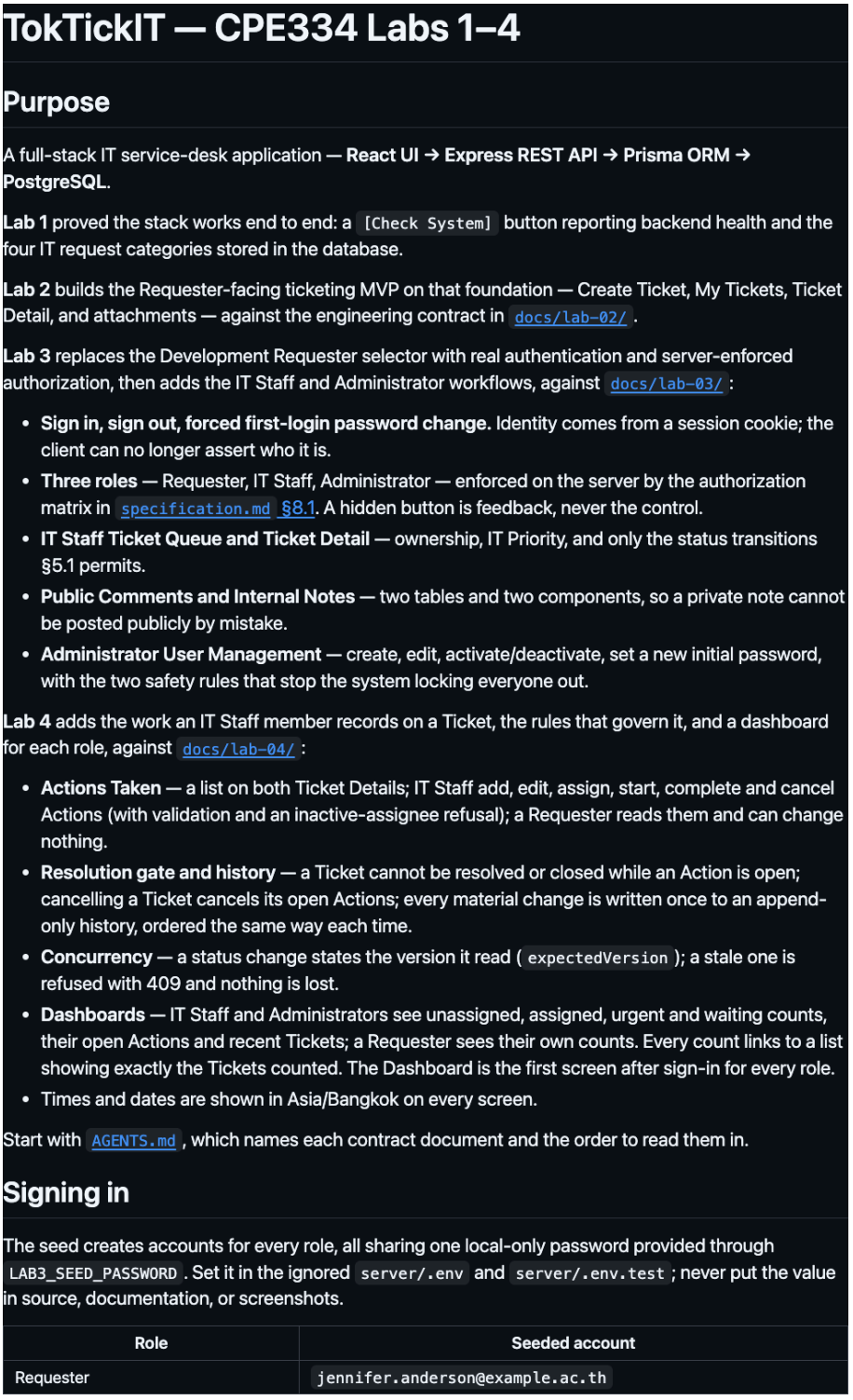
> RECENT-COMMIT VIEW — NOT COMPLETE HISTORY
>
> GitHub's Network graph shows recent commit ancestry, not a complete chronological history of every merged feature branch. The Pull Requests ledger above is the authoritative record of the Lab 4 PRs and merge SHAs.

Figure 1.3 — GitHub Network graph for Palapluem/toktickit_cpe334 after the Lab 4 release to main at d07fefe, captured 13 October 2026. Visible branches: feature/94-final-hardening-and-release, fix/lab4-review-readiness, lab4-staging and main.

Repository structure — Lab 4-relevant tracked paths

```
toktickit_cpe334/
├── client/
│   ├── src/                    React screens, routes and components
│   └── tests/lab-04/           Component and UI contract tests
├── server/
│   ├── src/                    Express actions, workflow and dashboards
│   ├── prisma/                 Schema, additive migrations and seed
│   ├── scripts/                Migration/recovery and performance smoke
│   └── tests/lab-04/           API, unit and concurrency tests
├── e2e/lab-04/                 Journeys, responsive checks and audit
├── artifacts/lab-04/screenshots/
├── docs/lab-04/
│   ├── specification.md
│   ├── api-spec.md
│   ├── tests.md
│   ├── ui-spec.md
│   ├── ai-use.md
│   └── reviewer.md
├── README.md
└── .gitignore
```

README — accepted staging snapshot



TokTickIT — CPE334 Labs 1–4

Purpose

A full-stack IT service-desk application — **React UI → Express REST API → Prisma ORM → PostgreSQL**.

Lab 1 proved the stack works end to end: a `[Check System]` button reporting backend health and the four IT request categories stored in the database.

Lab 2 builds the Requester-facing ticketing MVP on that foundation — Create Ticket, My Tickets, Ticket Detail, and attachments — against the engineering contract in `docs/lab-02/` .

Lab 3 replaces the Development Requester selector with real authentication and server-enforced authorization, then adds the IT Staff and Administrator workflows, against `docs/lab-03/` :

- **Sign in, sign out, forced first-login password change.** Identity comes from a session cookie; the client can no longer assert who it is.
- **Three roles** — Requester, IT Staff, Administrator — enforced on the server by the authorization matrix in `specification.md` §8.1. A hidden button is feedback, never the control.
- **IT Staff Ticket Queue and Ticket Detail** — ownership, IT Priority, and only the status transitions §5.1 permits.
- **Public Comments and Internal Notes** — two tables and two components, so a private note cannot be posted publicly by mistake.
- **Administrator User Management** — create, edit, activate/deactivate, set a new initial password, with the two safety rules that stop the system locking everyone out.

Lab 4 adds the work an IT Staff member records on a Ticket, the rules that govern it, and a dashboard for each role, against `docs/lab-04/` :

- **Actions Taken** — a list on both Ticket Details; IT Staff add, edit, assign, start, complete and cancel Actions (with validation and an inactive-assignee refusal); a Requester reads them and can change nothing.
- **Resolution gate and history** — a Ticket cannot be resolved or closed while an Action is open; cancelling a Ticket cancels its open Actions; every material change is written once to an append-only history, ordered the same way each time.
- **Concurrency** — a status change states the version it read (`expectedVersion`); a stale one is refused with 409 and nothing is lost.
- **Dashboards** — IT Staff and Administrators see unassigned, assigned, urgent and waiting counts, their open Actions and recent Tickets; a Requester sees their own counts. Every count links to a list showing exactly the Tickets counted. The Dashboard is the first screen after sign-in for every role.
- Times and dates are shown in Asia/Bangkok on every screen.

Start with `AGENTS.md` , which names each contract document and the order to read them in.

Signing in

The seed creates accounts for every role, all sharing one local-only password provided through `LAB3_SEED_PASSWORD` . Set it in the ignored `server/.env` and `server/.env.test` ; never put the value in source, documentation, or screenshots.

Role	Seeded account
Requester	`jennifer.anderson@example.ac.th`

Figure 1.4a — README GitHub Preview, segment 1/4 (overlapping; source unchanged at 876511f).

in source, documentation, or screenshots.

Role	Seeded account
Requester	`jennifer.anderson@example.ac.th`
IT Staff	`patricia.evans@example.ac.th`
Administrator	`margaret.hale@example.ac.th`

One seeded account (`david.lee@example.ac.th`) is deliberately left behind the must-change-password gate so that screen can be exercised without creating a user first.

Prerequisites

- Node.js 20+ and npm
- PostgreSQL running locally, with a disposable local database available for development (`toktickit_dev`) and one for automated tests (`toktickit_test`)

Installation

```
npm install               # root Playwright runner for Lab 2 E2E
cd server && npm install
cd ../client && npm install
```

Required Environment Variables

`.env.example` documents every variable; each package also has its own `.env.example` to copy from. Never commit the real `.env` files:

```
cp server/.env.example server/.env   # DATABASE_URL, PORT
cp client/.env.example client/.env   # VITE_API_BASE_URL
```

Lab 3 uses the local-only seed variable and adds two session variables, all documented in `.env.example` :

Variable	Purpose
`LAB3_SEED_PASSWORD`	Password used only to seed local/test accounts. Set the value in ignored env files; the seed refuses missing or too-short values.
`SESSION_COOKIE_SECURE`	`false` locally. A `Secure` cookie is never sent over plain HTTP, so setting it on a local server silently loses every session.
`CLIENT_ORIGIN`	The origin allowed to send the session cookie. Unset means "reflect the request's origin", which is what the split-port local setup needs — the server refuses to boot without it when `NODE_ENV=production` .

Running the Backend

```
cd server
npm run dev      # starts Express on http://localhost:3001 (tsx watch)
npm run build    # type-check + compile to dist/
```

Running the Frontend

```
cd client
npm run dev      # starts Vite on http://localhost:5173
npm run build    # production build
```

Figure 1.4b — README GitHub Preview, segment 2/4 (overlapping; source unchanged at 876511f).

```
cd client
npm run dev      # starts Vite on http://localhost:5173
npm run build    # production build
```

Database / Prisma Setup

Two databases: one for development, one for automated tests. They are separate on purpose — the test setup rewrites whatever it is pointed at, and that must never be the database holding demonstration data (`docs/lab-02/specification.md` §11.16).

```
createdb toktickit_dev
createdb toktickit_test

cd server
cp .env.example .env         # set DATABASE_URL → toktickit_dev and
LAB3_SEED_PASSWORD
cp .env.example .env.test    # set DATABASE_URL → toktickit_test and the same
LAB3_SEED_PASSWORD

npx prisma generate          # generate the Prisma Client
npx prisma migrate deploy    # apply migrations to the development database
npm run db:seed              # reference data — idempotent, safe to repeat
```

`DATABASE_URL` must point at a local, disposable PostgreSQL database. The seed password is required in the ignored env files and credentials are never committed — only `.env.example` is tracked.

The seed loads the reference data every screen depends on: four Categories, seven Related Systems, and five Development Requesters, one of which is deliberately inactive so the filtering rules (BR-10, BR-11) can be demonstrated rather than assumed.

Test Commands

```
cd server && npm test     # migrates + seeds toktickit_test, then runs the suite
cd client && npm test     # Vitest + Testing Library
```

`npm test` on the server prepares the test database first, so the suite is reproducible from a clean clone rather than depending on a database somebody prepared by hand. Use `npm run test:only` to skip that step when the test database is already current.

Server tests live in `server/tests/lab-01/` to `lab-04/` ; client tests under `client/tests/` . Every test cites the `FR` / `BR` / `AC` / `TC` / `STY` / `SEC-T` identifier it proves (`docs/lab-02/testing-contract.md` TCS-01).

Lab 3's authorization tests call the API **directly with the wrong role**, never through the UI — a test that drives the interface proves the button is hidden, not that the boundary holds. Before requesting review on anything touching auth, roles or user data, run the audit in `docs/lab-02/testing-contract.md` §7.

Lab 4 adds two scripts, both of which refuse any database whose name does not end in `_test` :

```
cd server
node scripts/verify-migration.mjs                # MIG-01, MIG-02: the Lab 4
migration on a populated Lab 3 database, and a restore
TEST_DATABASE_URL=… npx tsx scripts/perf-smoke.mjs  # PERF-01, PERF-02: dashboard
timing and query count; needs its own disposable database, not toktickit_test
```

Server tests that wait on a database lock use `holdTicketLock` and `waitUntilBlocked` (`server/tests/lab-04/lab4-fixtures.ts`) instead of sleeping.

E2E and responsive evidence

Issue #23 runs Playwright against a separate disposable database and starts its own API/client processes. Create a local database whose name ends in `_test` (for example `toktickit_e2e_test`), ensure `server/.env.test` contains the matching local credentials, then run from the repository root:

Figure 1.4c — README GitHub Preview, segment 3/4 (overlapping; source unchanged at 876511f).

Issue #23 runs Playwright against a separate disposable database and starts its own API/client processes. Create a local database whose name ends in `_test` (for example `toktickit_e2e_test`), ensure `server/.env.test` contains the matching local credentials, then run from the repository root:

```
npm run test:e2e
```

The runner applies migrations and the idempotent seed automatically. It uses the non-watch API command `tsx src/server.ts`, Vite's strict port 5174, and the isolated API port 3002. Lab 2's captures go to `artifacts/lab-02/screenshots/` and Lab 3's to `artifacts/lab-03/screenshots/` through a separate helper, so a Lab 3 run cannot overwrite submitted Lab 2 evidence. Set `E2E_DATABASE_URL` to override the `.env.test` value, but never point it at a development database. `DATABASE_URL` is a PostgreSQL URI, so percent-encode reserved credential characters such as `@`, `:`, `/`, and `%`.

Lab 4 evidence and demonstration

Lab 4's journeys are in `e2e/lab-04/`. Every one of them runs under an audit (`e2e/lab-04/audit.ts`): it fails on a console error, a failed request or an error response that the journey did not declare, and `audit.spec.ts` follows every link on every screen each role can reach. Run one at a time; the runner needs the API port 3002 and the client port 5174:

```
mkdir /tmp/toktickit-e2e.lock          # a simple lock when several agents share
one machine; rmdir it afterwards
npm run test:e2e                       # all labs; writes nothing that git tracks
CAPTURE_EVIDENCE=1 npm run test:e2e    # also writes Lab 4 screenshots under
artifacts/lab-04/screenshots/
```

Captures are written only when `CAPTURE_EVIDENCE=1` is set, and only for Lab 4; Lab 2 and Lab 3 captures are frozen. For evidence, point `E2E_DATABASE_URL` at a freshly created database ending in `_test`, so that the screens show the seed and nothing left over from earlier runs.

To see Lab 4 by hand: start the API and the client, then sign in as the seeded accounts above. As IT Staff (`patricia.evans@…`) open the Dashboard, follow "My open Actions" to a Ticket, add and complete an Action, try to resolve a Ticket that still has an open Action (it is not offered, and says why) and read its History. As the Requester (`jennifer.anderson@…`) open the same Ticket: the Actions and History are there, read-only.

Branch and Pull Request Rules

- `main` — stable release branch. Never develop directly on it.
- `lab2-staging`, `lab3-staging`, `lab4-staging` — per-sprint integration branches. Never develop directly on them.
- Each Issue is implemented on its own branch from the latest staging branch, including follow-up branches when peer review finds a release-candidate gap.
- Every feature Pull Request targets the sprint's staging branch, links its Issue through the Development panel, and requires peer review and passing tests before the reviewer merges it (merge commit, not squash/rebase).
- After all of a sprint's PRs are merged, Issues are closed and cards moved to Done, one release Pull Request promotes the staging branch to `main`.

Stacked branches. Where one Issue's branch is cut from the previous one rather than from staging, either delete each branch on merge so GitHub retargets the next Pull Request, or merge in **descending** order. Merging a stack in ascending order without deleting branches lands each PR in its already-merged parent, and nothing reaches staging — see `docs/lab-03/reviewer.md` §1.1.

Figure 1.4d — README GitHub Preview, segment 4/4 (overlapping; source unchanged at 876511f).

.gitignore — accepted staging snapshot

```
# Dependencies
node_modules/

# Environment variables — never commit real secrets
.env
.env.*
!.env.example

# Build output
dist/
build/
client/dist/
server/dist/

# Local course-report exports stay outside the product repository.
output/

# Generated Prisma client
server/src/generated/

# Test coverage
coverage/

# Playwright diagnostics (canonical responsive screenshots stay tracked)
artifacts/lab-02/playwright-results/
artifacts/lab-03/playwright-results/
artifacts/lab-04/playwright-results/
.playwright-cli/

# Editor / OS noise
.vscode/
.idea/
.DS_Store
Thumbs.db

# Logs
npm-debug.log*
*.log
```

Answer Part 2: Specification and Design Decisions

Question — How were Actions Taken, ticket workflow rules, and the two dashboards specified before implementation?

The approved engineering contract defines 28 functional requirements, 42 business rules and 41 acceptance criteria. An Action is one piece of work on a Ticket. Ticket Owner, Action assignee and Performed by remain separate identities. The contract specifies fields, permissions, lifecycle, transitions, metrics, ordering, migration/recovery and Product Definition of Done before implementation.

Actions move through PLANNED, IN_PROGRESS, COMPLETED or CANCELLED. Completion needs a Result; cancellation needs a reason. Terminal Actions cannot be edited. A Requester reads Actions on an owned Ticket but cannot mutate them. Resolution and closure are blocked by open Actions, while zero-Action legacy Tickets remain eligible under the existing role/status rules. Ticket cancellation atomically cancels open Actions. The Requester's 'problem appears resolved' remains a signal, not staff resolution.

Workflow writes state the Ticket version they read and reject stale requests with 409. Material changes append role-safe History with stable ordering. The migration is additive; recovery restores a dump into a disposable database. Seed re-runs preserve their recorded fields. SLA clocks, notifications, inventory, billing, approvals, BI and multi-tenancy are outside the approved scope.

The specification below is the accepted contract captured at lab4-staging @ 876511f; the same specification file is unchanged in main @ d07fefe. Product DoD statements are kept distinct from the final-main verification record.

Specification — full accepted contract at lab4-staging 876511f

API contract — authorization, request shapes and errors

Lab 4 Sprint Engineering Specification

Version: 1.0 — approved for implementation, 3 October 2026 **Status:** Approved by the student on 3 October 2026, after the four decision rounds recorded in §11. Implementation proceeds against this document. **Extends:** `docs/lab-03/specification.md`, which extends `docs/lab-02/specification.md`. Identifiers restart at 01 for this sprint; cross-lab references are written as `lab-03 BR-16`.

1. Sprint Goal

IT Staff record and finish the service work behind each Ticket as Actions Taken. A Ticket can be resolved only when no work is left open, every role gets a concise dashboard that leads straight to the matching detailed list, and every Lab 1–3 feature keeps working as one consistent Zen Green application.

2. Stakeholder Request Interpretation

The stakeholder wants the actual work on a Ticket to be planned, assigned and finished in the system, not remembered. An Action Taken is one line of that work. Three people stay distinct:

- the **Ticket Owner** coordinates the Ticket;
- the **assignee** is responsible for one Action;
- **Performed by** records who logged it.

They are often the same person, and the design never assumes they are.

The Requester's "problem appears resolved" signal stays a signal. Resolution remains a decision IT Staff make, and the backend refuses it while work is still open, whichever client asks.

Dashboards are entry points, not reports. Each number answers "where should I look?" and opens the list that produced it.

"Polish and harden" applies to the whole product, not only the new screens: every Lab 1–3 journey must still pass, and nothing obsolete may remain.

3. Scope

Included

- Actions Taken with the seven labsheet fields plus an assignee and a lifecycle.
- The resolution gate and the cancellation cascade.
- An append-only Ticket history.
- Optimistic concurrency on workflow writes.
- An IT Staff dashboard (also used by Administrators) and a Requester dashboard, both with drill-down.
- A `statusGroup` filter on both Ticket lists.
- Dashboard navigation and landing.
- The additive migration, the seed increment and a tested recovery procedure.
- Final regression and hardening of Labs 1–3.

Excluded

From labsheet §4.2: SLA clocks, escalation, on-call scheduling and breach notifications · email, SMS, LINE or push notifications · inventory, spare parts, purchasing and cost accounting · time-sheet billing and labour cost · approval workflows and e-signatures · BI tools, report builders and export warehouses · multi-tenancy and production-scale cloud operations.

Figure 2.1a — Specification GitHub Preview, segment 1/17 (overlapping; source unchanged at 876511f).

LINE or push notifications · inventory, spare parts, purchasing and cost accounting · time-sheet billing and labour cost · approval workflows and e-signatures · BI tools, report builders and export warehouses · multi-tenancy and production-scale cloud operations.

By decision (§11):

- No time-windowed metrics such as "last 7 days".
- No Administrator user-count cards.
- No deleting Actions or history events, and no editing COMPLETED or CANCELLED Actions.
- No attachment upload owned by an Action.
- No user-entered Action occurrence time.
- No change to the Lab 3 owner and IT Priority request shapes.
- No new Ticket statuses.

4. Functional Requirements

Actions Taken

- **FR-01** The system shall let IT Staff and Administrators create an Action Taken on a Ticket that is in a working status.
- **FR-02** The system shall list every Action Taken of a Ticket on IT Staff Ticket Detail and on the owning Requester's Ticket Detail.
- **FR-03** The system shall record Performed by and Action Date/Time automatically from the authenticated session and the server clock.
- **FR-04** The system shall let IT Staff and Administrators edit Description, Result, Follow-Up Required, Follow-up Note and Attachment Notes of an Action that is not terminal.
- **FR-05** The system shall let IT Staff and Administrators assign and reassign a non-terminal Action to an active IT Staff or Administrator user.
- **FR-06** The system shall let IT Staff and Administrators start, complete and cancel Actions through the lifecycle in §5.1.
- **FR-07** The system shall present Actions to Requesters read-only, without any mutation control.

Ticket workflow

- **FR-08** The system shall offer and accept only the Ticket transitions permitted for the caller's role and the Ticket's state (§5.2).
- **FR-09** The system shall refuse `RESOLVED` and `CLOSED` while any Action is open, including when a client bypasses the interface.
- **FR-10** The system shall cancel a Ticket's open Actions when the Ticket is cancelled.
- **FR-11** The system shall refresh the Ticket summary status after a successful change and keep the truthful status after a refused one.
- **FR-12** The system shall record an append-only history of material Ticket and Action changes.
- **FR-13** The system shall show that history according to role.
- **FR-14** The system shall detect a stale workflow update and refuse it without overwriting the newer change.

Dashboards

- **FR-15** The system shall provide an IT Staff dashboard, also used by Administrators, with the metrics in §5.3.
- **FR-16** The system shall show the current user's open Actions on the IT Staff dashboard.
- **FR-17** The system shall provide a Requester dashboard computed only from the Requester's own Tickets.
- **FR-18** The system shall show recently updated Tickets on both dashboards.

Figure 2.1b — Specification GitHub Preview, segment 2/17 (overlapping; source unchanged at 876511f).

- **FR-17** The system shall provide a Requester dashboard computed only from the Requester's own Tickets.
- **FR-18** The system shall show recently updated Tickets on both dashboards.
- **FR-19** The system shall link every metric to a list filtered by the same predicate.
- **FR-20** The system shall support `statusGroup=active` on the IT Staff Ticket Queue and My Tickets.
- **FR-21** The system shall make Dashboard the first navigation destination and the landing page after sign-in for every role.

Hardening and regression

- **FR-22** The system shall keep every Lab 1–3 function available to its permitted roles, unchanged except where §11 records a change.
- **FR-23** The system shall present loading, success, validation, empty, no-results, forbidden, not-found, conflict and safe-failure feedback for every new remote operation.
- **FR-24** The system shall prevent duplicate Actions caused by repeated clicks or network retry.
- **FR-25** The system shall keep entered form data after a recoverable failure.
- **FR-26** The system shall remove obsolete, placeholder or non-functional interface elements and console errors left from earlier labs.
- **FR-27** The system shall keep README setup, migration, seed, test and demonstration instructions current.
- **FR-28** The system shall present every new screen usably at desktop, tablet and mobile widths, operable by keyboard with visible focus.

5. Business Rules

Actions — ownership and roles

- **BR-01** An Action Taken belongs to exactly one Ticket. *(mandatory, labsheet §4.4)*
- **BR-02** The Ticket Owner coordinates the Ticket, but an Action Taken may be created by, assigned to, or completed by a different IT Staff member. Ticket Owner, Performed by and assignee are independent. *(mandatory, labsheet §4.4)*
- **BR-03** Only IT Staff and Administrators create or change Actions. A Requester never does.
- **BR-04** An Administrator has every IT Staff Action and dashboard capability. User management stays Administrator-only (continues `lab-03 §11.8`).
- **BR-05** The server sets these fields; none is accepted from the client, and none changes afterwards:
  - Performed by: the authenticated creator.
  - Action Date/Time: the server time of creation.
  - `completedBy` / `completedAt`.
  - `cancelledBy` / `cancelledAt`.

Assignee

- **BR-06** Every Action has exactly one assignee. It is required at creation, and the interface defaults it to the creator.
- **BR-07** An assignee must be an active IT Staff or Administrator user at the moment of assignment (parallel to `lab-03 BR-16`).
- **BR-08** Reassignment is permitted only while the Action is not terminal. An assignee who is later deactivated stays recorded and displayed.

Lifecycle and fields

- **BR-09** Action status is `PLANNED`, `IN_PROGRESS`, `COMPLETED` or `CANCELLED`. Only the transitions in §5.1 are permitted.

Figure 2.1c — Specification GitHub Preview, segment 3/17 (overlapping; source unchanged at 876511f).

Lifecycle and fields

- **BR-09** Action status is `PLANNED`, `IN_PROGRESS`, `COMPLETED` or `CANCELLED`. Only the transitions in §5.1 are permitted.
- **BR-10** `COMPLETED` and `CANCELLED` are terminal: no field edit, reassignment or transition afterwards.
- **BR-11** Completing an Action requires a non-blank Result.
- **BR-12** Cancelling an Action requires a reason of 1–500 characters.
- **BR-13** Description is required, 1–2000 characters. Result is optional until completion, at most 2000.
- **BR-14** Follow-up Note is required (1–2000) when Follow-Up Required is true. The server clears it when Follow-Up Required is false. A follow-up never blocks completion or resolution; further work is recorded as a new Action.
- **BR-15** Attachment Notes are optional, at most 500 characters: free text pointing to the Ticket's existing attachments. An Action has no upload of its own.
- **BR-16** Every text field is trimmed, and whitespace-only input counts as empty. Content is rendered as text, never as markup (continues `lab-03 BR-31`, `BR-32`).

Visibility

- **BR-17** A Requester sees every Action on their own Tickets, with all fields, read-only. People are shown by display name only. Private information belongs in Internal Notes.
- **BR-18** A Requester's call to an Action write operation is forbidden (403). Another Requester's Ticket is not found (404) (continues `lab-03 BR-14`, `BR-15`).

Ticket workflow

- **BR-19** Ticket statuses remain the eight values of `lab-03 BR-20`. Transitions remain `lab-03 §5.1`, constrained by BR-20 to BR-24 (§5.2).
- **BR-20** A Ticket may move to `RESOLVED` or `CLOSED` only when none of its Actions is `PLANNED` or `IN_PROGRESS`. The server enforces this inside the status-change transaction. *(mandatory, labsheet §4.5)*
- **BR-21** A Ticket with no Actions may be resolved or closed. Tickets created before Lab 4 keep their status and receive no backfilled Actions.
- **BR-22** When a Ticket becomes `CANCELLED`, each `PLANNED` or `IN_PROGRESS` Action becomes `CANCELLED` in the same transaction, with the reason "Ticket cancelled" and `cancelledBy` set to the user who cancelled the Ticket.
- **BR-23** Actions may be created, edited, assigned or transitioned only while the Ticket is `NEW`, `OPEN`, `IN_PROGRESS`, `WAITING_FOR_REQUESTER` or `REOPENED`. Otherwise they are read-only. Reopening a Ticket leaves terminal Actions as they are.
- **BR-24** A change to an Action never changes the Ticket's status.
- **BR-25** The Requester's "problem appears resolved" indication stays advisory and does not change the status (continues `lab-03 BR-22`, `BR-23`).

History and ordering

- **BR-26** Each material change writes one `TicketEvent` in the same transaction: status change, owner change, IT Priority change, Action created, updated, assigned, started, completed or cancelled. Cascaded cancellations are included.
- **BR-27** Ticket events are append-only. The application offers no update or delete, and the database rejects `UPDATE` and `DELETE` on the event table.
- **BR-28** IT Staff and Administrators see every event of a Ticket. A Requester sees only the status-change events of their own Tickets.

Figure 2.1d — Specification GitHub Preview, segment 4/17 (overlapping; source unchanged at 876511f).

- **BR-27** Ticket events are append-only. The application offers no update or delete, and the database rejects `UPDATE` and `DELETE` on the event table.
- **BR-28** IT Staff and Administrators see every event of a Ticket. A Requester sees only the status-change events of their own Tickets.
- **BR-29** Actions, Public Comments, Internal Notes and events are ordered by creation time ascending, then by identifier, so records with equal timestamps keep one stable order. Within one Ticket the events' `createdAt` values are also strictly increasing in commit order (§11.17).
- **BR-30** An event payload holds only identifiers, statuses, priorities and the names of changed fields. It never holds free text, credentials or personal data beyond the actor reference. Two structural values are allowed besides those: a `cascade` flag on `ACTION_CANCELLED` and a `cascadedActionCount` on the `STATUS_CHANGED` of a cancellation (§11.17).

Concurrency and retries

- **BR-31** Tickets and Actions carry an integer version.
  - A Ticket's version increments when its status, owner, IT Priority or resolution indication changes.
  - An Action's version increments on every change to it.
  - A Ticket status change states the Ticket's expected version. An Action edit, reassignment or transition states the Action's expected version.
  - A mismatch is refused as a conflict and changes nothing.
- **BR-32** Owner and IT Priority changes keep their Lab 3 request shape and need no expected version.
- **BR-33** Every Ticket status change and every Action creation locks the Ticket row for the length of its transaction, so BR-20 and BR-22 hold under concurrent requests.
- **BR-34** Action creation carries a client-generated request identifier, unique per Ticket. Repeating it returns the Action already created instead of creating another.
- **BR-35** Ticket field changes and Action writes update the Ticket's `updatedAt`, which never moves backwards: a request that waited for the Ticket lock and carries an earlier clock reading leaves the later value in place (§11.17). Public Comments and Internal Notes do not update it (Lab 3 behaviour).

Dashboards

- **BR-36** `ACTIVE` means `NEW`, `OPEN`, `IN_PROGRESS`, `WAITING_FOR_REQUESTER` or `REOPENED`.
- **BR-37** The backend computes every dashboard value from the same predicate that filters its drill-down list, so a count always equals the drill-down list's total.
- **BR-38** No metric depends on a calendar window. "Recently updated" means the five most recent `updatedAt` values, ties broken by identifier. Times are stored in UTC and shown in Asia/Bangkok.
- **BR-39** A Requester's dashboard covers only Tickets that Requester submitted.
- **BR-40** Dashboards return counts and bounded lists (at most 10 Actions and 5 Tickets), never whole collections. A zero is shown as 0 and still opens its list.

Migration and data

- **BR-41** The Lab 4 migration is additive. Every User, Session, Ticket, Attachment, Public Comment, Internal Note and reference row survives, with identical counts before and after.
- **BR-42** The seed remains idempotent and adds Actions, so that Tickets with zero, one and several Actions exist across the seeded statuses.

5.1 Action lifecycle

From	To	Who	Condition
— (create)	`PLANNED`	IT Staff, Administrator	Ticket in a working status (BR-23); active assignee (BR-07)
`PLANNED`	`IN_PROGRESS`	IT Staff, Administrator	—

Figure 2.1e — Specification GitHub Preview, segment 5/17 (overlapping; source unchanged at 876511f).

— (create)	`PLANNED`	Administrator	status (BR-23); active assignee (BR-07)
`PLANNED`	`IN_PROGRESS`	IT Staff, Administrator	—
`PLANNED`, `IN_PROGRESS`	`CANCELLED`	IT Staff, Administrator; system cascade (BR-22)	Reason 1–500 (BR-12)
`IN_PROGRESS`	`COMPLETED`	IT Staff, Administrator	Non-blank Result (BR-11)
`COMPLETED`, `CANCELLED`	—	—	Terminal (BR-10)

A Requester may make none of these moves. `PLANNED → COMPLETED` is refused: work is started before it is finished.

5.2 Ticket status transitions

The rows and roles are `lab-03 §5.1` unchanged.

- † means the BR-20 resolution gate applies.
- ‡ means the BR-22 cascade runs.

From	Requester may	IT Staff / Administrator may
`NEW`	`CANCELLED` ‡ (own Ticket)	`OPEN`, `IN_PROGRESS`, `CANCELLED` ‡
`OPEN`	—	`IN_PROGRESS`, `WAITING_FOR_REQUESTER`, `RESOLVED` †, `CANCELLED` ‡
`IN_PROGRESS`	—	`WAITING_FOR_REQUESTER`, `RESOLVED` †, `CANCELLED` ‡
`WAITING_FOR_REQUESTER`	—	`IN_PROGRESS`, `RESOLVED` †, `CANCELLED` ‡
`RESOLVED`	`REOPENED` (own Ticket)	`CLOSED` †, `REOPENED`
`CLOSED`	—	`REOPENED`
`REOPENED`	—	`IN_PROGRESS`, `WAITING_FOR_REQUESTER`, `RESOLVED` †, `CANCELLED` ‡
`CANCELLED`	—	— (terminal)

`RESOLVED → CLOSED` cannot normally meet an open Action, because BR-23 forbids new work on a resolved Ticket. The gate is still checked there, so data written outside the application cannot slip through.

5.3 Dashboard metric ledger

Predicates are written against the Ticket model. "Me" is the authenticated user. The seed-only values assume a database containing exactly the seed (§7): Staff values are as seen by Patricia Evans, Requester values by Jennifer Anderson. Tests compare against independent database counts, never against these constants.

ID	Label	Scope	Predicate	Drill-down	Seed-only value
MET-S01	Unassigned	IT Staff, Administrator	`ownerId IS NULL` and `ACTIVE`	Queue `ownerId=unassigned&statusGroup=active`	2

Figure 2.1f — Specification GitHub Preview, segment 6/17 (overlapping; source unchanged at 876511f).

					value
MET-S01	Unassigned	IT Staff, Administrator	`ownerId IS NULL` and `ACTIVE`	Queue `ownerId=unassigned&statusGroup=active`	2
MET-S02	Assigned to me	IT Staff, Administrator	`ownerId = me` and `ACTIVE`	Queue `ownerId=me&statusGroup=active`	1
MET-S03	Urgent	IT Staff, Administrator	`itPriority = URGENT` and `ACTIVE`	Queue `itPriority=URGENT&statusGroup=active`	3
MET-S04	Waiting for Requester	IT Staff, Administrator	`status = WAITING_FOR_REQUESTER`	Queue `status=WAITING_FOR_REQUESTER`	1
MET-S05	Tickets by status	IT Staff, Administrator	`status = X`, one count per status	Queue `status=X`	NEW 2 · OPEN 1 · IN_PROGRESS 2 · WAITING 1 · RESOLVED 1 · CLOSED 1 · REOPENED 1 · CANCELLED 1
MET-S06	My open Actions	IT Staff, Administrator	Action `assigneeId = me` and `status IN (PLANNED, IN_PROGRESS)`; total plus the 10 oldest	Each item opens its Ticket Detail	2
MET-S07	Recently updated	IT Staff, Administrator	all Tickets, top 5 by `updatedAt desc, id desc`	Ticket Detail; View all → Queue `sort=updatedAt:desc`	5 rows
MET-R01	Open	Requester	`requesterId = me` and `ACTIVE`	My Tickets `statusGroup=active`	2
MET-R02	Needs my attention	Requester	`requesterId = me` and `status = WAITING_FOR_REQUESTER`	My Tickets `status=WAITING_FOR_REQUESTER`	1
MET-R03	Resolved	Requester	`requesterId = me` and `status = RESOLVED`	My Tickets `status=RESOLVED`	0
MET-R04	Closed	Requester	`requesterId = me` and `status = CLOSED`	My Tickets `status=CLOSED`	0
MET-R05	Recently updated	Requester	`requesterId = me`, top 5 by `updatedAt desc, id desc`	Ticket Detail; View all → My Tickets `sort=updatedAt:desc`	3 rows

An Administrator sees the IT Staff dashboard with the same predicates. Under the seed, Margaret Hale's MET-S06 is 0, which demonstrates the zero state.

6. UI Specification Summary

The full specification is in `ui-spec.md`. Lab 2 and Lab 3 tokens, components, states, responsive rules and accessibility rules carry forward unchanged.

Figure 2.1g — Specification GitHub Preview, segment 7/17 (overlapping; source unchanged at 876511f).

The full specification is in `ui-spec.md`. Lab 2 and Lab 3 tokens, components, states, responsive rules and accessibility rules carry forward unchanged.

- **Shell:** Dashboard becomes the first navigation item for every role and therefore the landing page (`routes.ts homeFor`). The Requester routes to `/dashboard`; IT Staff and Administrators to `/staff/dashboard`.
- **New screens:** IT Staff Dashboard · Requester Dashboard.
- **Extended screens:**
  - IT Staff Ticket Detail gains Actions Taken (list, create mode, view/edit mode, assign, start, complete, cancel) and History.
  - Requester Ticket Detail gains a read-only Actions Taken list and status-only History.
  - The status control shows only permitted transitions and explains a blocked resolution.
- **New states:** conflict ("changed by someone else"), Ticket not workable, and resolution blocked by open Actions, beside the inherited loading, empty, no-results, validation, forbidden, not-found and safe-failure states.

7. Data Changes

All identifiers remain UUID, and timestamps remain `timestamptz` in UTC.

Model	Change
`ActionTaken`	**New.** See the field list below.
`TicketEvent`	**New.** `id`, `ticketId`, `actorId`, `type`, `actionId?`, `payload` (JSON, BR-30), `createdAt`
`Ticket`	Adds `version Int @default(1)` (BR-31). Existing rows take the default.
Every other model	Unchanged

`ActionTaken` fields:

- `id`, `ticketId`, `description`, `result?`, `followUpRequired`, `followUpNote?`, `attachmentNotes?`
- `status`, `assigneeId`, `performedById`
- `completedById?`, `completedAt?`, `cancelledById?`, `cancelledAt?`, `cancellationReason?`
- `requestId`, `version`, `createdAt` (the Action Date/Time), `updatedAt`

Enums:

- `ActionStatus { PLANNED, IN_PROGRESS, COMPLETED, CANCELLED }`
- `TicketEventType { STATUS_CHANGED, OWNER_CHANGED, IT_PRIORITY_CHANGED, ACTION_CREATED, ACTION_UPDATED, ACTION_ASSIGNED, ACTION_STARTED, ACTION_COMPLETED, ACTION_CANCELLED }`

Relationships:

- One `Ticket` has many `ActionTaken` and many `TicketEvent`.
- Each Action references four users by role: assignee, performer, completer and canceller, all named relations.
- Each event references its actor and optionally its Action.

Every foreign key is `ON DELETE RESTRICT`.

Indexes:

- `ActionTaken(ticketId, createdAt, id)` — the ordered Action list (BR-29)
- `ActionTaken(assigneeId, status)` — "My open Actions" (MET-S06)
- `ActionTaken(ticketId, requestId)` unique — idempotent creation (BR-34)
- `TicketEvent(ticketId, createdAt, id)` — ordered history

Figure 2.1h — Specification GitHub Preview, segment 8/17 (overlapping; source unchanged at 876511f).

- `ActionTaken(assigneeId, status)` — "My open Actions" (MET-S06)
- `ActionTaken(ticketId, requestId)` unique — idempotent creation (BR-34)
- `TicketEvent(ticketId, createdAt, id)` — ordered history
- `Ticket(updatedAt desc, id)` — the "Recently updated" lists (MET-S07, MET-R05)
- Lab 3's `Ticket(status, itPriority, createdAt)` and `Ticket(ownerId)` serve MET-S01 to MET-S05

Migration strategy.

- One additive migration creates the two enums, the two tables, the indexes and the append-only trigger, and adds `Ticket.version` with a default.
- No existing row is rewritten beyond that default, and no Action or event is backfilled (BR-21).
- Row counts from `server/scripts/row-counts.mjs` are compared before and after (BR-41).

Recovery.

- Prisma provides no down migration, so recovery is a verified restore.
- A disposable `_test` database at the Lab 3 state is dumped with `pg_dump -Fc`, migrated and checked.
- The dump is then restored with `pg_restore` into a second disposable database, whose counts must equal the pre-migration counts.
- Development data is never used.

Seed (idempotent, stable identifiers).

- The ten Lab 3 demo Tickets stay as they are.
- Eight Actions are added (BR-42):
  - `…900003` (OPEN, owner Patricia Evans): one PLANNED Action assigned to her.
  - `…900004` (IN_PROGRESS, owner Daniel Carter), three Actions:
    - COMPLETED, by and for Daniel;
    - IN_PROGRESS, created by Daniel and assigned to Olivia Reed;
    - PLANNED, created by Olivia and assigned to Daniel, with a follow-up.
  - `…900006` (RESOLVED): one COMPLETED and one CANCELLED Action.
  - `…900007` (CLOSED): one COMPLETED Action.
  - `…900010` (IN_PROGRESS, owner Margaret Hale): one IN_PROGRESS Action assigned to Patricia Evans. This is an Administrator-owned Ticket worked by IT Staff.
- `…900001`, `…900002`, `…900005`, `…900008` and `…900009` keep zero Actions.
- The seed writes no events. History starts with the first change made through the application.

Database-design decisions

1. **Append-only history is enforced by the database, not only by the absence of an API.** A `BEFORE UPDATE OR DELETE` trigger on `TicketEvent` raises an error. A future endpoint, script or agent mistake therefore cannot rewrite history, and the rule can be proved by a direct SQL test (BR-27). `TRUNCATE` is not blocked, on purpose (11.16): the trigger fires per row, and the test and E2E cleanups reset disposable `_test` databases by truncating the table. The application itself has no code path that truncates it.
2. **A version column plus a Ticket row lock, rather than either alone.**
   - The version detects a user acting on a screen that is out of date (BR-31).
   - The `SELECT … FOR UPDATE` on the Ticket serialises "create Action" against "resolve/close/cancel", which a version on the Ticket alone would not, because creating an Action does not change the Ticket's version (BR-33).
3. **References, not name snapshots.** Users are deactivated, never deleted (`lab-03 BR-37`). A `RESTRICT` reference therefore always resolves and always shows the current display name, without duplicating personal data into every Action.

Figure 2.1i — Specification GitHub Preview, segment 9/17 (overlapping; source unchanged at 876511f).

Action does not change the Ticket's version (BR-33).

3. **References, not name snapshots.** Users are deactivated, never deleted (`lab-03 BR-37`). A `RESTRICT` reference therefore always resolves and always shows the current display name, without duplicating personal data into every Action.
4. **Indexes follow the actual queries.** The Action list reads by Ticket in creation order, "My open Actions" reads by assignee and status, and the recency lists read by `updatedAt` . Each has a matching composite index, and the performance smoke test (AC-40) checks the result.

8. API Contract

Full shapes are in `api-spec.md` .

Capability	Endpoint	Roles
List a Ticket's Actions	`GET /api/tickets/:id/actions`	Requester (own), IT Staff, Administrator
Create Action	`POST /api/tickets/:id/actions`	IT Staff, Administrator
Edit / reassign Action	`PATCH /api/tickets/:id/actions/:actionId`	IT Staff, Administrator
Start / complete / cancel Action	`PATCH /api/tickets/:id/actions/:actionId/status`	IT Staff, Administrator
Ticket history	`GET /api/tickets/:id/history`	Requester (own, status events), IT Staff, Administrator
Change Ticket status	`PATCH /api/staff/tickets/:id/status` **— now requires `expectedVersion`** , gate and cascade	per §5.2
IT Staff dashboard	`GET /api/staff/dashboard`	IT Staff, Administrator
Requester dashboard	`GET /api/requester/dashboard`	Requester
Ticket lists	`GET /api/staff/tickets` , `GET /api/tickets` **— add `statusGroup=active`**	as Lab 3
Ticket detail	`GET /api/staff/tickets/:id` , `GET /api/tickets/:id` **— add `version` , `openActionCount`**	as Lab 3

Status codes: Lab 3's set is unchanged. Additions:

- **409** for a state that forbids an otherwise valid request:
  - `OPEN_ACTIONS_BLOCK_RESOLUTION`
  - `TICKET_NOT_WORKABLE`
  - `ACTION_TERMINAL`
  - `STALE_VERSION`
- **400** for:
  - `ASSIGNEE_NOT_ELIGIBLE`
  - `INVALID_ACTION_TRANSITION`

8.1 Authorization matrix additions

Every Lab 3 row is unchanged. An operation absent from the matrix is denied (`lab-03 BR-12`).

Figure 2.1j — Specification GitHub Preview, segment 10/17 (overlapping; source unchanged at 876511f).

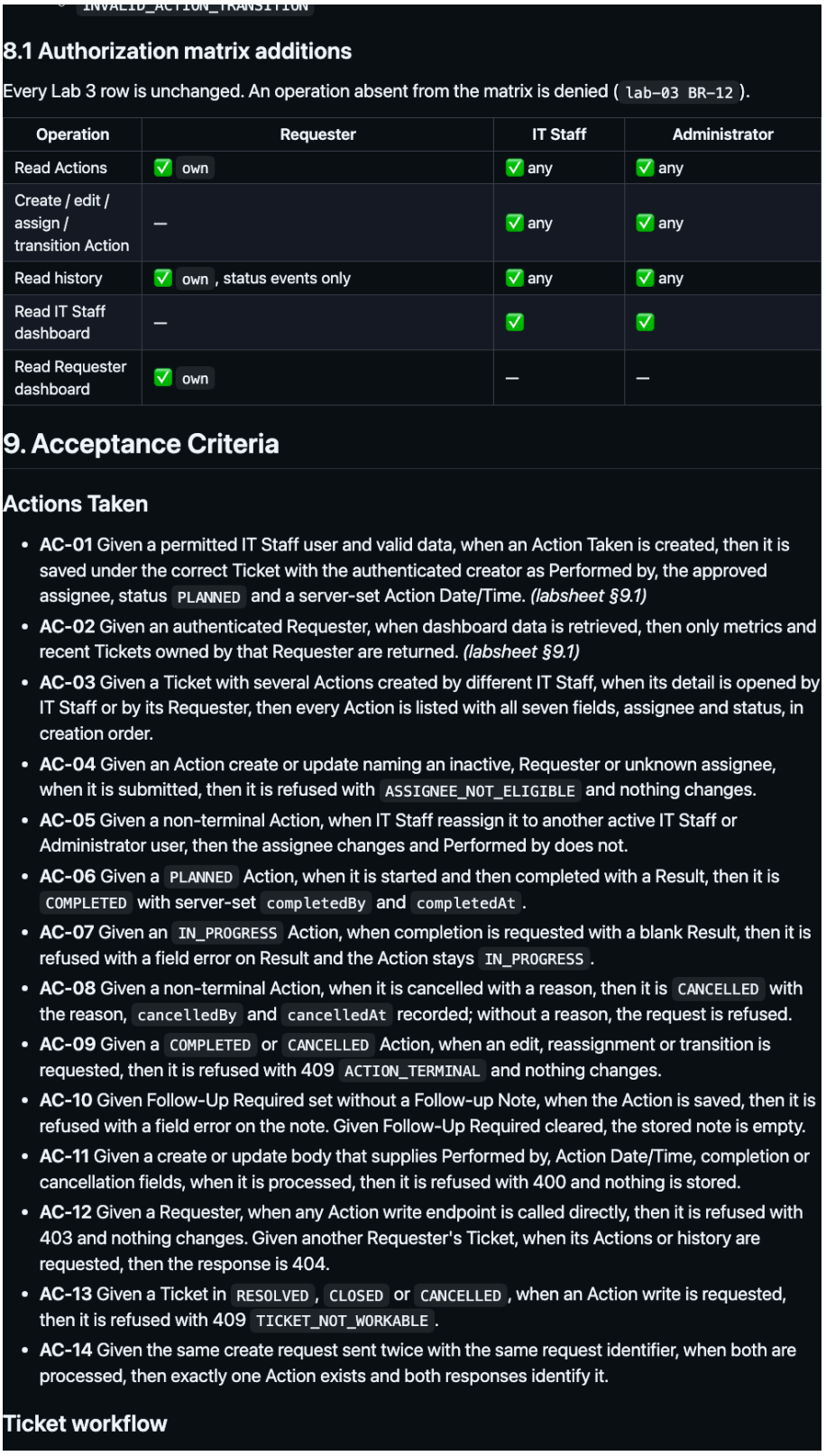


Figure 2.1k — Specification GitHub Preview, segment 11/17 (overlapping; source unchanged at 876511f).

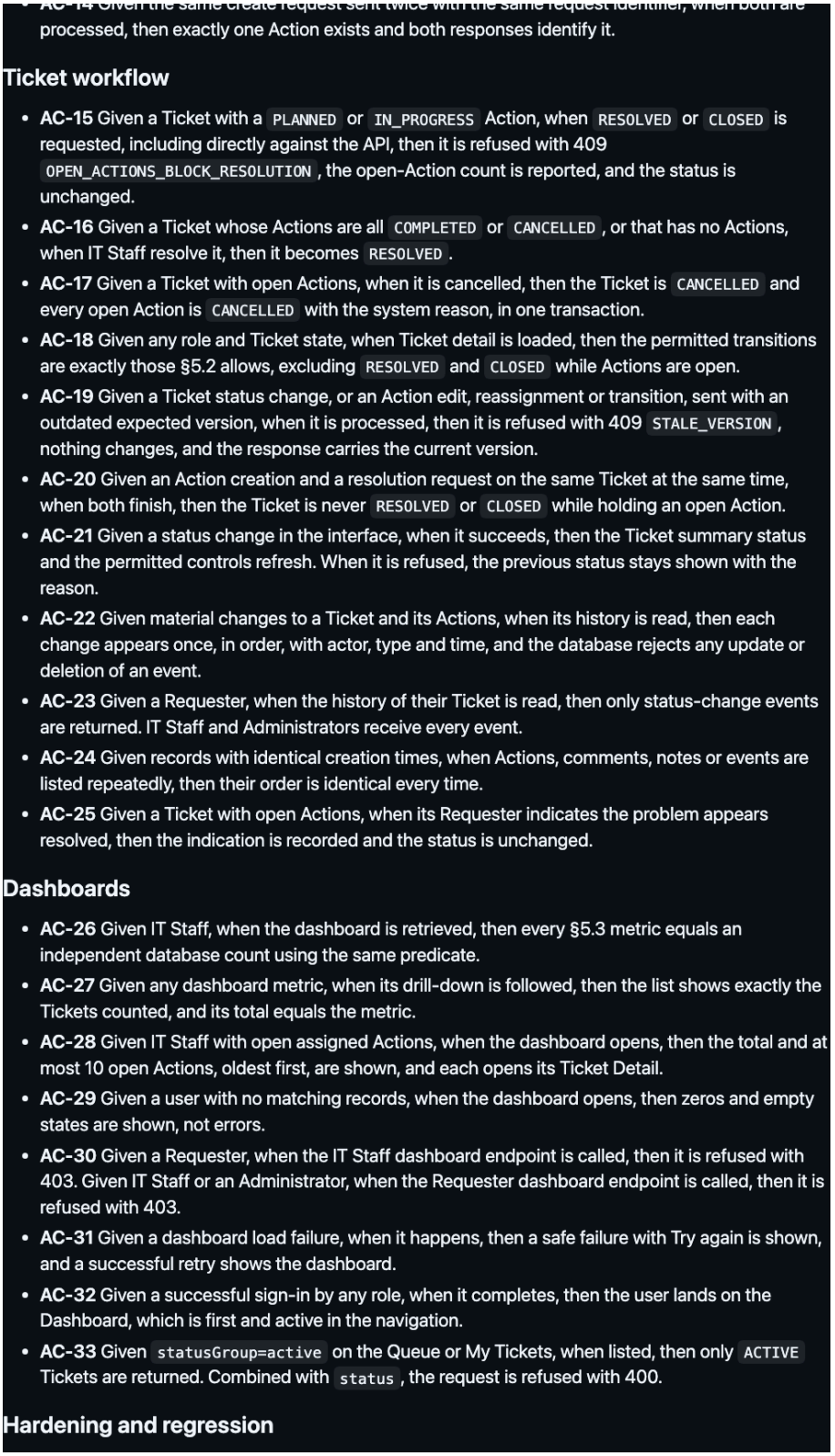


Figure 2.1l — Specification GitHub Preview, segment 12/17 (overlapping; source unchanged at 876511f).

- **AC-33** Given `statusGroup=active` on the Queue or My Tickets, when listed, then only `Active` Tickets are returned. Combined with `status`, the request is refused with 400.

Hardening and regression

- **AC-34** Given each new or changed screen at desktop, tablet and mobile widths, when rendered, then nothing is clipped or overlapping and the page does not scroll horizontally.
- **AC-35** Given keyboard-only use, when the Action form, status control, dialogs and dashboard drill-downs are operated, then every control is reachable with visible focus, and a dialog traps focus and returns it on close.
- **AC-36** Given a recoverable save failure or a conflict on the Action form, when it occurs, then the entered values remain and the user can retry.
- **AC-37** Given the complete Lab 1–3 automated suites, when run against the Lab 4 code, then they pass with no skipped, focused or disabled test.
- **AC-38** Given a database at the Lab 3 state, when the Lab 4 migration runs, then every pre-existing table keeps its row count and valid foreign keys, and restoring the pre-migration backup reproduces the original counts.
- **AC-39** Given the seed run twice, when counts are compared, then nothing is duplicated and Tickets with zero, one and several Actions exist.
- **AC-40** Given about 2,000 Tickets and 6,000 Actions locally, when each dashboard endpoint is called 20 times, then p95 latency is at most 300 ms and the number of queries does not grow with the data.
- **AC-41** Given the E2E journeys, when they run, then no console error, broken link, placeholder text or no-op control is observed.

Traceability from each criterion to its planned tests and evidence is maintained in `tests.md`.

10. Definition of Done

Part 1 — Product completion

- All approved scope in §3 is implemented, and nothing from the excluded list was added.
- Every acceptance criterion in §9 is linked to a passing automated test or recorded observable evidence.
- Every new protected operation in §8.1 has a test that calls it directly as each refused role and unauthenticated.
- BR-20, BR-22 and BR-33 are proved by real PostgreSQL tests, including a genuinely concurrent request pair.
- The migration preserves every row count, and the documented recovery restore has been executed successfully on a disposable database.
- The seed is idempotent across two runs.
- All server, client and E2E suites pass from the documented commands on the final `main`, with no skipped, focused or disabled test.
- Both production builds pass. Client lint has no errors, and any warning is disclosed.
- The performance smoke result (AC-40) is recorded with machine, dataset and command.
- No password, hash, session identifier or secret appears in any response, log, fixture, screenshot or committed file. `.env` and `.env.test` remain untracked.
- Screens conform to `ui-spec.md` and `style-contract.md`, and the §13 visual and accessibility checklist is completed with real results.
- An E2E run never rewrites committed Lab 2 or Lab 3 screenshots; evidence capture is explicit (`CAPTURE_EVIDENCE=1`).
- README, `AGENTS.md` and the test commands describe Lab 4 accurately.

Part 2 — Course delivery

Figure 2.1m — Specification GitHub Preview, segment 13/17 (overlapping; source unchanged at 876511f).

- README, `AGENTS.md` and the test commands describe Lab 4 accurately.

Part 2 — Course delivery

- Issues exist on the six-column board before their implementation and move through it truthfully, one active Issue at a time.
- Each Issue is implemented on its own branch and enters `lab4-staging` through a reviewed Pull Request linked through the Development panel. The reviewer merges, and the author answers every comment.
- One release Pull Request goes from `lab4-staging` to `main`, opened only after the staging gate passes.
- No Pull Request exists only to record that another Pull Request merged.
- `docs/lab-04/` contains `specification.md`, `tests.md`, `ui-spec.md`, `api-spec.md`, `reviewer.md` and `ai-use.md`.
- The report itself is governed separately and is not part of the product's Definition of Done.

11. Assumptions and Decisions

All decisions below were approved by the student on 3 October 2026 in four rounds, each answered "ตามที่แนะนำ" ("as recommended"). The student's private planning notes record the exact wording.

11.1 Actions have an assignee and a lifecycle the labsheet field list does not name. Labsheet §4.1 lists seven fields. Part 6 and example AC-01, however, require assignment, status transitions, completion, cancellation and inactive-assignee rejection. The lifecycle `PLANNED → IN_PROGRESS → COMPLETED`, with cancellation, is taken from the System-Level SDS (pp. 7, 12) because it is the course's own design baseline for service work.

11.2 Any active IT Staff or Administrator may act on any non-terminal Action. Lab 3 lets every IT Staff member act on every Ticket. Restricting completion to the assignee would block work when that person is away, and would multiply the permission tests, without a stakeholder need. Accountability comes from Performed by, `completedBy`, `cancelledBy` and the history.

11.3 Action Date/Time is the server's creation time. Labsheet §8.3 names it "Action create date/time". A user-entered occurrence time would need rules for past and future values that nothing in the request asks for.

11.4 Requesters see every Action field. Labsheet §8.3 says Requesters "will see all Actions Taken items". Internal Notes already provide a private channel, and the form says so.

11.5 The resolution gate means "no open work", not "work was recorded". Requiring at least one completed Action would need an exception for every legacy Ticket, and would turn trivial or duplicate Tickets into busywork.

11.6 Ticket cancellation cascades to open Actions without a new mandatory reason. The SDS (p. 11) requires a cancellation reason and the cascade. The cascade is adopted, because open work on a cancelled Ticket would pollute "My open Actions". The new reason field is not adopted, because it would change the Lab 3 request shape and screens for every role.

11.7 Actions are writable only while the Ticket is in a working status, and never change the Ticket status. This keeps BR-20 true by construction and avoids hidden status changes that the "permitted transitions only" rule (labsheet §8.4) could not show.

11.8 History is a new append-only `TicketEvent`. `lab-02 §11.10` deferred the SDS event log. Lab 4's Part 7 asks for append-only behaviour and stable ordering in the workflow itself, so the deferral ends here, scoped to the events in BR-26.

11.9 Optimistic concurrency is required on status changes and Action writes only. Labsheet §6.1 asks that a user never unknowingly overwrite another's workflow change. The Lab 3 owner and IT Priority endpoints keep their request shape to protect the existing contract and tests, but their changes increment the version, so a stale status change after them is still detected. The status endpoint's new required `expectedVersion` is a recorded change to `lab-03` `api-spec.md` §8; its Lab 3 UI and tests are updated in the same increment.

Figure 2.1n — Specification GitHub Preview, segment 14/17 (overlapping; source unchanged at 876511f).

Priority endpoints keep their request shape to protect the existing contract and tests, but their changes increment the version, so a stale status change after them is still detected. The status endpoint's new required `expectedVersion` is a recorded change to `lab-03` `api-spec.md` §8; its Lab 3 UI and tests are updated in the same increment.

11.10 Dashboards have no time-windowed metrics. Every labsheet example can be expressed as a state count or a "five most recent" list. That removes calendar-boundary and time-zone edge cases from correctness while keeping the dashboards useful.

11.11 Both Ticket lists gain `statusGroup=active`. Lab 2 and Lab 3 accept a single `status`. An "Open" metric spans five statuses, and its drill-down must show exactly the counted Tickets (BR-37).

11.12 Dashboard is the landing page for every role, and Administrators reuse the IT Staff dashboard. Labsheet §8.1 calls the dashboard "a concise operational starting point", and §4.6 allows the reuse. Three Lab 3 assertions about where a user lands after signing in or changing a password change accordingly, and nothing else in the Lab 1 to 3 tests changes: the E2E AUTH-03 (Requester lands on My Tickets, now Dashboard), the E2E ADMIN-04 (after the forced password change the page showed the Ticket Queue, now the Dashboard), and the client tests `Login.test.tsx` and `ChangePassword.test.tsx` (the Requester's route after sign-in). The first version of this section named only AUTH-03; the other two were found when Issue #93 ran the full regression (5 October 2026).

11.13 Evidence capture is explicit. Every Lab 3 E2E run rewrote 48 committed screenshots. From Lab 4, a normal run writes only to ignored runner output, and captures go to `artifacts/lab-04/screenshots/` only with `CAPTURE_EVIDENCE=1`.

11.14 Quality targets are local and measurable. The performance smoke test (AC-40) runs on the student's Mac against a disposable dataset. It is reported as a local measurement, not as production capacity.

11.15 Supersessions of the System-Level SDS recorded by this sprint.

- Ticket vocabulary and the Requester's cancel and reopen rights follow `lab-03 BR-20` and `§5.1`, not D-02.
- Ticket cancellation carries no new reason field (11.6).
- Reopening goes to `REOPENED` (Lab 3), not to In Progress or New.
- The API prefix `/api` and validation status 400 follow `lab-02 §11.3–§11.4`.
- Attachments remain on the local filesystem (`lab-02 §11.5`).

11.16 Gaps in this contract found while building the Action API (#90), recorded in the Pull Request that closes them. Each is covered by a test; the test IDs are in `tests.md`.

1. **An edit that changes nothing is a success, not an error.** When every supplied field equals the stored value and the assignee is unchanged, `PATCH /api/tickets/:id/actions/:actionId` answers 200 with the unchanged Action: no new version and no event (BR-31 counts changes). A request that supplies no editable field at all stays 400 `VALIDATION_FAILED`. Test API-04, API-10.
2. **Event payloads are stored as identifiers.** BR-30 already limits a payload to identifiers, statuses, priorities and field names. The `details` examples in `api-spec.md` §2 show display names because the history endpoint (#92) resolves identifiers to current display names when it reads, so a renamed person shows the current name. Test API-18.
3. **An Action addressed under the wrong Ticket is not found.** Any Action write or status change whose `:actionId` does not belong to `:id`, does not exist, or is not a UUID answers 404 `ACTION_NOT_FOUND`, the same in every case, and changes nothing (BR-01). The code is new in `api-spec.md` §1. Test API-14.
4. **The seed has eight Actions, not nine.** The list in §7 always summed to eight (1 + 3 + 2 + 1 + 1) and the seed and its test agree; the word "Nine" was a drafting error, corrected in #95. A second seed run now also leaves every Action column as it was, including `updatedAt`. Test MIG-04.

Figure 2.1o — Specification GitHub Preview, segment 15/17 (overlapping; source unchanged at 876511f).

api-spec.md §1. Test API-14.

4. **The seed has eight Actions, not nine.** The list in §7 always summed to eight (1 + 3 + 2 + 1 + 1) and the seed and its test agree; the word "Nine" was a drafting error, corrected in #95. A second seed run now also leaves every Action column as it was, including `updatedAt`. Test MIG-04.
5. **`TRUNCATE` of the history table is allowed.** The append-only trigger rejects row `UPDATE` and `DELETE` (BR-27) but not `TRUNCATE`, so that disposable test databases can be reset (§7, decision 1). Test MIG-05 proves the trigger rejects `UPDATE` and `DELETE`, leaves the row alone, and still allows the reset.

11.17 Decisions made while building the Ticket workflow (#92), and what the independent audit changed. Each is covered by a test; the IDs are in `tests.md`.

1. **Events of one Ticket keep their commit order.** Every writer of events holds the Ticket row lock (BR-33). A change starts its events at its own clock reading or 1 ms after the Ticket's latest event, whichever is later, and spaces the events of one change 1 ms apart. `createdAt` is therefore strictly increasing within a Ticket, whatever the server clock does and however long a cancellation cascade ran. A burst can end up to about 1 ms per event ahead of the wall clock; the screen shows minutes. This replaces the earlier rule that spaced only the events of one change, which let a later change sort before the tail of an earlier cascade (AC-22). Test WF-15.
2. **Events store identifiers, and names are resolved when history is read.** An event for a person who was renamed later shows the current name; history is not a snapshot of names (BR-30).
3. **A status request for the status the Ticket already has is `400 INVALID_STATUS_TRANSITION`.** A repeat of a request that succeeded, with the old `expectedVersion`, is `409 STALE_VERSION`. Neither moves the version, `updatedAt` or the history. Test WF-13.
4. **An owner or IT Priority request that changes nothing is a success without effect:** 200, no new version, no event and no change to `updatedAt` (BR-31, BR-35). Test WF-13.
5. **Claiming a `NEW` Ticket is one write.** The owner and the move to `OPEN` happen together: one version increment, `OWNER_CHANGED` then `STATUS_CHANGED`.
6. **A status request is checked in this order:** body (400), Ticket (404), `expectedVersion` (409), transition for the role (400 or 403), the gate (409). A missing `expectedVersion` is therefore 400 even for an unknown Ticket. Test WF-14.
7. **The status response is the summary** (`id`, `status`, `version`, `updatedAt`, `permittedTransitions`, `cancelledActionCount`); the screen reloads the Ticket for the rest.
8. **History is a new authorization operation, `history:read`:** own Ticket with status events only for a Requester, any Ticket for staff (§8.1).
9. **Two structural values widen BR-30's list:** `cascade` (a boolean) and `cascadedActionCount` (a count, present on every cancellation, including 0). Neither carries text or personal data.
10. **Both Ticket details add `version` and `openActionCount`.** The Requester can already read the Actions, so this discloses nothing staff-only.
11. **The Ticket's `updatedAt` never moves backwards.** Every writer holds the Ticket lock and keeps the later of its own clock reading and the stored `updatedAt`, so "recently updated" (BR-38) follows the order the changes were committed in, as the history does. Test WF-16.
12. **The Requester's "appears resolved" indication is written under the Ticket lock, like every other change.** It reads the status after taking the lock, so a Ticket that was cancelled or closed while the request waited is refused with `409 TICKET_CLOSED` and nothing is stored (BR-33). It moves the version (BR-31) and stamps `updatedAt` with the instant of the request, or keeps the later instant that a write already stored, never backwards (BR-35); so `updatedAt` equals `requesterResolvedAt` unless a later write is already there. It writes no event (BR-26 lists none). Tests WF-17.
13. **Every screen shows times and dates in Asia/Bangkok, including the Lab 2 and Lab 3 screens (BR-38).** This replaces the wording of `lab-02 §11.13` that a date is "rendered in the requester's local time". Its reason, that the year in a Ticket Number and the date beside it must not contradict each other, holds on a device in any zone only if the display zone is fixed. The list column that shows a date alone can differ from the device's date near midnight. Test UI-26.

11.18 Decisions made while building the Actions Taken screen (#91), and what the independent audit changed. Each is covered by a test; the IDs are in `tests.md`.

Figure 2.1p — Specification GitHub Preview, segment 16/17 (overlapping; source unchanged at 876511f).

shows a date alone can differ from the device's date near midnight. Test UI-26.

11.18 Decisions made while building the Actions Taken screen (#91), and what the independent audit changed. Each is covered by a test; the IDs are in `tests.md`.

1. **A retry after a lost answer keeps what was typed (AC-36).** A save that fails without an answer leaves it unknown whether the server stored it. The retry resends the first entries under their `requestId` (BR-34), which either finds the Action already made or makes it, and then saves any later changes onto that Action as an update. It never replaces the user's changes with the stored first entries, and never creates a second Action. Tests UI-23, E2E-07.
2. **A form open when the Ticket stops taking Actions is read-only (BR-23).** Its entries stay visible and Save is withdrawn; the server still refuses the write. The Actions are loaded again whenever the Ticket status changes, so a cancellation's cascade shows at once (BR-22). Tests UI-24, E2E-08.
3. **Focus rules (AC-35).** The first invalid field receives the focus on submit, including after a server refusal and in the cancel dialog. Closing a panel, form or dialog gives the focus back to its trigger, or to the details heading when the trigger is gone. A dialog whose controls are all disabled by a save in flight holds the focus itself. Tests UI-20 to UI-22, A11Y-02.
4. **Times use one formatter, in Asia/Bangkok (api-spec §1).** Both the Actions and the History use it, so a device in another zone shows the same time. Tests UI-19, E2E-09.
5. **Requester controls depend on the role, not on the Action's state.** A Requester sees no control on a Planned or In progress Action either; terminal Actions alone proved nothing. Test UI-25.

11.19 Decisions made while hardening and releasing (#94). Each is covered by a test; the IDs are in `tests.md`.

1. **A link to an Action names the Action's own row (AC-28, AC-41).** The dashboard's "My open Actions" linked to `#actions`, an anchor that did not exist; the console and link audit (E2E-06) found it. The link is now `/staff/tickets/<id>#action-<actionId>`, and the Actions Taken section scrolls once to a row (or to itself) named in the address after its list has loaded, because a single-page application does not scroll to an anchor that did not exist at load. Test UI-27.
2. **Every Lab 4 journey is audited (AC-41, FR-26).** A journey fails on a console error, a page error, a failed request or an error response that it did not declare; a journey that causes one on purpose (a refusal, a conflict, an injected failure) declares it, with the reason, in its own test. E2E-06 also follows every link on every screen each role can reach. One declaration is common to all: a signed-out visit asks who is signed in and is answered `401` (`lab-03 api-spec` §2), which Chromium logs as an error; it is declared, not removed, because the answer is part of the Lab 3 contract. A declaration names the status or failure, the method and the exact address, and licenses only the one console line it causes (UNIT-09). The audit does not click controls, so a control that does nothing is caught only where a journey uses it and asserts its effect (`tests.md` §7).
3. **`/system-check` and `/style-guide` are kept (FR-26).** They are Lab 1 and Lab 2 screens that work, are named in those contracts and are covered by their tests; they are not placeholders. A Requester or IT Staff member does not reach them from any navigation.
4. **A not-found state has no heading of its own.** "Ticket not found" and a foreign Ticket's "Could not load Ticket details" are messages inside the shell (Lab 3). They stay as they are; REG-04 captures both beside the forbidden and failure states (ui-spec §11 row 16).
5. **The dashboards read their counts in separate queries, not in one snapshot.** The five dashboard queries run together, not in a transaction, so a count and the list beside it can differ by a write made in between. Accepted for a dashboard; the drill-down always shows the Tickets of the moment it is opened.
6. **A cancellation cascade stamps each Action's own `updatedAt` with the request's instant, without the never-backwards rule.** BR-35 covers the Ticket only. Recorded in `tests.md` §7.

Figure 2.1q — Specification GitHub Preview, segment 17/17 (overlapping; source unchanged at 876511f).

Answer Part 3: Test Design and Traceability

Question — How do planned tests trace the Lab 4 acceptance criteria to test results from the tree promoted to main?

The accepted test plan contains 118 named rows. Each row identifies its requirement or business rule, expected result, actual test-file path and recorded status. Unit, API, component, migration/recovery, performance and E2E checks provide complementary coverage. Hiding a UI control alone does not prove server-side authorization. Red-phase and mutation checks help show whether tests detect the intended faults.

The 41 acceptance criteria are mapped to the accepted test plan. The independent release-gate run was executed on product-release main at d07fefe (tree 91a8f1a); PR #103's review gate at 1141738 covered the same tree and remains a separate verification record. The current tracked tests.md at main 6d38208 is rendered below in 30 overlapping L2 segments; its release-status record is current, while the executed test results remain tied to d07fefe. This mapping demonstrates plan-to-test traceability rather than replacing the rendered source.

Actual test and supporting-source path keys

Key	Actual test / supporting file or directory
T01	client/src
T02	client/tests/lab-02
T03	client/tests/lab-03/ChangePassword.test.tsx
T04	client/tests/lab-03/Login.test.tsx
T05	client/tests/lab-03/StyleContract.test.tsx
T06	client/tests/lab-04/ActionsTaken.test.tsx
T07	client/tests/lab-04/AppNavigation.test.tsx
T08	client/tests/lab-04/BangkokTime.test.tsx
T09	client/tests/lab-04/RequesterDashboard.test.tsx
T10	client/tests/lab-04/StaffDashboard.test.tsx
T11	client/tests/lab-04/TicketWorkflow.test.tsx
T12	client/tests/lab-04/dateTime.test.ts
T13	e2e/lab-02
T14	e2e/lab-03
T15	e2e/lab-04/actions-taken-flow.spec.ts
T16	e2e/lab-04/actions-taken-recovery.spec.ts

Key	Actual test / supporting file or directory
T17	e2e/lab-04/audit-rules.ts
T18	e2e/lab-04/audit.spec.ts
T19	e2e/lab-04/audit.ts
T20	e2e/lab-04/dashboards.spec.ts
T21	e2e/lab-04/regression.spec.ts
T22	e2e/lab-04/ticket-resolution.spec.ts
T23	server/scripts/perf-smoke.mjs
T24	server/scripts/verify-migration.mjs
T25	server/tests/lab-01
T26	server/tests/lab-04/action-rules.unit.test.ts
T27	server/tests/lab-04/actions-taken.api.test.ts
T28	server/tests/lab-04/audit-rules.unit.test.ts
T29	server/tests/lab-04/dashboard-metrics.unit.test.ts
T30	server/tests/lab-04/migration-seed.api.test.ts
T31	server/tests/lab-04/requester-dashboard.api.test.ts
T32	server/tests/lab-04/staff-dashboard.api.test.ts
T33	server/tests/lab-04/ticket-workflow.api.test.ts
T34	server/tests/lab-04/transition-spec.unit.test.ts
T35	server/tests/lab-04/workflow-rules.unit.test.ts

Acceptance-criterion traceability

AC	Planned test IDs	File keys / verification scope
AC-01	API-01, E2E-01, E2E-10	T27, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-02	SEC-05, DASH-07, UI-16	T31, T09. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-03	API-02, API-16, UI-01, E2E-01	T27, T06, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).

AC	Planned test IDs	File keys / verification scope
AC-04	API-03, E2E-01	T27, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-05	API-04, E2E-01	T27, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-06	API-05, E2E-01	T27, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-07	API-06, UNIT-03, UI-04	T27, T26, T06. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-08	API-07, UI-05, E2E-01	T27, T06, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-09	API-08, UI-06	T27, T06. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-10	API-09, UNIT-02, UI-02, E2E-01, E2E-10	T27, T26, T06, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-11	API-10	T27. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-12	SEC-01, SEC-02, UI-08, E2E-01	T27, T33, T06, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-13	API-11, UI-07, E2E-02	T27, T06, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-14	API-12, UI-09	T27, T06. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-15	WF-01, WF-02, E2E-02	T33, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-16	WF-03, WF-04, E2E-02	T33, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-17	WF-05, E2E-02	T33, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-18	UNIT-04, WF-06, UI-11	T35, T33, T11. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-19	API-13, WF-07, WF-14, E2E-03	T27, T33, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).

AC	Planned test IDs	File keys / verification scope
AC-20	WF-08	T33. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-21	UI-12, E2E-02	T11, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-22	WF-09, SEC-06, MIG-05, E2E-02	T33, T30, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-23	WF-10, SEC-02, UI-13, E2E-02	T33, T27, T11, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-24	UNIT-05, WF-11	T35, T33. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-25	WF-12	T33. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-26	DASH-01, DASH-07, E2E-04	T32, T31, T20. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-27	DASH-02, DASH-08, UI-14, E2E-04	T32, T31, T10, T20. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-28	DASH-03, UI-14, UI-27, E2E-04	T32, T10, T06, T20. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-29	DASH-04, DASH-09, UI-15, UI-16, E2E-04	T32, T31, T10, T09, T20. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-30	SEC-04, E2E-05	T32, T20. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-31	UI-15, UI-17, E2E-05	T10, T09, T20. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-32	UI-18, E2E-04	T07, T20. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-33	DASH-10	T31. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-34	RESP-01, E2E-10	T20, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-35	UI-05, A11Y-01	T06, T15. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-36	UI-10, E2E-03	T06, T22. Suite PASS @ tree 91a8f1a (run at d07fefe).

AC	Planned test IDs	File keys / verification scope
AC-37	REG-01, REG-02, REG-03	T25, T02, T13, T14, T04, T03. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-38	MIG-01, MIG-02, MIG-03	T24, T30. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-39	MIG-04	T30. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-40	PERF-01, PERF-02, DASH-06	T23, T32. Suite PASS @ tree 91a8f1a (run at d07fefe).
AC-41	E2E-06, UNIT-09	T19, T18, T28, T17. Suite PASS @ tree 91a8f1a (run at d07fefe). No-op detection is partial and journey-specific.

Exact-main and reviewer verification

Check	Staging record	Exact-main @ d07fefe
Server	771/771; default and shuffle seed 33	771/771; 40 files; default + shuffle seed 33
Client	328/328 in 34 files	328/328; 34 files
Full Chromium E2E	64/64; fresh disposable database	64/64; 0 failed / skipped; exact-main run @ d07fefe; isolated DB (reviewer gate @ 1141738 on tree 91a8f1a)
Firefox / WebKit / Edge Lab 4	70/72; two WebKit A11Y-01 failures; none skipped	Exact-main run @ d07fefe: 70/72; 2 WebKit A11Y-01 Tab-walk failures; 0 skipped
Migration / recovery	11/11 checks	11/11
Both builds	Pass	Server + client: PASS
Client lint	0 errors; 3 known warnings: REQUESTER_B unused, 2 SessionContext refresh warnings	Exit 0; 0 errors; 3 known warnings

The results below are the independently recorded release-gate run executed on product-release main at d07fefe (tree 91a8f1a). PR #103's release gate at 1141738 records verification of that identical tree as a separate review-stage record. Appendix A, nested under

Part 3, presents complete compact output for the default-order server and client suites, builds, lint, migration/recovery, and Chromium E2E; shuffled-server and three-browser outputs are explicitly excerpts. Database URLs are redacted, and suppression of application log lines by --silent is disclosed. The three-browser run recorded 70 passed, 2 failed (WebKit A11Y-01 Tab walks), and 0 skipped. The 30 images below render tracked tests.md at main 6d38208; they show the final documentation state, not a rerun of the tests at 6d38208.

Performance smoke — one invocation

```
The exact-main performance smoke was one invocation on d07fefe. Historical
one-run values at 2fa4b97 were 3.8 ms (staff) and 1.59 ms (Requester);
neither run is a benchmark or capacity claim.
Dataset: 2,000 Tickets and 6,000 Actions; 20 requests per endpoint
Staff dashboard p95: 3.26 ms (single run on main @ d07fefe)
Requester dashboard p95: 1.39 ms (single run on main @ d07fefe)
Query count: staff 7 → 7; Requester 2 → 2
Conclusion: single local smoke; not a benchmark or capacity claim
```

A single local invocation does not establish production latency or stability. Independent SQL counts verify metric predicates; timing alone does not prove authorization or filtering.

Lab 4 Test Plan and Results

Version: 1.3 — approved for implementation on 3 October 2026; results updated on 14 October 2026 **Status:** The implementation was released to `main` by PR #103 (merge commit `d07fefe`, 13 October 2026), including the review-readiness follow-up of PR #104. The full gate was run on that exact `main` commit on 14 October (§6.4). Every result is tied to the candidate it actually ran against; §6 names the runs, commands and outputs. Anything not run says so. **Standard:** `docs/lab-02/testing-contract.md` (TC, TCS, TDT and red-phase rules) applies unchanged. Every test names the identifier it proves (TCS-01).

1. Test Strategy

Level	Proves	Tool	Runs against
Unit	Pure rules: Action lifecycle, field validation, gate-aware transitions, ordering, event payloads, metric predicates	Vitest	No database
API integration	Handler + validation + Prisma + PostgreSQL over HTTP	Vitest + Supertest	`toktickit_test`
Security / authorization	Every new operation called **directly** as each refused role and unauthenticated	Vitest + Supertest	`toktickit_test`
Workflow	Every §5.2 edge, the gate, the cascade, history and concurrency, including genuinely parallel requests	Vitest + Supertest	`toktickit_test`
Migration / recovery / seed	Lab 3-state data survives the migration; the backup restores; the seed is idempotent	Script + Vitest	Disposable `_test` databases
Performance smoke	Dashboard p95 and constant query count on a local 2,000-Ticket dataset	Script	Disposable `_test` database
UI component and style	Rendering, validation, states, focus and token compliance; network mocked	Vitest + Testing Library	jsdom
Responsive, accessibility, E2E	Whole journeys at three viewports and the boundary widths	Playwright	`toktickit_e2e_test`
Regression	Every Lab 1–3 suite, after every change	All of the above	As each suite defines

Techniques (testing contract §2):

- equivalence partitions for roles and assignees;
- boundary values for every length limit;
- decision tables for the gate-aware transition matrix and dashboard predicates;
- state transitions for both lifecycles;
- error guessing for double submission, stale screens, concurrent resolve-versus-create and cross-Ticket identifiers.

TDD. Each implementation Issue commits its failing tests first (`test:`), with captured assertion failures, then the implementation (`feat:` / `fix:`) and the passing run (testing contract §5). A seam needed by a test is a stub returning a neutral value, never an import error.

Figure 3.1a — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 1/30).

TDD. Each implementation issue commits its failing tests first (`test:`), with captured assertion failures, then the implementation (`feat:` / `fix:`) and the passing run (testing contract §5). A seam needed by a test is a stub returning a neutral value, never an import error.

Isolation.

- Suites run one at a time on `_test` databases, never `toktickit_dev` .
- **TC-025 — Test-harness transport:** test-created HTTP servers bind only to `127.0.0.1` , and each Supertest request is answered by the specific app instance passed to it, not another local process sharing a port.
- E2E runs write evidence only when `CAPTURE_EVIDENCE=1` (ui-spec §10). After every run, `git status -- artifacts` must show only intended Lab 4 captures.
- A UI failure, loading or empty state produced by API interception is labelled *fixture*. Backend behaviour is proved by the API tests, not by those captures.

2. Planned Tests

Unit

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
UNIT-01	Unit	BR-09, BR-10	Every Action (from, to) pair	Only §5.1 moves are allowed; terminal states allow none	`server/tests/lab-04/action-rules.unit.test.ts`	PASS (#98)
UNIT-02	Unit	BR-12–BR-16, AC-10	Boundaries: Description 0/1/2000/2001 · Result 2000/2001 · Follow-up Note conditional, 1/2000/2001 · Reason 0/1/500/501 · Attachment Notes 500/501 · whitespace-only	Correct field error per case; stored values trimmed	`server/tests/lab-04/action-rules.unit.test.ts`	PASS (#98)
UNIT-03	Unit	BR-11, AC-07	Effective Result at completion (stored, supplied, blank)	Completion allowed only with a non-blank effective Result	`server/tests/lab-04/action-rules.unit.test.ts`	PASS (#98)
UNIT-04	Unit	BR-20, BR-23, AC-18	Decision table: role × Ticket status × open-Action count	`RESOLVED` / `CLOSED` omitted exactly when the count > 0; blocked list reports the count	`server/tests/lab-04/workflow-rules.unit.test.ts`	PASS (#92)
UNIT-05	Unit	BR-29, AC-24	Ordering comparator with equal	Deterministic `(createdAt, id)`	`server/tests/lab-04/workflow-`	PASS (#92)

Figure 3.1b — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 2/30).

UNIT-05	Unit	BR-29, AC-24	Ordering comparator with equal timestamps	Deterministic `(createdAt, id)` order	`server/tests/lab-04/workflow-rules.unit.test.ts`	PASS (#92)
UNIT-06	Unit	BR-30	Event payload builder for each event type	Only identifiers, statuses, priorities and field names; no free text	`server/tests/lab-04/workflow-rules.unit.test.ts`	PASS (#92)
UNIT-07	Unit	BR-36, BR-37	`ACTIVE` group and each metric's query object	Exactly five statuses; each query reproduces its predicate	`server/tests/lab-04/dashboard-metrics.unit.test.ts`	PASS (9/9; mutation check failed as expected when `REOPENED` was removed)
UNIT-08	Unit	BR-19–BR-22, AC-18	The §5.2 table, read as text from `specification.md`, against `transitions.ts`, the gate in `workflowRules.ts` and `transition-oracle.ts`, for each of the eight statuses and three roles; the † and ‡ marks and the own-Ticket marker	Every cell equal in all four places; † is exactly what the gate holds back while work is open; ‡ marks every move into `CANCELLED`; the Administrator column is the IT Staff column	`server/tests/lab-04/transition-spec.unit.test.ts`	PASS (#92)
UNIT-09	Unit	AC-41, FR-26	The audit's allowance rules: a declared response, a declared failed request, a declared console line, each with a near miss (another status, method, address, a second console line, another page error)	A declared event passes with the one browser console line it causes and nothing else; every near miss is reported	`server/tests/lab-04/audit-rules.unit.test.ts` (it tests `e2e/lab-04/audit-rules.ts`)	PASS (server suite, §6; eight injected faults each fail it)

Test-harness transport

Test ID	Type	Requirement	What it tests	Expected result	Test file	Final
			The server binds to loopback	The observed		

Figure 3.1c — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 3/30).

TC-025	Infrastructure	Isolate disposable Supertest servers from other local listeners	The server binds to loopback only; 400 uniquely tagged requests each reach the app instance supplied to Supertest	The observed bind address is `127.0.0.1`; every response carries the nonce of its own app	`server/tests/lab-04/test-transport.unit.test.ts`	PASS (local review-readiness run, §6)

API — Actions Taken

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
API-01	API	AC-01, BR-05	IT Staff creates a valid Action	201; `PLANNED`; Performed by = caller; assignee as sent; server `actionAt`; verified by reading the row back	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-02	API	AC-03, BR-02	Staff A owns the Ticket, Staff B creates an Action assigned to Staff C; list	All fields returned; owner, performer and assignee all differ; creation order	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-03	API	AC-04, BR-07	Assignee inactive, a Requester, unknown, or deactivated after the list was fetched	400 `ASSIGNEE_NOT_ELIGIBLE`; nothing stored. Also the labsheet Part 6 inactive-assignee rejection: cases "an inactive IT Staff user" and "deactivated after the list was fetched"; reassigning to an ineligible user is API-04	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-04	API	AC-05, BR-08	Reassign a non-terminal Action	Assignee changed; Performed by unchanged; version +1; `ACTION_ASSIGNED` written	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-05	API	AC-06	`PLANNED → IN_PROGRESS → COMPLETED` with a Result	`COMPLETED`; `completedBy` / `completedAt` set by the server	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-06	API	AC-07, BR-11	Complete with a blank Result	400 with a field error on `result`; status unchanged	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-07	API	AC-08, BR-12	Cancel with and without a reason	`CANCELLED` with reason, `cancelledBy`, `cancelledAt` / 400	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)

Figure 3.1d — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 4/30).

API-07	API	AC-08, BR-12	Cancel with and without a reason	`cancelledBy`, `cancelledAt` / 400	`04/actions-taken.api.test.ts`	PASS (#98)
API-08	API	AC-09, BR-10	Edit, reassign and transition `COMPLETED` and `CANCELLED` Actions	409 `ACTION_TERMINAL`; row unchanged	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-09	API	AC-10, BR-14	Follow-up required without a note; follow-up later cleared	400 field error / stored note becomes null	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-10	API	AC-11, BR-05	Body supplies Performed by, `createdAt`, completion fields, `status` or `version`	400 `VALIDATION_FAILED`; nothing stored	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-11	API	AC-13, BR-23	Every Action write on `RESOLVED`, `CLOSED` and `CANCELLED` Tickets	409 `TICKET_NOT_WORKABLE`	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-12	API	AC-14, BR-34	Same `requestId` sent twice, sequentially and in parallel	Exactly one Action; both responses carry its id; the repeat returns 200	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-13	API	AC-19, BR-31	Action edit and transition with an outdated `expectedVersion`	409 `STALE_VERSION` with `currentVersion`; nothing changes	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-14	API	BR-01	An `actionId` of another Ticket used under this Ticket's path	404; nothing changes	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-15	API	BR-35	Ticket `updatedAt` after Action writes and after a comment	Changed by Action writes; unchanged by the comment	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-16	API	AC-03, BR-17	Requester reads their own Ticket's Actions	Requester view: display names only, no ids, emails or `isActive`, including the names of who completed or cancelled	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
			Create an Action while another transaction holds	The create is known to be blocked (a lock wait is visible in the database), finishes with	`server/tests/lab-`	

Figure 3.1e — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 5/30).

API-17	API	BR-33	while another transaction holds the Ticket row, once with no change and once after the holder moved the Ticket to `RESOLVED`	The create is known to be blocked (a lock wait is visible in the database), finishes with 201 after the release, or reads the Ticket after the lock and answers 409 `TICKET_NOT_WORKABLE`	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
API-18	API	BR-26, BR-30	One event per Action write; an event that cannot be written	Identifiers and field names only, no free text; the Action write is rolled back	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)

Security / authorization — called directly, never through the UI

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
SEC-01	Security	AC-12, BR-18	Requester calls create, edit and status on their own Ticket's Actions	403 `FORBIDDEN`; database unchanged	`server/tests/lab-04/actions-taken.api.test.ts`	PASS (#98)
SEC-02	Security	AC-12, AC-23	Requester reads Actions and history of another Requester's Ticket	404, identical to a missing Ticket	`server/tests/lab-04/actions-taken.api.test.ts` (Actions) · `server/tests/lab-04/ticket-workflow.api.test.ts` (history)	PASS (#98, #92)
SEC-03	Security	lab-03 BR-12	Every new endpoint without a session	401 for each	`server/tests/lab-04/actions-taken.api.test.ts` (Actions) · `server/tests/lab-04/ticket-workflow.api.test.ts` (history) · `server/tests/lab-04/staff-dashboard.api.test.ts`, `server/tests/lab-04/requester-dashboard.api.test.ts` (dashboards)	PASS (#98 Actions, #92 history and status, #93 dashboards)
SEC-04	Security	AC-30	Requester → IT Staff dashboard; IT Staff and Administrator	403 for each	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS (full server suite at 876511f:

Figure 3.1f — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 6/30).

SEC-04	Security	AC-30	dashboard; IT Staff and Administrator → Requester dashboard	403 for each	`server/tests/lab-04/staff-dashboard.api.test.ts`	server suite at 876511f: 771/771)
SEC-05	Security	AC-02, BR-39	Requester dashboard with two Requesters' Tickets present	Counts and rows from the caller's Tickets only; no foreign ids	`server/tests/lab-04/requester-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771; ownership mutation detected)
SEC-06	Security	AC-22, BR-27	Direct SQL `UPDATE` and `DELETE` on an event row	Both rejected by the trigger; row unchanged	`server/tests/lab-04/ticket-workflow.api.test.ts`	PASS (#92)
SEC-07	Security	BR-30; testing contract §7	Action, history and dashboard responses scanned for hashes, session ids, other users' emails	None present	`server/tests/lab-04/actions-taken.api.test.ts` (Actions) · `server/tests/lab-04/ticket-workflow.api.test.ts` (history) · `server/tests/lab-04/staff-dashboard.api.test.ts` , `server/tests/lab-04/requester-dashboard.api.test.ts` (dashboards)	PASS (#98 Actions, #92 history, #93 dashboards)

Workflow

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	
WF-01	Workflow	AC-15, BR-20	`RESOLVED` requested directly with a `PLANNED` , then an `IN_PROGRESS` , Action	409 `OPEN_ACTIONS_BLOCK_RESOLUTION` with `openActionCount` ; status unchanged	`server/tests/lab-04/ticket-workflow.api.test.ts`	
WF-02	Workflow	AC-15, BR-20	`CLOSED` requested on a `RESOLVED` Ticket holding an open Action inserted directly into the database	409; the gate holds against out-of-band data	`server/tests/lab-04/ticket-workflow.api.test.ts`	

Figure 3.1g — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 7/30).

			directly into the database		
WF-03	Workflow	AC-16	Resolve when every Action is terminal	`RESOLVED` ; `STATUS_CHANGED` written	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-04	Workflow	AC-16, BR-21	Resolve a zero-Action Ticket and a seeded legacy Ticket	`RESOLVED`	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-05	Workflow	AC-17, BR-22	Cancel a Ticket with two open and one completed Action; repeat with a failure injected after the cascade	Ticket `CANCELLED` ; two Actions cancelled with the system reason and `cancelledBy` ; completed one untouched / with the injected failure, nothing persisted	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-06	Workflow	AC-18, BR-19	Every §5.2 cell for Requester, IT Staff and Administrator, against the table transcribed in `transition-oracle.ts`	Each permitted edge succeeds; each absent edge is refused (400 or 403, as in Lab 3); a wrong edge in the code fails the test	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-07	Workflow	AC-19, BR-31	Status change with the version read before an owner change	409 `STALE_VERSION` ; the owner change is kept	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-08	Workflow	AC-20, BR-33	Create-Action and resolve sent concurrently, repeated 20 times; a status change blocked behind a held lock, with and without a change the holder commits	Never `RESOLVED` with an open Action; in each pair, one request is refused or ordered after the other; the blocked request is known to wait (a lock wait is visible in the database) and reads the version after the lock (409 `STALE_VERSION`)	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-09	Workflow	AC-22, BR-26	A scripted sequence of status, owner, priority and Action changes	One event per change, with the right type, actor and order	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-		AC-23, BR-	The same history read by its	Requester gets `STATUS_CHANGED`	`server/tests/lab-`

Figure 3.1h — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 8/30).

WF-10	Workflow	AC-23, BR-28	The same history read by its Requester and by IT Staff	Requester gets `STATUS_CHANGED` only; IT Staff get every event	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-11	Workflow	AC-24, BR-29	Actions, comments, notes and events inserted with identical `createdAt`	Identical order across five repeated reads	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-12	Workflow	AC-25, BR-25	Requester resolution indication on a Ticket with open Actions	Indication recorded; status unchanged; version +1	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-13	Workflow	BR-32	Owner and IT Priority changes in the Lab 3 request shape	Succeed without a version; version +1; event written. A change to what is already stored, an Action edit that changes nothing and a same-status request leave `version`, events and `updatedAt` as they were	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-14	Workflow	AC-19, BR-31	Status change without `expectedVersion`	400 `VALIDATION_FAILED`	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-15	Workflow	AC-22, BR-29	A claim then a priority change in one millisecond; a change 5 ms into a 40-Action cancellation burst; a change after the clock went back	Events keep commit order and `createdAt` is strictly increasing, in the database and in the History response	`server/tests/lab-04/ticket-workflow.api.test.ts`
WF-16	Workflow	BR-35, BR-38	Every kind of change (owner, IT Priority, status, Action create, edit and move) carrying an instant earlier than the Ticket's `updatedAt`; one carrying a later instant	`updatedAt` keeps the later value; a later instant moves it forward	`server/tests/lab-04/ticket-workflow.api.test.ts`
			The Requester		

Figure 3.1i — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 9/30).

			instant			
WF-17	Workflow	AC-25, BR-25, BR-31, BR-33, BR-35	The Requester indication while another transaction holds the Ticket and cancels or closes it; with an earlier clock; with a later one	Waits and records (control); 409 `TICKET_CLOSED` and nothing stored once the Ticket closed; `updatedAt` never moves back and equals `requesterResolvedAt`	`server/tests/lab-04/ticket-workflow.api.test.ts`	

Dashboards and lists

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
DASH-01	API	AC-26, BR-37	MET-S01–S05 against independent, hand-written SQL `COUNT` queries over a fixture set, including urgent Tickets in each terminal status	Every value equal; terminal urgent Tickets excluded from the active urgent count	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)
DASH-02	API	AC-27	Queue called with each metric's returned `query`, including an urgent Ticket in each terminal status	`totalItems` equals the count; the id set equals the SQL set and excludes terminal urgent Tickets	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)
DASH-03	API	AC-28	12 open Actions assigned to me, plus terminal ones and others'	Total 12; 10 items, oldest first; terminal and others' excluded	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771; fixture owns all 12 open Actions)
DASH-04	API	AC-29	Administrator with no open Actions; statuses with zero Tickets	Zeros and empty arrays; all eight statuses present; 200	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)
DASH-	API	BR-38	Recently updated, including equal	Top five by `updatedAt`	`server/tests/lab-04/staff-`	PASS (full server suite at

Figure 3.1j — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 10/30).

DASH-05	API	BR-38	Recently updated, including equal `updatedAt` values	Top five by `updatedAt desc, id desc`	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)
DASH-06	API	AC-40, BR-40	Response size and query count with 50 vs 500 Tickets	Same query count; at most 5 and 10 rows	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)
DASH-07	API	AC-02, AC-26, BR-38	MET-R01–R04 against independent SQL with `requesterId`; all eight statuses represented; recent list checked with more than five Tickets	Every value equal; only own Tickets; exactly the five newest fixture Tickets in descending `updatedAt` order	`server/tests/lab-04/requester-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771; ownership mutation check failed as expected)
DASH-08	API	AC-27	My Tickets called with each Requester metric's `query`; all eight statuses represented	`totalItems` equals the count and the returned id set exactly matches each metric predicate	`server/tests/lab-04/requester-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)
DASH-09	API	AC-29	Requester with no Tickets	`totalTickets = 0`; all counts 0; empty list	`server/tests/lab-04/requester-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)
DASH-10	API	AC-33, FR-20	`statusGroup=active` on both lists; combined with `status`; unknown value	Only `ACTIVE` Tickets / 400 / 400	`server/tests/lab-04/requester-dashboard.api.test.ts`	PASS (full server suite at 876511f: 771/771)

Migration, recovery, seed, performance

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
MIG-	Migration	AC-38, BR-	Disposable database at the Lab 3 state, migrated to	Identical counts for every pre-existing	`server/scripts/verify-migration.mjs` (output	PASS (#98)

Figure 3.1k — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 11/30).

MIG-01	Migration	AC-38, BR-41	the Lab 3 state, migrated to Lab 4; counts per table and FK checks	counts for every pre-existing table; no orphaned reference	`server/scripts/verify-migration.mjs` (output retained)	PASS (#98)
MIG-02	Recovery	AC-38	`pg_dump -Fc` before migrating, then `pg_restore` into a new disposable database	Restored counts equal the pre-migration counts	`server/scripts/verify-migration.mjs`	PASS (#98)
MIG-03	Migration	AC-38, BR-21	Legacy Tickets after migration	Status unchanged; zero Actions; `version = 1`	`server/tests/lab-04/migration-seed.api.test.ts`	PASS (#98)
MIG-04	Seed	AC-39, BR-42	Seed run twice	No duplicates, every Action column unchanged by the second run; Tickets with 0, 1 and several Actions exist	`server/tests/lab-04/migration-seed.api.test.ts`	PASS (#98)
MIG-05	Migration	BR-27	Direct SQL `UPDATE` and `DELETE` on an event row; then `TRUNCATE`	Both rejected by the trigger and the row is unchanged; the reset by `TRUNCATE` still works	`server/tests/lab-04/migration-seed.api.test.ts`	PASS (#98)
			The helper the lock	Settles for the first, and refuses the second		

Figure 3.1l — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 12/30).

LOCK-01	Test helper	BR-33	The helper the lock tests wait on, with a waiter on the Ticket and with a waiter on another table	the first, and refuses the second with a clear message, so a lock test cannot pass because of an unrelated wait	`server/tests/lab-04/lock-helpers.api.test.ts`	PASS (#92)
PERF-01	Performance	AC-40	2,000 Tickets / 6,000 Actions; each dashboard endpoint called 20 times	p95 ≤ 300 ms; machine, dataset and command recorded	`server/scripts/perf-smoke.mjs`	PASS — Darwin 27.0 Apple M5, Node v26 2,000 synthetic Tick 6,000 Actions + see baseline; p95 staff 4 requester 1.86 ms o 3.8 ms and 1.59 ms Oct (§6) (`server` : `./node_modules/.b scripts/perf-smoke` disposable `toktickit_codex93_`
PERF-02	Performance	AC-40, BR-40	Query count at 200 vs 2,000 Tickets	Equal	`server/scripts/perf-smoke.mjs`	PASS — staff 7→7, requester 2→2 queries/request (`./node_modules/.b scripts/perf-smoke` disposable `toktickit_codex93_`

UI component

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	F
UI-01	UI	AC-03	Actions list	Every field rendered in creation order; " (inactive)" marker; table ≥ 992 px, cards below	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-02	UI	AC-10, BR-13–BR-15	Create form: required, conditional and boundary inputs	Field messages; API not called while invalid; note hidden and cleared when unticked	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9

Figure 3.1m — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 13/30).

			boundary inputs	and cleared when unticked		
UI-03	UI	BR-06, BR-07	Assignee control	Defaults to the current user; options are `assignableOwners` only	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-04	UI	AC-07	Complete without a Result	Field error under Result, focus moved there; no request	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-05	UI	AC-08, AC-35	Cancel dialog	Reason required; focus trapped; Escape closes; focus returns to the trigger	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-06	UI	AC-09	Terminal Action	Read-only, with no Edit or status buttons	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-07	UI	AC-13	Ticket not workable	Banner replaces Add Action; no edit controls	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-08	UI	AC-12, BR-17	Requester view	Read-only; no controls; display names only	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-09	UI	FR-24, AC-14	Busy state, double click and retry	Save disabled while in flight; one request per click burst; retry reuses `requestId`	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-10	UI	AC-36	Save failure and `STALE_VERSION`	Entered values kept; conflict or failure message; save possible again	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-11	UI	AC-18, AC-21	Status control	Only permitted transitions; blocked helper text shows the open count	`client/tests/lab-04/TicketWorkflow.test.tsx`	PAS (#9
UI-12	UI	AC-21	Status outcomes	Success refreshes badge and controls and announces; 409 open-Actions and stale messages keep the previous status	`client/tests/lab-04/TicketWorkflow.test.tsx`	PAS (#9

Figure 3.1n — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 14/30).

				and stale messages keep the previous status		
UI-13	UI	AC-23	History	IT Staff see every type; Requester sees status changes only; empty state	`client/tests/lab-04/TicketWorkflow.test.tsx`	PAS (#92
UI-14	UI	AC-27, AC-28	Staff dashboard content	Cards link with the returned `query`; open-Actions list and "10 oldest" footer	`client/tests/lab-04/StaffDashboard.test.tsx`	PAS (incl in cl suit 328
UI-15	UI	AC-29, AC-31	Staff dashboard states	Loading; failure → Try again → recovered; forbidden; zero values	`client/tests/lab-04/StaffDashboard.test.tsx`	PAS (incl in cl suit 328
UI-16	UI	AC-02, AC-29	Requester dashboard content	Cards and links; attention cue as text; first-use empty state	`client/tests/lab-04/RequesterDashboard.test.tsx`	PAS (incl in cl suit 328
UI-17	UI	AC-31	Requester dashboard states	Loading; failure → Try again; forbidden	`client/tests/lab-04/RequesterDashboard.test.tsx`	PAS (incl in cl suit 328
UI-18	UI	AC-32, FR-21	Navigation per role	Dashboard first and active; post-login redirect lands on it	`client/tests/lab-04/AppNavigation.test.tsx`	PAS 3 ro E2E in ro acti navi link; inclu in cl suit 328
UI-19	UI	api-spec §1, BR-38	History and Actions times and the formatter on devices set to UTC, America/Los_Angeles, Asia/Bangkok and Pacific/Kiritimati	03:12 UTC reads 10:12 and the date moves exactly at 17:00 UTC	`client/tests/lab-04/dateTime.test.ts`, `client/tests/lab-04/TicketWorkflow.test.tsx`, `client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#92 #91

Figure 3.1o — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 15/30).

			Asia/Bangkok and Pacific/Kiritimati		`04/ActionsTaken.test.tsx`	
UI-20	UI	AC-10, ui-spec §9	An invalid create, edit, server-refused and cancel-dialog submission	The first invalid field in display order has the focus, once the controls are enabled again	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-21	UI	ui-spec §9	Close the details, cancel and save the create form, discard and save the edit form, cancel an Action	The focus returns to View, Add Action or Edit; after a cancellation it lands on the details heading, not on the page	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-22	UI	AC-35	Tab and Shift+Tab in the cancel dialog while its save is in flight and after it fails	The dialog takes the focus when its last control is disabled, and the focus never leaves it	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-23	UI	AC-14, AC-36, BR-34	A create whose answer is lost, entries changed, then Try again; the update failing; no change; a refused first attempt	The first entries are resent under their key, then the changes are saved as an update; one Action; the update alone is retried	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-24	UI	AC-13, BR-22, BR-23	The Ticket becomes Resolved or Cancelled while the create form, the edit form or the cancel dialog is open; a status change	The forms are read-only with their text kept and no Save; the dialog closes; the Actions are loaded again	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-25	UI	AC-12, BR-17	A Requester opens a Planned and an In progress Action	Only Close; no Edit, Start, Complete, Cancel Action or textbox	`client/tests/lab-04/ActionsTaken.test.tsx`	PAS (#9
UI-26	UI	api-spec §1, BR-38	Staff and Requester Ticket Detail, the staff queue, comments, attachments, My Tickets (a date alone) and the Create Ticket confirmation on devices set to UTC	Each shows the Bangkok time or date, including a date that falls on the next day	`client/tests/lab-04/BangkokTime.test.tsx`	PAS (#9

Figure 3.1p — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 16/30).

			confirmation on devices set to UTC and America/Los_Angeles	date that falls on the next day		
UI-27	UI	AC-28, AC-41	The dashboard's open-Action link, and an address that names an Action, the section, or something else, once the list has loaded and again after it is reloaded	The link ends `#action-<actionId>`; the named row (or the section) is scrolled to once; any other address is left to the browser	`client/tests/lab-04/StaffDashboard.test.tsx`, `client/tests/lab-04/ActionsTaken.test.tsx`	PAS (clie suit four inje faul eac it: n onc gua nev scr scr for add the `#ac` link)
UI-28	UI	AC-31, ui-spec §11 row 10	Press Refresh on the staff dashboard while the request is held	A polite live region says "Refreshing dashboard." and is empty again afterwards	`client/tests/lab-04/StaffDashboard.test.tsx`	PAS (clie suit four inje faul eac it)

UI style

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
STYLE-01	UI style	STY-001–STY-003	Colour literals and Bootstrap colour utilities across all of `client/src`	Literals only in the `:root` token block; no colour utilities	`client/tests/lab-03/StyleContract.test.tsx` (existing; already scans every `client/src` file)	PASS (client suite, §6)
STYLE-02	UI style	STY-018, STY-019	Action status badges	Text label for each of the four values, from the single badge mapping	`client/tests/lab-04/ActionsTaken.test.tsx`	PASS (#91)
STYLE-03	UI style	STY-026	Every new input	A programmatically associated label	`client/tests/lab-04/ActionsTaken.test.tsx`	PASS (#91)

Figure 3.1q — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 17/30).

STYLE-03	UI style	STY-026	Every new input	A programmatically associated label	`client/tests/lab-04/ActionsTaken.test.tsx`	PASS (#91)
STYLE-04	UI style	STY-012	Validation placement	Message below its field, error token, input text not red	`client/tests/lab-04/ActionsTaken.test.tsx`	PASS (#91)
STYLE-05	UI style	STY-009, STY-010	System fields	Rendered on the read-only surface, never as inputs	`client/tests/lab-04/ActionsTaken.test.tsx`	PASS (#91)

Responsive, accessibility and E2E

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
E2E-01	E2E	AC-01–AC-10, AC-12, BR-02	Daniel owns the Ticket; Olivia creates several Actions, assigns Patricia, edits, starts, completes, cancels; inactive Thomas is refused; the Requester sees everything read-only	Each step visible and persisted; captured in `actions-taken/`	`e2e/lab-04/actions-taken-flow.spec.ts`	PASS (#91
E2E-02	E2E	AC-13, AC-15–AC-17, AC-21–AC-23	Resolve blocked → work completed → resolved; cancel with cascade; history for IT Staff vs Requester	Each state visible; status refreshes; each state captured at 1280, 834 and 390 px in `ticket-workflow/`, with no horizontal scroll	`e2e/lab-04/ticket-resolution.spec.ts`	PASS (#92
E2E-03	E2E	AC-19, AC-36	Two browser sessions change the same Action and the same Ticket status	Conflict message; entered values kept; save succeeds after review	`e2e/lab-04/ticket-resolution.spec.ts`	PASS (#92 #91)
			Both dashboards: values equal the API and SQL; metric	Matching values and filtered rows; every visible		PASS (focu mobile-targ

Figure 3.1r — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 18/30).

E2E-04	E2E	AC-26–AC-29, AC-32, AC-34	values equal the API and SQL; metric queries reproduce the lists; Requester needs-attention opens filtered My Tickets with matching rows; every visible dashboard link is measured at 390 px	values and filtered rows; every visible dashboard link is at least 44 px; current captures show the filtered Requester list	`e2e/lab-04/dashboards.spec.ts`	PASS (focu mobile-targ check 2/2; E2E 64/64; every visibl Staff and Requester dashboard measures ≥ px at 390 p
E2E-05	E2E	AC-30, AC-31	Cross-role dashboard routes; load failure by interception (*fixture*) and retry	Forbidden state; failure then recovery	`e2e/lab-04/dashboards.spec.ts`	PASS (2 te failure/emp captures explicitly na `fixture`)
RESP-01	Responsive	AC-34	Dashboards and Ticket Detail with Actions at 1280, 834 and 390 px; widths 767/768/991/992; 2000-character content; every visible dashboard link measured at 390 px	No clipping or overlap; no horizontal page scroll; every visible dashboard link is at least 44 px high	`e2e/lab-04/dashboards.spec.ts` , `e2e/lab-04/actions-taken-flow.spec.ts`	PASS: full E 64/64; ever visible link both dashboards measures ≥ px at 390 p Ticket Deta dashboards pass listed widths
A11Y-01	Accessibility	AC-35	Keyboard-only: Action form, cancel dialog, status control, dashboard drill-downs	Every control reachable; focus visible; dialog focus trap and return	`e2e/lab-04/actions-taken-flow.spec.ts`	PASS (#91 Action form cancel dial #93: dashb drill-down reachable b Tab with vis `:focus-visible`); WebKit Tab order is a k limit (§7); r screen reac used
			Console errors, failed requests and broken	No console error, page error, failed request or		PASS (all 2 tests of the `e2e/lab-0` specs run u the audit, f them are th

Figure 3.1s — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 19/30).

E2E-06	E2E	AC-41, FR-26	Console errors, failed requests and broken links collected across every Lab 4 journey, plus a walk that follows every link on every screen a signed-out visitor, a Requester, an IT Staff member and an Administrator can reach	error, page error, failed request or undeclared error response; no link that leads nowhere; no anchor without its element; no placeholder text	`e2e/lab-04/audit.ts` (the fixture every Lab 4 spec runs under), `e2e/lab-04/audit.spec.ts`	the audit, f them are th walks of `audit.spe` the audit fo a broken an fixed under 27; a conso error, placeholde and an emp link each fa control that does nothin not detecte §7)
E2E-07	E2E	AC-14, AC-36	The server commits a create and its answer is lost (real response aborted); the Description is changed; Try again	One Action, with the changed text, created once and updated once	`e2e/lab-04/actions-taken-recovery.spec.ts`	PASS (#91
E2E-08	E2E	AC-13, BR-23	The Ticket is resolved from the same page while Add Action is open	The form keeps its text, is disabled, has no Save, and nothing is stored	`e2e/lab-04/actions-taken-recovery.spec.ts`	PASS (#91
E2E-09	E2E	api-spec §1	A device set to America/Los_Angeles opens a Ticket with an Action	Every time in the list and the History equals the Asia/Bangkok rendering of the API's value	`e2e/lab-04/actions-taken-recovery.spec.ts`	PASS (#91
E2E-10	E2E	AC-34, AC-35, AC-10, AC-12	One Ticket with a Planned, an In progress, a Completed and a Cancelled Action, at 1280, 834 and 390 px: the list (button and badge stay on one line in the table), the Add form, an	No horizontal page scroll; the error beside Description with the focus on it; no control for the	`e2e/lab-04/actions-taken-flow.spec.ts`	PASS (#91

Figure 3.1t — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 20/30).

10	E2E	AC-12	and badge stay on one line in the table), the Add form, an empty save, the edit panel, and the Requester's read-only view	no control for the Requester; captured in `actions-taken/`	`taken-flow.spec.ts`	PASS (#9
A11Y-02	Accessibility	AC-35, ui-spec §9	Real Chromium: invalid follow-up note; close the details; Tab inside a busy cancel dialog	The note has the focus; View has it back; Tab and Shift+Tab never leave the dialog; the heading has the focus after the cancellation	`e2e/lab-04/actions-taken-recovery.spec.ts`	PASS (#91

Regression

Test ID	Type	Requirement / BR / AC	What it tests	Expected result	Test file	Final
REG-01	Regression	AC-37	Full Lab 1–3 server suites. The only planned edit is the status request body gaining `expectedVersion` (specification §11.9)	All pass; nothing skipped	`server/tests/lab-01`, `lab-02`, `lab-03`	PASS (full server suite §6; it runs `lab-01` to `lab-04` wi nothing skipped)
REG-02	Regression	AC-37	Full Lab 2–3 client suites	All pass	`client/tests/lab-02`, `lab-03`	PASS (full client suite, §6)
REG-03	Regression	AC-32, AC-37, FR-21	Lab 2–3 E2E specs and Lab 3 sign-in/password-change flows. Besides AUTH-03, the three affected legacy landing assertions (ADMIN-04, Login, Change Password) now expect	All pass with capture disabled; no tracked screenshot modified	`e2e/lab-02`, `e2e/lab-03`, `client/tests/lab-03/Login.test.tsx`, `client/tests/lab-03/ChangePassword.test.tsx`	PASS (full E 59/59 with #98, #100 and #101 merged in; client suite

Figure 3.1u — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 21/30).

03		37, FR-21	Login, Change Password) now expect Dashboard, as required by Lab 4's role home; `/tickets` remains the plain My Tickets route.	tracked screenshot modified	`client/tests/lab-03/ChangePassword.test.tsx`	merged in; client suite 325/325)
REG-04	Regression	FR-22	Representative Part 8 journeys: sign-in through the screen, My Tickets, Ticket Detail with attachments and a comment, the Queue, an Internal Note (hidden from the Requester), a missing Ticket and another Requester's Ticket, User Management	Each journey succeeds; no horizontal scroll at 1280, 834 and 390 px; captured in `regression/`	`e2e/lab-04/regression.spec.ts`	PASS (one journey, 8 states at 3 widths = 24 captures in `regressio`

3. Acceptance-Criterion Traceability

Every criterion maps to at least one planned test and, where the rubric asks for visible proof, to a planned capture (`artifacts/lab-04/screenshots/…`).

Checked on 8 October 2026 for the dashboard criteria (AC-02, AC-26 to AC-33, AC-40): every test listed has a row in §2, its file exists, and the file contains the test ID (SEC-04 and SEC-05 are inside the `SEC-03/04` and `SEC-03/05/07` blocks).

Twelve tests prove a business or style rule rather than a criterion, and are traced through their Requirement column instead:

- business rules: UNIT-01 (BR-09/10), UNIT-06 (BR-30), UNIT-07 (BR-36/37), API-14 (BR-01), API-15 (BR-35), DASH-05 (BR-38), WF-13 (BR-32), UI-03 (BR-06/07);
- requirements: REG-04 (FR-22);
- style rules: STYLE-01, STYLE-03 and STYLE-04 (STY rules).

AC	Planned tests	Planned visible evidence
AC-01	API-01, E2E-01, E2E-10	`actions-taken/create-form-{desktop,tablet,mobile}` , `list-{desktop,tablet,mobile}`
AC-02	SEC-05, DASH-07, UI-16	`requester-dashboard/desktop`
AC-	API-02, API-16, UI-01,	`actions-taken/list-{desktop,tablet,mobile}` , `requester-read-only-`

Figure 3.1v — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 22/30).

02	16	
AC-03	API-02, API-16, UI-01, E2E-01	`actions-taken/list-{desktop,tablet,mobile}`, `requester-read-only-{desktop,tablet,mobile}`
AC-04	API-03, E2E-01	`actions-taken/inactive-assignee-refused`
AC-05	API-04, E2E-01	`actions-taken/edit-reassign-{desktop,tablet,mobile}`
AC-06	API-05, E2E-01	`actions-taken/started`, `completed`
AC-07	API-06, UNIT-03, UI-04	`actions-taken/complete-result-required`
AC-08	API-07, UI-05, E2E-01	`actions-taken/cancel-dialog`, `cancelled`
AC-09	API-08, UI-06	`actions-taken/completed` (no controls)
AC-10	API-09, UNIT-02, UI-02, E2E-01, E2E-10	`actions-taken/follow-up-required`, `description-required-{desktop,tablet,mobile}`
AC-11	API-10	API test output
AC-12	SEC-01, SEC-02, UI-08, E2E-01	`actions-taken/requester-read-only-{desktop,tablet,mobile}`; API test output
AC-13	API-11, UI-07, E2E-02	`actions-taken/ticket-not-workable`
AC-14	API-12, UI-09	API test output
AC-15	WF-01, WF-02, E2E-02	`ticket-workflow/resolve-blocked-{desktop,tablet,mobile}`
AC-16	WF-03, WF-04, E2E-02	`ticket-workflow/resolved-after-work-complete-{desktop,tablet,mobile}`
AC-17	WF-05, E2E-02	`ticket-workflow/cancel-cascade-{desktop,tablet,mobile}`
AC-18	UNIT-04, WF-06, UI-11	`ticket-workflow/resolve-blocked-{desktop,tablet,mobile}`
AC-19	API-13, WF-07, WF-14, E2E-03	`ticket-workflow/stale-status-{desktop,tablet,mobile}`, `actions-taken/conflict`
AC-20	WF-08	API test output (concurrent runs)
AC-21	UI-12, E2E-02	`ticket-workflow/resolved-after-work-complete-{desktop,tablet,mobile}`

Figure 3.1w — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 23/30).

AC-21	UI-12, E2E-02	`ticket-workflow/resolved-after-work-complete-{desktop,tablet,mobile}`
AC-22	WF-09, SEC-06, MIG-05, E2E-02	`ticket-workflow/history-staff-{desktop,tablet,mobile}` ; trigger test output
AC-23	WF-10, SEC-02, UI-13, E2E-02	`ticket-workflow/history-requester-{desktop,tablet,mobile}`
AC-24	UNIT-05, WF-11	API test output
AC-25	WF-12	API test output
AC-26	DASH-01, DASH-07, E2E-04	Dashboard capture beside the SQL count output
AC-27	DASH-02, DASH-08, UI-14, E2E-04	`staff-dashboard/drilldown-unassigned` , `requester-dashboard/drilldown-open`
AC-28	DASH-03, UI-14, UI-27, E2E-04	`staff-dashboard/desktop`
AC-29	DASH-04, DASH-09, UI-15, UI-16, E2E-04	`staff-dashboard/admin-no-open-actions` , `requester-dashboard/first-use-empty-fixture`
AC-30	SEC-04, E2E-05	`staff-dashboard/forbidden-requester`
AC-31	UI-15, UI-17, E2E-05	`staff-dashboard/failure-fixture` , `requester-dashboard/failure-fixture`
AC-32	UI-18, E2E-04	Dashboard captures show the active navigation item
AC-33	DASH-10	API test output
AC-34	RESP-01, E2E-10	Desktop, tablet and mobile captures of every new screen: `actions-taken/` , `staff-dashboard/` , `requester-dashboard/` , `ticket-workflow/` , `regression/`
AC-35	UI-05, A11Y-01	Focus-ring capture; keyboard-walk notes in the checklist
AC-36	UI-10, E2E-03	`actions-taken/conflict` , `save-failure-fixture`
AC-37	REG-01, REG-02, REG-03	Full runner output of the release candidate (§6)
AC-38	MIG-01, MIG-02, MIG-03	Script output: counts before, after and restored
AC-39	MIG-04	Seed-twice test output
AC-40	PERF-01, PERF-02, DASH-06	Performance-smoke output with machine and dataset

Figure 3.1x — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 24/30).

AC-40	PERF-01, PERF-02, DASH-06	Performance-smoke output with machine and dataset
AC-41	E2E-06, UNIT-09	Console and link audit output (§6); a no-op control is covered only where a journey uses the control (§7)

4. Coverage of the labsheet's named files

Labsheet path (§12)	Planned tests
`server/tests/lab-04/actions-taken.api.test.ts`	API-01–API-18, SEC-01, and the Actions part of SEC-02, SEC-03 and SEC-07
`server/tests/lab-04/ticket-workflow.api.test.ts`	WF-01–WF-17, SEC-02, SEC-03, SEC-06, SEC-07
`server/tests/lab-04/requester-dashboard.api.test.ts`	DASH-07–DASH-10, SEC-05
`server/tests/lab-04/staff-dashboard.api.test.ts`	DASH-01–DASH-06, SEC-04
`client/tests/lab-04/StaffDashboard.test.tsx`	UI-14, UI-15, UI-27 (the link), UI-28
`client/tests/lab-04/RequesterDashboard.test.tsx`	UI-16, UI-17
`client/tests/lab-04/ActionsTaken.test.tsx`	UI-01–UI-10, UI-19–UI-25, UI-27 (the scroll), STYLE-02–STYLE-05
`client/tests/lab-04/TicketWorkflow.test.tsx`	UI-11–UI-13, UI-19
`client/tests/lab-04/dateTime.test.ts`	UI-19
`client/tests/lab-04/BangkokTime.test.tsx`	UI-26
`e2e/lab-04/actions-taken-flow.spec.ts`	E2E-01, E2E-03 (Actions), RESP-01 (Ticket Detail with Actions), E2E-10, A11Y-01
`e2e/lab-04/ticket-resolution.spec.ts`	E2E-02, E2E-03
`e2e/lab-04/dashboards.spec.ts`	E2E-04, E2E-05, RESP-01

Supporting files beyond the minimum:

- `server/tests/lab-04/transition-oracle.ts` (the status table of §5.2, shared by UNIT-04 and WF-06);
- `server/tests/lab-04/lock-helpers.api.test.ts` (LOCK-01, the lock-wait helper behind API-17, WF-08 and WF-17);
- `e2e/lab-04/actions-taken-recovery.spec.ts` (E2E-07–E2E-09, A11Y-02);
- `server/tests/lab-04/action-rules.unit.test.ts`, `workflow-rules.unit.test.ts`, `transition-spec.unit.test.ts`, `audit-rules.unit.test.ts`, `dashboard-metrics.unit.test.ts` and `migration-seed.api.test.ts` (MIG-03–MIG-05);
- `server/tests/lab-04/test-transport.unit.test.ts` (TC-025; the loopback and request-routing guard behind the #96 suite-stability fix);
- `client/tests/lab-04/AppNavigation.test.tsx`;
- `e2e/lab-04/audit.ts`, `e2e/lab-04/audit-rules.ts` (UNIT-09) and `e2e/lab-04/audit.spec.ts` (E2E-06), and `e2e/lab-04/regression.spec.ts` (REG-04);

Figure 3.1y — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 25/30).

- `client/tests/lab-04/AppNavigation.test.tsx`;
- `e2e/lab-04/audit.ts`, `e2e/lab-04/audit-rules.ts` (UNIT-09) and `e2e/lab-04/audit.spec.ts` (E2E-06), and `e2e/lab-04/regression.spec.ts` (REG-04);
- `server/scripts/verify-migration.mjs` (MIG-01, MIG-02) and `server/scripts/perf-smoke.mjs` (PERF-01, PERF-02);
- `client/tests/lab-03/StyleContract.test.tsx` (STYLE-01, existing).

Every path above exists on `lab4-staging`.

5. Test Commands and Isolation

These commands exist today (verified on `a429f29`):

```
cd server && npm test        # migrate deploy + seed toktickit_test, then Vitest in one worker
cd server && npm run test:only
cd client && npm test
npm run test:e2e             # repository root; migrates and seeds toktickit_e2e_test; API :3002, client :
cd server && npm run build
cd client && npm run build
cd client && npm run lint    # oxlint
node scripts/row-counts.mjs .env.test   # from server/; prints table counts, never the connection string
```

Added by the contract increment (#89) and verified on 3 October 2026:

```
CAPTURE_EVIDENCE=1 npm run test:e2e   # writes Lab 4 captures under artifacts/lab-04/screenshots/ only
TEST_DATABASE_URL=… npm test          # from server/; a fresh database whose name ends in _test, or the ru
```

- A default `npm run test:e2e` writes nothing tracked: 40 passed with and without `CAPTURE_EVIDENCE=1`, and `git status -- artifacts` stayed clean both times. Lab 2 and Lab 3 captures are frozen.
- Runner output moved to the ignored `artifacts/lab-04/playwright-results/`.

Cross-browser run (Chromium is the only project in `playwright.config.ts`; browsers installed once with `npx playwright install firefox webkit`). A throwaway config, not committed, next to it:

```
// playwright.cross-browser.config.ts
import { devices } from '@playwright/test'
import base from './playwright.config'
export default {
  ...base,
  projects: [
    { name: 'firefox', use: { ...devices['Desktop Firefox'] } },
    { name: 'webkit', use: { ...devices['Desktop Safari'] } },
    { name: 'msedge', use: { ...devices['Desktop Edge'], channel: 'msedge' } }, // the installed Microsoft
  ],
}
```

```
npx playwright test -c playwright.cross-browser.config.ts e2e/lab-04   # under the E2E lock
```

Both scripts the plan listed now exist: `server/scripts/verify-migration.mjs` (#90, run as `node scripts/verify-migration.mjs` from `server/`) and `server/scripts/perf-smoke.mjs` (#93, needs its own disposable database).

For evidence, the E2E runner is pointed at a freshly created database ending in `_test` (`E2E_DATABASE_URL`), so a screenshot

Figure 3.1z — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 26/30).

migration.mjs from server/) and server/scripts/perf-smoke.mjs (#96, needs its own disposable database).

For evidence, the E2E runner is pointed at a freshly created database ending in `_test` (`E2E_DATABASE_URL`), so a screenshot shows the seed and nothing left by earlier runs; the database is dropped afterwards.

Suites always run sequentially. `toktickit_dev` is never a target.

6. Final Results

Both runs were made on the student's Mac (macOS Darwin 27.0.0, Apple M5, Node v26.7.0, PostgreSQL 18.6), on freshly created databases whose names end in `_test` , one run at a time, with `/tmp/toktickit-e2e.lock` held for every E2E run. The runner output is kept in the private working record (`_private/lab-04/WORK-RECORD.md`), which is not part of the repository.

1. The integrated staging candidate, `lab4-staging` `b5d75ed` (the five feature Pull Requests merged), 12 October 2026.

Check	Result
Server suite, default order and `--sequence.shuffle --sequence.seed=33`	758/758 both times
Client suite	34 files, 325/325
Server and client production builds	both pass
Client lint	3 known warnings (an unused `REQUESTER_B` in a Lab 2 test; two react-refresh warnings in `SessionContext.tsx`), 0 errors
`node scripts/verify-migration.mjs`	all 11 checks (MIG-01, MIG-02)
Chromium E2E, whole repository	59/59; `git status -- artifacts` clean
`e2e/lab-04` on Firefox, WebKit and Microsoft Edge 154	55 of 57 (as it then stood); the two failures are A11Y-01 on WebKit (§7)

2. The release-candidate code tree: `feature/94-final-hardening-and-release` at `5685095` , 12 October 2026 (re-run after the review of #102 changed the audit; the run at `2fa4b97` gave the same results except 758 server tests). Its product-code tree is the one now integrated into `lab4-staging` at `876511f` . PR #102's final head `ab53f8a` and merge changed documentation only relative to code head `5685095` .

Check	Command	Result
Server suite, default order	recreate `toktickit_test` , then `cd server && npm test`	771/771 (758 plus the 13 cases of UNIT-09)
Server suite, shuffled	the same with `--sequence.shuffle --sequence.seed=33`	771/771
Client suite	`cd client && npm test`	34 files, 328/328
Builds	`cd server && npm run build` , `cd client && npm run build`	both pass
Lint	`cd client && npm run lint`	the same 3 known warnings, 0 errors
Migration and recovery	`cd server && node scripts/verify-migration.mjs`	all 11 checks
E2E, Lab 2 to Lab 4,	`E2E_DATABASE_URL` → a fresh	

Figure 3.1aa — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 27/30).

recovery	`migration.mjs`	all 11 checks
E2E, Lab 2 to Lab 4, Chromium	`E2E_DATABASE_URL` → a fresh database, `npx playwright test`	64/64 in 2.9 min; `git status -- artifacts` clean
`e2e/lab-04` on Firefox, WebKit and Microsoft Edge	the throwaway config of §5, a fresh database	70 of 72; the two failures are A11Y-01's Tab walks on WebKit (§7)
Performance smoke	`cd server && ./node_modules/.bin/tsx scripts/perf-smoke.mjs`, its own database	2,000 Tickets and 6,000 Actions: p95 3.8 ms (staff) and 1.59 ms (Requester). The query counts of PERF-02 (7 and 2 per request, the same at 200 and at 2,000 Tickets) are those measured on 5 October. A local measurement (specification §11.14)

Nothing was skipped, focused or disabled. Not run: a screen reader (§7).

3. Review-readiness follow-up, branch `fix/lab4-review-readiness`, product-code/dependency head `0ba3abc`, based on `lab4-staging@876511f` (13 October 2026). It was approved and merged as PR #104 into `lab4-staging` (`1141738`) before the release PR #103 was merged, so it is part of the release. Its follow-up changes broaden the dashboard mobile link-size assertions, fix the two dashboard footer links to meet 44 px, refresh audited dependency versions, and correct the stale test-harness identifier. This section's documentation is a later docs-only commit; the tested product tree is `0ba3abc`.

Check	Command	Result
Server suite, default and shuffled	`cd server && npm test`; then the same with `--sequence.shuffle --sequence.seed=33`	771/771 on each run; server code was unchanged after these runs
Client suite	`cd client && npm test`	34 files, 328/328 on Vite 8.3.4 and Vitest 4.1.11
Builds	`cd server && npm run build`; `cd client && npm run build`	both pass; the client build was repeated on Vite 8.3.4
Lint	`cd client && npm run lint`	exit 0; the same 3 existing warnings, 0 errors
Migration / recovery	`cd server && node scripts/verify-migration.mjs`	all 11 checks pass
Transport guard	`cd server && npx vitest run tests/lab-04/test-transport.unit.test.ts`	2/2 pass (TC-025)
Chromium E2E, whole repository	`npm run test:e2e` with `E2E_DATABASE_URL` naming a newly created dedicated `_test` database and `CAPTURE_EVIDENCE=0`	64/64; tracked artifacts clean; database dropped and verified absent
Cross-browser, Lab 4	`npm run test:e2e -- -c playwright.cross-browser.config.ts e2e/lab-04` using a throwaway config and a new dedicated `_test` database	Firefox 24/24; Microsoft Edge 154 24/24; WebKit 22/24. The only failures are the two known A11Y-01 Tab walks on WebKit (§7); the database was dropped and verified absent
Dependency audit	`npm audit --audit-level=low` and `npm audit --omit=dev --audit-level=low`, in both `server/` and `client/`	0 vulnerabilities in all four scans
Prisma toolchain	`npx prisma validate`; `npm run`	schema validation and client generation pass with

Figure 3.1ab — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 28/30).

	`server/` and `client/`	
Prisma toolchain	`npx prisma validate` ; `npm run prisma:generate`	schema validation and client generation pass with the scoped `@prisma/config` override

Dependency versions in this candidate: Multer 2.4.0; Vite 8.3.4 (manifest floor `^8.2.4`); Vitest 4.1.11; Prisma 7.9.1's scoped `deepmerge-ts` override 8.0.2 and `mysql2` override 3.24.5. The overrides are limited to Prisma's dependency branches; the app uses PostgreSQL.

Observed warning, not a test failure: the PostgreSQL `pg@8.22.0` client emitted a deprecation warning about calling `client.query()` while another query is executing, through the Prisma PostgreSQL adapter. It appeared during server/E2E runs and was not traced to a specific application operation; it was not suppressed or represented as fixed.

4. The exact final `main` commit `d07fefe` (tree `91a8f1a`), 14 October 2026. PR #103 was merged into `main` on 13 October at 12:47 UTC. The full gate was then run on that commit itself: a detached worktree at `d07fefe23dffafd7dd8257d125c97ffbed4c7250` , a clean `npm ci` in the root, `server/` and `client/` , `npx prisma generate` , a newly created `_test` database for every suite (dropped afterwards), one run at a time, the E2E lock held. Its tree is identical to `lab4-staging` at `1141738` (`git rev-parse` of both gives `91a8f1a`), the commit on which the reviewer ran the same gate before approving #103.

Check	Command	Result
Server suite, default order	a new `_test` database, `cd server && npm test`	40 files, 771/771
Server suite, shuffled	the same with `--sequence.shuffle --sequence.seed=33`	771/771
Client suite	`cd client && npm test`	34 files, 328/328
Builds	`cd server && npm run build` , `cd client && npm run build`	both pass
Lint	`cd client && npm run lint`	exit 0; the same 3 known warnings, 0 errors
Migration and recovery	`cd server && node scripts/verify-migration.mjs`	all 11 checks passed
E2E, Lab 2 to Lab 4, Chromium	`E2E_DATABASE_URL` → a new `_test` database, `npx playwright test`	64/64 in 2.1 min; `git status --ignored artifacts` clean
`e2e/lab-04` on Firefox, WebKit and Microsoft Edge	the throwaway config of §5, a new database	70 of 72 in 4.5 min; the two failures are A11Y-01's Tab walks on WebKit (§7); Firefox and Edge: none
Performance smoke	`perf-smoke.mjs` on its own database, run alone on the machine	p95 3.26 ms (staff) and 1.39 ms (Requester); a local measurement (specification §11.14)
Dependency audit	`npm audit --audit-level=low` in `server/` and in `client/`	0 vulnerabilities in each

Nothing was skipped, focused or disabled. Not run: a screen reader (§7). The documentation pull request that records this run changes documentation only, so it does not alter the tree this section ran on.

7. Known Limitations

Figure 3.1ac — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 29/30).

7. Known Limitations

- **Cross-browser coverage is ad hoc.** The suite runs on Chromium. A run of `e2e/lab-04` on the #91 head also passed on Firefox (12 of 12), on the student's everyday browser Microsoft Edge 154 (12 of 12, run twice, the second time by an independent audit), and on WebKit except A11Y-01's Tab walk: WebKit's default Tab order skips checkboxes and buttons (Safari's "Press Tab to highlight each item" setting), so that walk is Chromium and Firefox only. The dialog and focus tests of A11Y-02 pass on WebKit. No screen reader has been used: the labsheet does not ask for one, and ui-spec §11 row 21 rests on the accessibility-tree assertions named there.
- **The My Tickets table scrolls sideways inside its card when a row is wide.** Since Lab 2 the list sits in a container that scrolls horizontally (`.my-tickets__table-container`). A row whose status is `WAITING_FOR_REQUESTER` makes the table 1219 to 1231 px wide (measured on `d07fefe`) in a card 1150 px wide at 1280 and 1600 px, so the Last Updated column is reached by scrolling the table; the same holds at 1024 and 834 px (ui-spec §11 row 11). The two Requester drill-down captures (`drilldown-open` and `drilldown-needs-attention` , 1600 × 1000) therefore cut that column at the card's edge: they show the filter and the rows, not the last column. The page itself does not scroll sideways at the three widths of REG-04. Recorded, not changed.
- **A very long name in the page header is not covered.** A person whose own display name is 120 characters with no break widens the shell header (Lab 3's shell, found while testing RESP-01); the Actions area wraps such a name. Recorded, not changed.
- The performance smoke result is a local measurement on one Mac, not production capacity (specification §11.14).
- Dashboard and Action failure, loading and empty captures that need a broken server are produced by API interception and labelled *fixture*. The API tests prove the backend behaviour.
- History begins with Lab 4. Seeded and legacy Tickets have no events for earlier changes (specification §7).
- **Server-suite instability, found and fixed in #96 (PR #97).** Before that fix, full runs of the unchanged Lab 1–3 server suite on fresh `_test` databases failed intermittently: in default order 4 of 5 runs failed, with 1 to 22 tests each time and a different test every time. The cause was test transport, not shared state. Supertest's throwaway servers listened on every interface but were called on `127.0.0.1` , so a local program holding the same loopback port could answer. Statuses 401, 403 and 404 from other programs, and socket errors, were observed. The fix is merged in `lab4-staging` (#97); TC-025 guards the loopback binding and verifies that each disposable server answers its own requests.
- **A11Y-01's Tab walks fail on WebKit, twice.** WebKit's default Tab order skips checkboxes and buttons, so the Action form's walk and the dashboard drill-down walk cannot reach them by Tab (Safari's "Press Tab to highlight each item" changes this). Firefox, Microsoft Edge and Chromium pass both. The dashboard walk was first run on WebKit on 12 October.
- **A cancellation cascade stamps each Action's own `updatedAt` with the request's instant,** without the never-backwards rule that BR-35 applies to the Ticket (reviewer's note on #100, 11 October). Nothing reads an Action's `updatedAt` for ordering.
- **The dashboard's counts and lists are separate queries, not one snapshot** (specification §11.19, item 5).
- **A signed-out visit logs a 401 in the browser console.** Asking who is signed in is answered 401 by the Lab 3 contract; the audit declares it (specification §11.19, item 2).
- **`/system-check` and `/style-guide` are kept** (specification §11.19, item 3).
- **A "not found" state has no heading of its own** (specification §11.19, item 4).
- **No screen reader was used.** The accessibility evidence is keyboard operation, visible focus and assertions on names, roles and states (ui-spec §11, rows 8 to 10 and 21). An automated check does not say whether the speech is understandable.
- **Evidence images were opened, not all read.** 86 captures exist; an image of every kind of screen was opened at phone width, and the width and overflow assertions cover the rest (ui-spec §11, row 11).
- **The audit does not detect a no-op control** (reviewer's note on #102, 12 October). E2E-06 finds console and network errors, empty links, missing anchors and placeholder text; it does not click every button. A control that does nothing is caught only where a journey uses it and asserts what it does (the Actions Taken, workflow, dashboard and regression journeys do so for their own controls). AC-41's "no-op control" is therefore proved in part.

Figure 3.1ad — Rendered tests.md at GitHub Preview in overlapping L2 viewport segments; source at main 6d38208 (segment 30/30).

Release-gate verification summary

```
Verified 14 Oct 2026 (+07); run at d07fefe on tree 91a8f1a = main @ d07fefe
Server: npm test 771/771 (40 files); npm test -- --sequence.shuffle
--sequence.seed=33 771/771 (40 files).
Client: npm test; 328/328 (34 files).
E2E: npm run test:e2e; Chromium 64/64; 0 failed/skipped; isolated DB.
Builds: server + client PASS. Lint: exit 0, 0 errors, 3 warnings (2
SessionContext refresh; unused REQUESTER_B).
Migration/recovery: 11/11. Three-browser E2E on main @ d07fefe: 70/72 (70
passed, 2 WebKit A11Y-01 failures, 0 skipped). No screen reader.
```

Appendix A — Compact main-test output

This appendix records complete compact output and clearly labeled excerpts from the independent verification on main @ d07fefe (tree 91a8f1a), run 14 October 2026 (+07). “Complete output” is used only for suites reproduced in full below; the shuffled-order and three-browser entries are excerpts. Database connection URLs are redacted. The source notes when --silent suppresses application log lines.

```
COMPLETE OUTPUT — SERVER SUITE, DEFAULT ORDER

# Server suite, default order
# commit: d07fefe23dffafd7dd8257d125c97ffbed4c7250  tree: 91a8f1aa22ff064d83084fc0d11e535b4414001a
# date: 2026-10-14 21:48:27 +0700  node v26.7.0  postgres 18.6 (Homebrew)  Darwin arm64
# command: cd server && npm test -- --silent   (a new _test database; --silent hides the
application's own log lines)
# output below is complete; only database connection URLs are redacted


> server@0.0.0 test
> npm run db:test:setup && vitest run --silent


> server@0.0.0 db:test:setup
> node scripts/prepare-test-db.mjs

◇ injected env (3) from .env.test // tip: ⌘ suppress logs { quiet: true }
Preparing test database "toktickit_cmpsrv_test"...
Datasource "db": PostgreSQL database "toktickit_cmpsrv_test", schema "public" at "localhost:5432"

4 migrations found in prisma/migrations

Applying migration `20260805110504_add_category`
Applying migration `20260830000000_lab2_uuid_schema_and_reference_data`
Applying migration `20260912000000_lab3_users_roles_sessions`
Applying migration `20261004000000_lab4_actions_taken_and_history`

The following migration(s) have been applied:

migrations/
  └─ 20260805110504_add_category/
    └─ migration.sql
  └─ 20260830000000_lab2_uuid_schema_and_reference_data/
    └─ migration.sql
  └─ 20260912000000_lab3_users_roles_sessions/
    └─ migration.sql
  └─ 20261004000000_lab4_actions_taken_and_history/
    └─ migration.sql

All migrations have been successfully applied.
Seeded 4 categories, 7 related systems, 10 users (1 active administrator), 10 tickets, 9 public
comments, 5 internal notes, 8 actions.
```

```
Test database ready.

 RUN  v4.1.11 /private/tmp/claudemain3/server


 Test Files  40 passed (40)
      Tests  771 passed (771)
   Start at  21:48:29
   Duration  23.03s (transform 261ms, setup 806ms, import 1.48s, tests 18.69s, environment 1ms)


--- stderr ---
Loaded Prisma config from prisma.config.ts.

Prisma schema loaded from prisma/schema.prisma.
┌─────────────────────────────────────────────────────────┐
│  Update available 7.9.1 -> 8.0.0-rc.22                  │
│                                                         │
│  This is a major update - please follow the guide at    │
│  https://pris.ly/d/major-version-upgrade                │
│                                                         │
│  Run the following to update                            │
│    npm i --save-dev prisma@latest                       │
│    npm i @prisma/client@latest                          │
└─────────────────────────────────────────────────────────┘
(node:91796) DeprecationWarning: Calling client.query() when the client is already executing a query
is deprecated and will be removed in pg@9.0. Use async/await or an external async flow control
mechanism instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:91916) DeprecationWarning: Calling client.query() when the client is already executing a query
is deprecated and will be removed in pg@9.0. Use async/await or an external async flow control
mechanism instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:91924) DeprecationWarning: Calling client.query() when the client is already executing a query
is deprecated and will be removed in pg@9.0. Use async/await or an external async flow control
mechanism instead.
(Use `node --trace-deprecation ...` to show where the warning was created)

EXCERPT — SERVER SUITE, SHUFFLED ORDER
# commit: d07fefe23dffafd7dd8257d125c97ffbed4c7250  tree: 91a8f1aa22ff064d83084fc0d11e535b4414001a
# date: 2026-10-14 21:48:52 +0700  node v26.7.0  postgres 18.6 (Homebrew)  Darwin arm64
# command: cd server && npm test -- --silent --sequence.shuffle --sequence.seed=33
      Running tests with seed "33"
 Test Files  40 passed (40)
      Tests  771 passed (771)
   Duration  22.20s (transform 249ms, setup 772ms, import 1.41s, tests 18.07s, environment 1ms)

COMPLETE OUTPUT — CLIENT SUITE

# Client suite
# commit: d07fefe23dffafd7dd8257d125c97ffbed4c7250  tree: 91a8f1aa22ff064d83084fc0d11e535b4414001a
# date: 2026-10-14 21:49:16 +0700  node v26.7.0  postgres 18.6 (Homebrew)  Darwin arm64
# command: cd client && npx vitest run --silent
# output below is complete; only database connection URLs are redacted


 RUN  v4.1.11 /private/tmp/claudemain3/client


 Test Files  34 passed (34)
      Tests  328 passed (328)
   Start at  21:49:17
   Duration  21.47s (transform 263ms, setup 1.15s, import 851ms, tests 11.35s, environment 6.29s)

COMPLETE OUTPUT — BUILDS AND LINT

# Builds and lint
# commit: d07fefe23dffafd7dd8257d125c97ffbed4c7250  tree: 91a8f1aa22ff064d83084fc0d11e535b4414001a
# date: 2026-10-14 21:49:38 +0700  node v26.7.0  postgres 18.6 (Homebrew)  Darwin arm64
# command: cd client && npm run build; cd ../server && npm run build; cd ../client && npm run lint
# output below is complete; only database connection URLs are redacted

## client: npm run build
```

```
> client@0.0.0 build
> tsc -b && vite build

vite v8.3.4 building client environment for production...
transforming...
✓ 53 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.45 kB │ gzip:  0.29 kB
dist/assets/index-dg5PFwEG.css  261.64 kB │ gzip: 35.90 kB
dist/assets/index-DU6GtQ3a.js   338.15 kB │ gzip: 99.26 kB

✓ built in 364ms
exit 0

## server: npm run build

> server@0.0.0 build
> tsc -p tsconfig.json

exit 0

## client: npm run lint

> client@0.0.0 lint
> oxlint

tests/lab-02/MyTickets.test.tsx:29:7: warning eslint(no-unused-vars): Variable 'REQUESTER_B' is
declared but never used. Unused variables should start with a '_'. help: Consider removing this
declaration.
src/context/SessionContext.tsx:56:17: warning react(only-export-components): Fast refresh only works
when a file only exports components. Use a new file to share constants or functions between
components.
src/context/SessionContext.tsx:66:17: warning react(only-export-components): Fast refresh only works
when a file only exports components. Use a new file to share constants or functions between
components.
exit 0

COMPLETE OUTPUT — MIGRATION AND RECOVERY

# Migration and recovery
# commit: d07fefe23dffafd7dd8257d125c97ffbed4c7250  tree: 91a8f1aa22ff064d83084fc0d11e535b4414001a
# date: 2026-10-14 21:49:41 +0700  node v26.7.0  postgres 18.6 (Homebrew)  Darwin arm64
# command: cd server && node scripts/verify-migration.mjs
# output below is complete; only database connection URLs are redacted

Lab 3 state loaded in toktickit_mig_1791989381900_before_test: User=10 Session=3 Category=4
RelatedSystem=7 Ticket=24 Attachment=6 PublicComment=30 InternalNote=12 TicketNumberSequence=1
PASS  the Lab 4 migration exists and was applied — 20261004000000_lab4_actions_taken_and_history
PASS  MIG-01 every pre-existing table keeps its row count — User=10 Session=3 Category=4
RelatedSystem=7 Ticket=24 Attachment=6 PublicComment=30 InternalNote=12 TicketNumberSequence=1
PASS  MIG-01 every pre-existing table keeps its contents
PASS  MIG-01 all foreign keys are validated
PASS  the ActionTaken and TicketEvent tables exist
PASS  legacy Tickets have no Actions or events and start at version 1
PASS  the TicketEvent append-only trigger exists
PASS  schema.prisma matches the migrated database (no drift)
PASS  MIG-02 restoring the pre-migration dump reproduces the Lab 3 counts — User=10 Session=3
Category=4 RelatedSystem=7 Ticket=24 Attachment=6 PublicComment=30 InternalNote=12
TicketNumberSequence=1
PASS  MIG-02 the restored contents are identical
PASS  MIG-02 migrating the restored database again gives the same result
All 11 checks passed.
exit 0

COMPLETE OUTPUT — FULL CHROMIUM E2E LIST

# E2E, Lab 2 to Lab 4, Chromium (Playwright, list reporter)
# commit: d07fefe23dffafd7dd8257d125c97ffbed4c7250  tree: 91a8f1aa22ff064d83084fc0d11e535b4414001a
# date: 2026-10-14 21:49:44 +0700  node v26.7.0  postgres 18.6 (Homebrew)  Darwin arm64
# command: npx playwright test --reporter=list   (E2E_DATABASE_URL: a new _test database)
# output below is complete; only database connection URLs are redacted
```

```
Datasource "db": PostgreSQL database "toktickit_cmpchr_e2e_test", schema "public" at
"localhost:5432"

4 migrations found in prisma/migrations

Applying migration `20260805110504_add_category`
Applying migration `20260830000000_lab2_uuid_schema_and_reference_data`
Applying migration `20260912000000_lab3_users_roles_sessions`
Applying migration `20261004000000_lab4_actions_taken_and_history`

The following migration(s) have been applied:

migrations/
  └─ 20260805110504_add_category/
    └─ migration.sql
  └─ 20260830000000_lab2_uuid_schema_and_reference_data/
    └─ migration.sql
  └─ 20260912000000_lab3_users_roles_sessions/
    └─ migration.sql
  └─ 20261004000000_lab4_actions_taken_and_history/
    └─ migration.sql

All migrations have been successfully applied.
Seeded 4 categories, 7 related systems, 10 users (1 active administrator), 10 tickets, 9 public
comments, 5 internal notes, 8 actions.
◇ injected env (0) from .env.test // tip: ⌘ override existing { override: true }
Reset credentials on 10 seeded accounts in toktickit_cmpchr_e2e_test; 1 left behind the must-change
gate.

Running 64 tests using 1 worker

Cleaned 1 E2E ticket(s).
  ✓   1 [chromium] › e2e/lab-02/attachment-lifecycle.spec.ts:11:5 › E2E-03 covers add, download,
soft-remove, and blocked attachment (1.9s)
Cleaned 1 E2E ticket(s).
  ✓   2 [chromium] › e2e/lab-02/requester-isolation.spec.ts:5:5 › E2E-02 isolates tickets between
signed-in users (1.7s)
Cleaned 1 E2E ticket(s).
  ✓   3 [chromium] › e2e/lab-02/requester-ticket-flow.spec.ts:5:5 › E2E-01 completes requester
creation and My Tickets journey (1.4s)
  ✓   4 [chromium] › e2e/lab-02/responsive.spec.ts:11:5 › RESP-01 renders Create Ticket at desktop,
tablet, and mobile (634ms)
Cleaned 1 E2E ticket(s).
  ✓   5 [chromium] › e2e/lab-02/responsive.spec.ts:24:5 › RESP-02 renders My Tickets at desktop,
tablet, and mobile (1.7s)
Cleaned 1 E2E ticket(s).
  ✓   6 [chromium] › e2e/lab-02/responsive.spec.ts:48:5 › RESP-03 renders Ticket Detail at desktop,
tablet, and mobile (1.6s)
  ✓   7 [chromium] › e2e/lab-03/authentication.spec.ts:30:5 › AUTH-01 · AC-34 captures the sign-in
screen at three viewports (407ms)
  ✓   8 [chromium] › e2e/lab-03/authentication.spec.ts:43:5 › AUTH-02 refuses wrong, unknown, and
inactive accounts with one safe message (902ms)
  ✓   9 [chromium] › e2e/lab-03/authentication.spec.ts:60:5 › AUTH-03 signs in and reaches
Dashboard (368ms)
  ✓  10 [chromium] › e2e/lab-03/authentication.spec.ts:77:5 › AUTH-04 · AC-34 captures the
mandatory Change Password screen (657ms)
  ✓  11 [chromium] › e2e/lab-03/authentication.spec.ts:100:5 › AUTH-05 refuses every other route
until the password is changed (450ms)
  ✓  12 [chromium] › e2e/lab-03/authentication.spec.ts:116:5 › AUTH-06 logs out and cannot return
without signing in again (512ms)
  ✓  13 [chromium] › e2e/lab-03/comments-and-notes.spec.ts:46:5 › THREAD-01 captures both threads,
visually distinct (451ms)
  ✓  14 [chromium] › e2e/lab-03/comments-and-notes.spec.ts:67:5 › THREAD-02 posts to each thread
through its own composer (2.3s)
  ✓  15 [chromium] › e2e/lab-03/comments-and-notes.spec.ts:88:5 › THREAD-03 captures what the
Requester sees, and does not (395ms)
  ✓  16 [chromium] › e2e/lab-03/comments-and-notes.spec.ts:107:5 › THREAD-04 refuses the Requester
directly, revealing no count (286ms)
  ✓  17 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:58:5 › QUEUE-01 · AC-34 captures the
queue at three viewports (598ms)
  ✓  18 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:86:5 › QUEUE-02 shows ownership and the
resolution marker (303ms)
```

```
  ✓  19 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:99:5 › QUEUE-03 captures the no-results
state (1.2s)
  ✓  20 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:116:5 › QUEUE-04 · fixture-backed UI
evidence captures the true empty state (442ms)
  ✓  21 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:156:5 › QUEUE-05 refuses a Requester with
the forbidden state (367ms)
  ✓  22 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:170:5 › QUEUE-07 · AC-19 preserves the
live CANCELLED and Assigned to me filter check (628ms)
  ✓  23 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:189:5 › DETAIL-01 · AC-34 captures the
staff Ticket Detail at three viewports (839ms)
  ✓  24 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:206:5 › DETAIL-02 claims an unassigned
Ticket and moves it to OPEN (634ms)
  ✓  25 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:226:5 › DETAIL-03 offers only permitted
transitions (520ms)
  ✓  26 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:238:5 › DETAIL-04 refuses the moves
nobody and no role may make (449ms)
  ✓  27 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:263:5 › DETAIL-05 captures the refusal a
Requester can actually see (695ms)
  ✓  28 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:289:5 › DETAIL-06 lets the Requester say
the problem appears resolved (529ms)
Cleaned 1 E2E ticket(s).
  ✓  29 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:304:5 › DETAIL-07 carries an uploaded
attachment into Staff Ticket Detail (2.0s)
  ✓  30 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:329:5 › DETAIL-08 reassigns a Ticket to
another active owner (564ms)
  ✓  31 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:351:5 › DETAIL-09 shows a safe load
failure and recovers with Try again (669ms)
  ✓  32 [chromium] › e2e/lab-03/staff-ticket-flow.spec.ts:367:5 › QUEUE-06 · fixture-backed UI
evidence demonstrates 21-ticket pagination (441ms)
  ✓  33 [chromium] › e2e/lab-03/user-administration.spec.ts:81:5 › ADMIN-01 · AC-34 captures the
user list at three viewports (586ms)
  ✓  34 [chromium] › e2e/lab-03/user-administration.spec.ts:99:5 › ADMIN-02 captures the create
dialogue and creates a user (656ms)
  ✓  35 [chromium] › e2e/lab-03/user-administration.spec.ts:128:5 › ADMIN-03 captures the
duplicate-email conflict (482ms)
  ✓  36 [chromium] › e2e/lab-03/user-administration.spec.ts:144:5 › ADMIN-04 edits a user, resets
its password, and proves the next-login gate (1.4s)
  ✓  37 [chromium] › e2e/lab-03/user-administration.spec.ts:234:5 › ADMIN-05 captures
self-deactivation disabled with its reason (484ms)
  ✓  38 [chromium] › e2e/lab-03/user-administration.spec.ts:258:5 › ADMIN-06 captures the
last-Administrator refusal (461ms)
  ✓  39 [chromium] › e2e/lab-03/user-administration.spec.ts:283:5 › ADMIN-07 captures search and
role-filtered results (671ms)
  ✓  40 [chromium] › e2e/lab-03/user-administration.spec.ts:313:5 › ADMIN-08 captures the forbidden
state for a Requester (313ms)
Cleaned 1 E2E ticket(s).
  ✓  41 [chromium] › e2e/lab-04/actions-taken-flow.spec.ts:31:5 › E2E-01 · AC-01 to AC-10, AC-12 ·
Olivia works a Ticket Daniel owns, and the Requester reads it all (4.9s)
Cleaned 1 E2E ticket(s).
  ✓  42 [chromium] › e2e/lab-04/actions-taken-flow.spec.ts:154:5 › E2E-01 · AC-13 · a Ticket that
is not workable shows the banner instead of Add Action (1.6s)
Cleaned 1 E2E ticket(s).
  ✓  43 [chromium] › e2e/lab-04/actions-taken-flow.spec.ts:165:5 › E2E-03 · AC-14 · AC-19 · AC-36 ·
a conflict and a save failure keep what was typed (2.0s)
Cleaned 1 E2E ticket(s).
  ✓  44 [chromium] › e2e/lab-04/actions-taken-flow.spec.ts:208:5 › RESP-01 · AC-34 · Ticket Detail
with Actions fits every width, with 2000-character Description, Follow-up Note and Result,
500-character Attachment Notes and a 120-character name (11.8s)
Cleaned 1 E2E ticket(s).
  ✓  45 [chromium] › e2e/lab-04/actions-taken-flow.spec.ts:281:5 › E2E-10 · AC-34 · AC-35 · the Add
form, its validation, the edit panel and the Requester view fit desktop, tablet and phone (7.9s)
Cleaned 1 E2E ticket(s).
  ✓  46 [chromium] › e2e/lab-04/actions-taken-flow.spec.ts:367:5 › A11Y-01 · AC-35 · the Action
form and the cancel dialog work from the keyboard alone (1.7s)
Cleaned 1 E2E ticket(s).
  ✓  47 [chromium] › e2e/lab-04/actions-taken-recovery.spec.ts:22:5 › E2E-07 · AC-14 · AC-36 · a
retry after a lost response keeps what was changed meanwhile (1.9s)
Cleaned 1 E2E ticket(s).
  ✓  48 [chromium] › e2e/lab-04/actions-taken-recovery.spec.ts:53:5 › E2E-08 · BR-23 · AC-13 · a
form left open when the Ticket is resolved can no longer be saved (1.6s)
Cleaned 1 E2E ticket(s).
```

```
  ✓  49 [chromium] › e2e/lab-04/actions-taken-recovery.spec.ts:72:5 › A11Y-02 · AC-35 · ui-spec §9
· the first invalid field takes the focus, a closed panel gives it back, a busy dialog keeps it
(2.0s)
Cleaned 1 E2E ticket(s).
  ✓  50 [chromium] › e2e/lab-04/actions-taken-recovery.spec.ts:132:7 › E2E-09 · api-spec §1 · times
are Bangkok time on a device in another zone › shows the Actions and the history in Asia/Bangkok
(1.5s)
  ✓  51 [chromium] › e2e/lab-04/audit.spec.ts:51:5 › E2E-06 · AC-41 · FR-26 · a signed-out visitor
reaches the sign-in and the two public pages without an error (3.9s)
  ✓  52 [chromium] › e2e/lab-04/audit.spec.ts:63:7 › E2E-06 · AC-41 · FR-26 · the Requester follows
every link from the first screens without an error (7.3s)
  ✓  53 [chromium] › e2e/lab-04/audit.spec.ts:63:7 › E2E-06 · AC-41 · FR-26 · the IT Staff follows
every link from the first screens without an error (17.0s)
  ✓  54 [chromium] › e2e/lab-04/audit.spec.ts:63:7 › E2E-06 · AC-41 · FR-26 · the Administrator
follows every link from the first screens without an error (16.3s)
  ✓  55 [chromium] › e2e/lab-04/dashboards.spec.ts:110:5 › E2E-04 · AC-32 · all roles land on their
first Dashboard navigation item after sign-in (941ms)
  ✓  56 [chromium] › e2e/lab-04/dashboards.spec.ts:129:5 › E2E-04 · AC-26–AC-29, AC-34 · staff
dashboard matches SQL, drills down, and keeps every mobile link target large enough (467ms)
  ✓  57 [chromium] › e2e/lab-04/dashboards.spec.ts:158:5 › E2E-04 · AC-02, AC-26–AC-29, AC-34 ·
Requester dashboard matches SQL, filters My Tickets, and keeps every mobile link target large enough
(772ms)
  ✓  58 [chromium] › e2e/lab-04/dashboards.spec.ts:203:5 › E2E-04/05 · AC-29–AC-31 · zero,
forbidden and recoverable failure states (786ms)
  ✓  59 [chromium] › e2e/lab-04/dashboards.spec.ts:247:5 › E2E-05 · AC-31 · staff dashboard loading
and failure fixtures recover safely (470ms)
  ✓  60 [chromium] › e2e/lab-04/dashboards.spec.ts:282:5 › A11Y-01 · AC-35 · keyboard reaches a
descriptive dashboard drill-down with visible focus (291ms)
Cleaned 1 E2E ticket(s).
  ✓  61 [chromium] › e2e/lab-04/regression.spec.ts:22:5 › REG-04 · FR-22 · sign-in, My Tickets,
Ticket Detail with an attachment and a comment, the Queue, an Internal Note and User Management
(7.6s)
Cleaned 1 E2E ticket(s).
  ✓  62 [chromium] › e2e/lab-04/ticket-resolution.spec.ts:23:5 › E2E-02 · AC-15 to AC-18, AC-21 to
AC-23 · resolving is blocked by open work, then allowed, and it is all in the history (1.9s)
Cleaned 1 E2E ticket(s).
  ✓  63 [chromium] › e2e/lab-04/ticket-resolution.spec.ts:85:5 › E2E-02 · AC-17 · cancelling a
Ticket asks first and cancels its open Actions with it (1.5s)
Cleaned 1 E2E ticket(s).
  ✓  64 [chromium] › e2e/lab-04/ticket-resolution.spec.ts:107:5 › E2E-03 · AC-19 · AC-21 · a stale
status change is refused, the latest status is shown, and a retry succeeds (1.6s)

  64 passed (2.2m)

--- stderr ---
Loaded Prisma config from prisma.config.ts.

Prisma schema loaded from prisma/schema.prisma.
(the application server's own [WebServer] log lines are not reproduced in this file)
git status -- artifacts after the run (empty = clean):

EXCERPT — LAB 4 THREE-BROWSER E2E, MAIN @ d07fefe
Summary: 70 passed, 2 failed, 0 skipped.
    [webkit] › e2e/lab-04/actions-taken-flow.spec.ts:367:5 › A11Y-01 · AC-35 · the Action form and
the cancel dialog work from the keyboard alone
    [webkit] › e2e/lab-04/dashboards.spec.ts:282:5 › A11Y-01 · AC-35 · keyboard reaches a
descriptive dashboard drill-down with visible focus
```

Per-file test counts

These derived counts summarize the suite totals and all 14 Chromium E2E spec files; they complement, rather than replace, the complete outputs above.

Scope / file	Files / result
Server (default; shuffled seed 33)	40 files; 771/771 each
Client	34 files; 328/328

Chromium E2E (whole repository)	14 spec files; 64/64
Lab 4 three-browser E2E	70 passed; 2 WebKit A11Y-01 failed; 0 skipped
e2e/lab-02/attachment-lifecycle.spec.ts	1
e2e/lab-02/requester-isolation.spec.ts	1
e2e/lab-02/requester-ticket-flow.spec.ts	1
e2e/lab-02/responsive.spec.ts	3
e2e/lab-03/authentication.spec.ts	6
e2e/lab-03/comments-and-notes.spec.ts	4
e2e/lab-03/staff-ticket-flow.spec.ts	16
e2e/lab-03/user-administration.spec.ts	8
e2e/lab-04/actions-taken-flow.spec.ts	6
e2e/lab-04/actions-taken-recovery.spec.ts	4
e2e/lab-04/audit.spec.ts	4
e2e/lab-04/dashboards.spec.ts	6
e2e/lab-04/regression.spec.ts	1
e2e/lab-04/ticket-resolution.spec.ts	3

Answer Part 4: AI Use with Reflection

Question — Which AI-assisted steps materially changed the work, and how did the student verify or reject the agent's output?

The ten examples below are selected from the AI-use record I confirmed (main @ 6d38208). Entry 1 is verbatim; entries 2–10 are faithful English summaries or reconstructions of recorded requests, decisions, and approved follow-ups—not word-for-word quotations. I checked each against its source task brief and recorded outcome. Model names follow the available session evidence: Claude entries are recorded or marked unknown; the Codex entry is GPT-5 (deployment suffix not exposed). I reviewed and confirmed the first-person reflections on 13 October 2026.

AI-use record and student-confirmed reflections

Tool / period	Purpose	Model name
Claude Code / 3 October 2026	Orientation, decisions and specification	`claude-opus-5-5` (session setting)
Claude Code / 3–5 October 2026	Coding work on #90, #91 and #92; test-transport fix for #96/#97; audit fixes through 5 October	`claude-sonnet-5-5` (session setting)
Claude Code / 6–12 October 2026	#94 final hardening and release-candidate verification	unknown (setting not recorded)
Claude Code / 14–16 October 2026	Final-main verification of `d07fefe`; PR #105 documentation corrections; read-only report/layout audits; preparation of partner-repository PR reviews #105 (third round) and #106	`claude-sonnet-5-5` (stated by the session)
Codex / October 2026	Independent audits, dashboards and report editing	`GPT-5` (deployment suffix not exposed)

Partner-repository reviews

Claude Code helped me prepare reviews of PR #101 (12 October), #105 (third round) and #106 (15 October) in my partner's repository. It compared each diff with the partner's contract, ran the test suites against fresh databases, used fault injection to assess whether the tests detected the injected defects, and drafted review text. I reviewed the drafts, chose the verdicts, and posted the reviews from my account. I merged PRs #101 and #105; PR #106 was approved but remained unmerged at the recorded review.

Selected Prompt 1

Prompt or decision input	What the agent did and the outcome	My Reflection
`Read _private/lab-04/MASTER-PROMPT.md and _private/lab-04/prompts/00-orientation.md in full. Perform only the read-only orientation, report findings, and stop before implementation or GitHub changes.`	Reviewed the labsheet, workflow guide, inherited contracts and both repositories without making changes. Confirmed that live main was at `af59d34`, identified that each E2E run rewrote 48 tracked Lab 3 screenshots, and documented conflicts between the system-design specification and Labs 3/4. The baseline gate passed, and the agent presented the first set of decisions before implementation.	The orientation-only boundary proved useful: it exposed the screenshot-overwrite risk and inherited-contract conflicts before implementation. I want to keep source and workspace checks ahead of coding, because an agent's first assumption can otherwise become expensive rework.

Selected Prompt 2

Prompt or decision input	What the agent did and the outcome	My Reflection
`Review the source-backed recommendations and trade-offs. Incorporate my approved decisions on the Action lifecycle and assignment, the resolution gate, role-based field visibility, History, concurrency, and dashboard metrics into the Lab 4 contracts. Estimate whether the work can be completed in three to four days, including peer-review time.`	The agent presented evidence-backed options and trade-offs across the six contract areas; I approved the selected behaviors. It also used actual Lab 3 review latency to estimate a schedule that included peer-review time.	Deciding the lifecycle, permissions, history and metrics before implementation made trade-offs visible instead of letting the code choose them. Using Lab 3 review latency also made the schedule more realistic by accounting for peer review, not just coding time.

Selected Prompt 3

Prompt or decision input	What the agent did and the outcome	My Reflection
`Before further server-test work, prepare the local environment by retaining only toktickit_dev, toktickit_test, and toktickit_e2e_test; remove the 24 obsolete databases. Then investigate Issue #96 without assuming shared database state is the cause. Reproduce the failure with isolated file runs, measure loopback requests without database access, identify the process answering on the port, and add a regression test that fails before the fix. Run the full suite 20 times and report the evidence, root cause, fix, and verification; update PR #95 and Issue #96 only when supported by the results.`	Removed 24 obsolete databases. During work on #96, the initial shared-state explanation was recorded in PR #95. Isolated file runs and 24,000 requests without using a database instead showed another local process answering on the same loopback port. The agent added a failing regression test, ran the suite 20 times with 412/412 passing each time, and corrected PR #95 and Issue #96.	The first explanation was plausible but wrong. Isolated runs and measurements without a database identified a competing listener. I learned to treat a root-cause claim as a hypothesis until a controlled check reproduces it.

Selected Prompt 4

Prompt or decision input	What the agent did and the outcome	My Reflection
`While the other work awaits peer review, proceed with Issue #92 within the approved Ticket-workflow and History scope. Write and run failing tests before implementation. Keep changes within that scope; if the contract leaves behavior undefined, stop and ask before changing expectations.`	Implemented the Ticket resolution gate, cancellation cascade, versioned status changes, and History test-first. All 18 intentionally injected faults were caught by the resulting tests.	The red-green sequence and 18/18 detected faults made the workflow tests more convincing. It also reminded me that a green suite alone is not enough: I still need to compare test coverage with every approved transition.

Selected Prompt 5

Prompt or decision input	What the agent did and the outcome	My Reflection
`Audit PR #98 against the approved Lab 4 contract. Do not change code or GitHub.` `Check the Action lifecycle and limits, assignee rules, requestId idempotency, version checks and event ordering, Requester-visible fields, append-only History, and seed idempotency. Probe authorization, cross-Ticket access, concurrency, stale or replayed writes, refusal side effects, error leakage, migration safety, and weak tests.`  `In an isolated worktree with a disposable _test database, run at least eight independent mutations one at a time and restore each afterward. Run the server suite single-worker; skip E2E and migration verification. Report conformance, test quality, probe and mutation results, suite/build results, cleanup, and unverified items. Separate evidence from inference.`	Codex found no confirmed code defect, but its mutations exposed a test gap: the Requester view of who completed or cancelled an Action was untested, and rerunning the seed changed updatedAt despite a test expecting no change. Claude Code added failing tests first, fixed both gaps, and reran the relevant suites.	The audit found no confirmed implementation defect, yet mutation checks exposed missing Requester-visibility coverage and a seed-idempotence gap. I learned that review should challenge the tests as well as the code, especially where data could be exposed or silently changed.

Selected Prompt 6

Prompt or decision input	What the agent did and the outcome	My Reflection
`Independently audit the #92/#91 stack against the approved contract without changing product code or either branch.` `Review Ticket transitions and races, cancellation, version checks and History ordering, Actions Taken forms and states, Requester privacy, keyboard and focus behavior, and the Lab 3 landing assertions affected by dashboard routing. Use an isolated worktree and disposable _test database. Apply at least ten independent mutations, including three client mutations, and restore each. Run E2E only under the shared lock, then re-audit the follow-up fixes.`  `Report requirement-to-code/test coverage, probes and mutation results, test-quality findings, suite results, and unverified items. Separate observed evidence from inference, then clean up the worktree, database, and lock.`	Codex reproduced seven defects, including lost form input after a failed retry, editable controls after a Ticket closed, unstable History ordering, and focus and time-zone faults; it also identified two test gaps. Claude Code addressed the findings test-first and ran six server and 16 client fault injections; new browser tests failed against the old code. The re-audit classified nine findings as fixed or partly fixed and identified three additional issues.	The independent audit made UI and workflow risks concrete, including lost input, stale editing, event ordering, focus and time-zone behavior. I treated its findings as claims to reproduce, which helped separate confirmed problems, partial findings and issues revealed during re-audit.

Selected Prompt 7

Prompt or decision input	What the agent did and the outcome	My Reflection
`Disclose AI assistance in the report and working record, but do not list an AI tool as a contributor in commits, pull requests, Issues, or comments. Preserve already-merged history.`	Stopped adding AI-credit lines to commits and pull requests and documented the rule for both agents. With the student's approval, removed those lines from the open PR #98 commits without changing their code tree. Merged history remained untouched so the SHAs cited by Lab 3 evidence stayed valid. AI use is disclosed in the report and working records instead.	I want to disclose AI use honestly without listing an AI tool as a repository contributor. Keeping that disclosure in the report and work record preserved the authorship rule without rewriting merged history.

Selected Prompt 8

Prompt or decision input	What the agent did and the outcome	My Reflection
`Evaluate each audit finding separately rather than applying the recommendations as a batch. Reproduce the database-lock issue; measure the timestamp behavior with an appropriate clock; and ask me to resolve any conflict between the two contracts. Implement only the changes I approve, test-first, and verify that each test fails for the intended reason.`	The agent assessed three findings before acting: it reproduced the database-lock issue; partly confirmed the timestamp claim after measuring a 12 ms change caused by a different clock; and identified a conflict between two contracts for me to resolve. I chose Bangkok time for every screen. All accepted changes were test-first. One test initially passed for the wrong reason—a lazily started database call—so the test was corrected before the result counted.	The findings needed different responses: reproduce the lock issue, measure the timestamp claim, and bring the contract conflict back to me. I learned to judge recommendations individually and to check why a test passed before trusting the result.

Selected Prompt 9

Prompt or decision input	What the agent did and the outcome	My Reflection

`Implement Issue #93—the Staff and Requester dashboards—in an isolated worktree, test-first, and within the approved contract. Update only the three Lab 3 landing assertions made obsolete by dashboard routing, then remove the temporary dashboard-landing workaround.`
 	
`Keep the dashboard implementation on the existing PR #99. When later release-readiness review identifies a defect, reproduce it with a test before fixing it; refresh a screenshot only after verifying its heading, status, and rows. Rerun the full suites, builds, and lint. Do not open a duplicate PR for the same dashboard work.`
	Implemented the dashboards test-first. The first full E2E run was 45/46 because a Lab 3 forced-password landing assertion still expected Ticket Queue. Three landing assertions were updated red-first; two client tests failed while navigation still went to /tickets, and the temporary detour was removed. Later review found dashboard links only 18 px high at 390 px and a stale filtered-list capture. The follow-up added checks requiring every visible dashboard link to be at least 44 px and refreshed the capture only after verifying its heading, status, and rows. On the updated candidate: server 771/771, client 328/328, full E2E 64/64, focused link checks 2/2; both builds passed and lint exited 0 with three existing warnings.	The initial regression exposed an outdated Lab 3 route expectation, not a reason to keep a navigation workaround. Later, targeted checks confirmed two dashboard-evidence defects before they were fixed. I learned to change only behavior that fails a reproducible check and to verify that screenshots show the asserted state.

Selected Prompt 10

Prompt or decision input	What the agent did and the outcome	My Reflection
`On the existing PR #99 branch, strengthen tests for gaps G1–G5; the review identifies fixture weaknesses, not a confirmed production-query defect. Add contrasting fixtures for terminal and urgent Staff work and Requester Tickets in all eight statuses. For each of the five query mutants, show the targeted test fail with the mutant and pass with production code. Rerun the server and client suites, both builds, lint, and full E2E on the isolated database. Record results and clean up. Do not open another PR or change production code without evidence.`	Added fixtures for terminal/urgent Staff work and Requester Tickets in all eight statuses; the recent-list assertion requires exactly the newest five. Each of the five intentional query mutants failed its targeted API tests, while production code passed. The server and client suites, both builds, and full E2E passed (46/46); lint had three existing warnings. One E2E command initially targeted the shared test database, reseeded its accounts, and was interrupted. Codex disclosed the incident in handoff 011; Claude Code later inspected the database, removed the single leftover E2E Ticket, and reran E2E. The follow-up is recorded in WORK-RECORD.md.	The five surviving dashboard-query mutants came from fixtures too weak to distinguish incorrect predicates. Strengthening the fixtures and making each mutant fail showed me why realistic contrasting data matters. The earlier E2E database-target mistake also reinforced checking database isolation before a run.

Overall reflection

I used the specification and coding agents for different parts of the work. The specification phase helped turn inherited requirements into explicit decisions before implementation; the coding phase produced tests, changes and evidence against those decisions. I still owned the choices and approvals, and the reviewer—not the author—performed the merges.

The strongest safeguard was independent verification. A confident initial diagnosis was wrong; another audit claim was only partly supported; and mutation checks uncovered tests that looked complete but still let faults survive. Reproducing findings against the named source taught me to accept evidence, not confidence. I also learned to distinguish staging-candidate results from main results and to label historical snapshots clearly rather than presenting them as final evidence.

For the next lab, I would verify test isolation before every run, preserve a concise decision log and check each test against the contract. AI assistance can speed up analysis and implementation, but responsibility for decisions, review and honest reporting remains mine.

Answer Part 5: Working IT Staff Dashboard UI

Question — Does the IT Staff dashboard show accurate work and the required loading, empty, error, and responsive states?

IT Staff and Administrators enter the staff Dashboard. Cards show a label, value and descriptive link; the resulting Queue filter follows the same predicate. My open Actions concerns the authenticated assignee, not the Ticket Owner or logger. Recent Tickets are ordered by updatedAt descending with id as a stable tie-breaker.

Behaviour	Predicate / destination
Unassigned	ownerId is null and ACTIVE → Queue ownerId=unassigned, statusGroup=active
Assigned to me	ownerId equals authenticated user and ACTIVE → Queue ownerId=me, statusGroup=active
Urgent	IT priority URGENT and ACTIVE → Queue itPriority=URGENT, statusGroup=active
Waiting for Requester	status WAITING_FOR_REQUESTER → matching Queue status
Tickets by status	One count per actual status → Queue status=X
My open Actions	assigneeId equals authenticated user; PLANNED or IN_PROGRESS; total and 10 oldest → Ticket Detail
Recently updated	Five newest Tickets; updatedAt DESC, id DESC → Ticket Detail / sorted Queue

DASH-01–DASH-06 and SEC-04 compare endpoint results with independent SQL counts and permission boundaries. Terminal urgent Tickets in the fixtures catch an incorrect ACTIVE filter. The positive Unassigned drill-down has a matching Queue capture. Loading and failure captures are fixture-backed, not natural production incidents.

Staff query and independent SQL-comparison tests

The refreshed dashboard screenshots were recorded on 5 October at accepted staging 876511f and refreshed through PR #104; source assets are tracked on main @ d07fefe. Each figure links to the complete capture. Loading and failure states are fixture-backed.

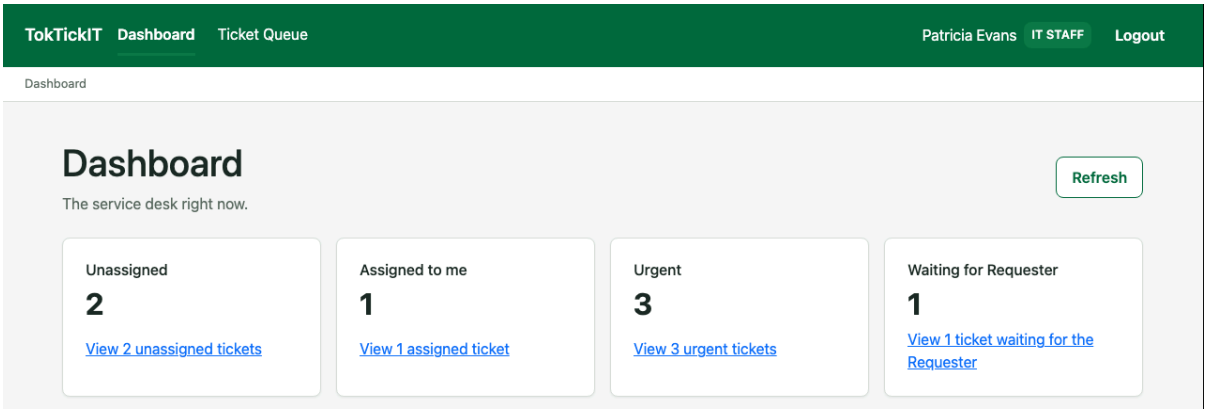


Figure 5.1 — IT Staff Dashboard at desktop width (1,280 px). Crop shows the title and all four summary cards; full capture.

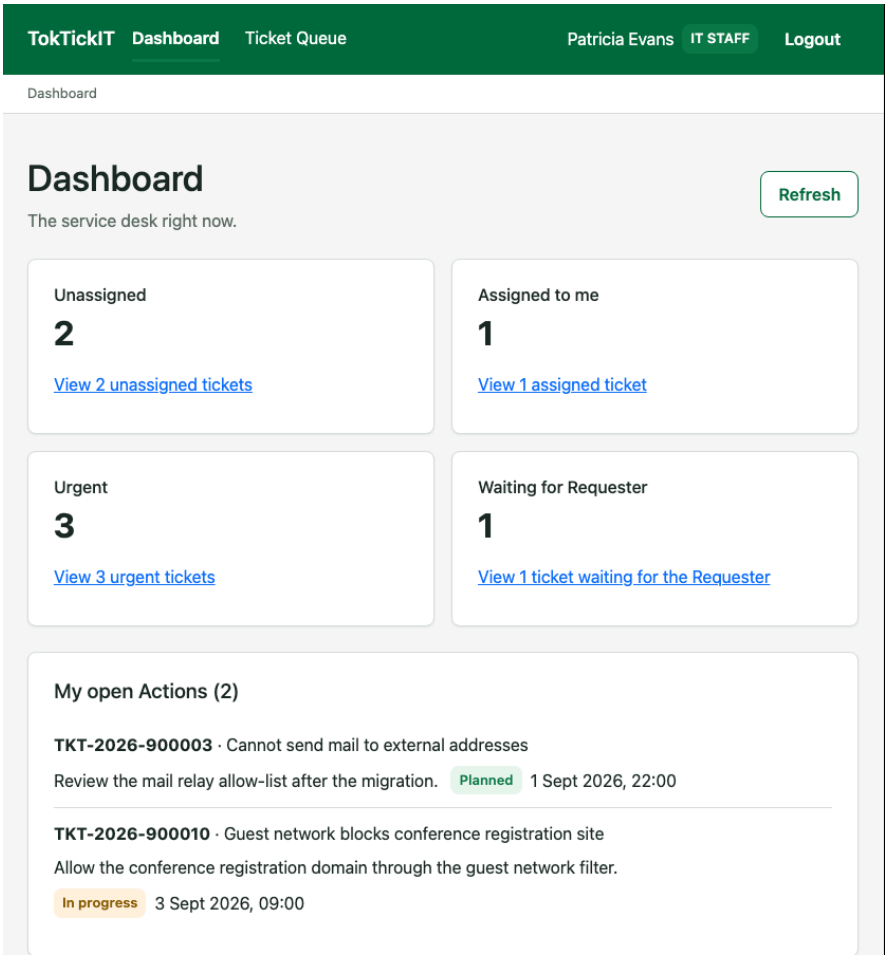


Figure 5.2 — IT Staff Dashboard at tablet width (834 px). Crop shows the metrics and My open Actions; full capture.

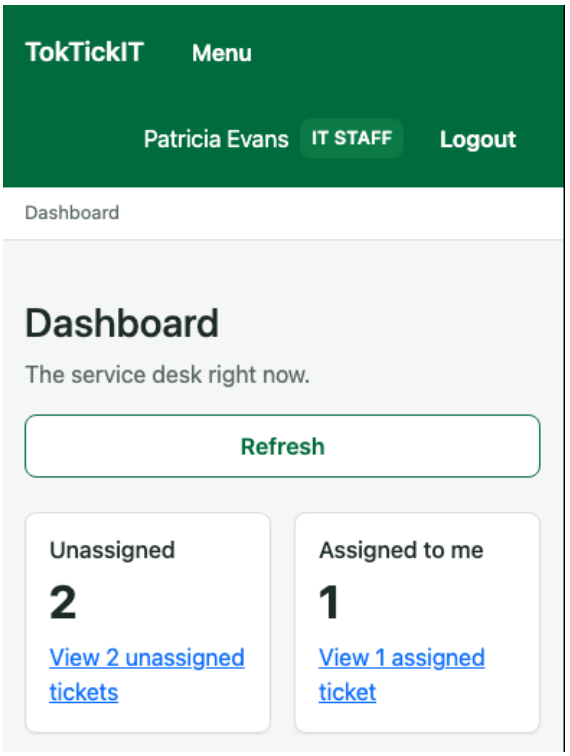


Figure 5.3 — IT Staff Dashboard at mobile width (390 px). Top crop shows navigation and the first two metric cards; the linked full capture shows the remaining dashboard panels.

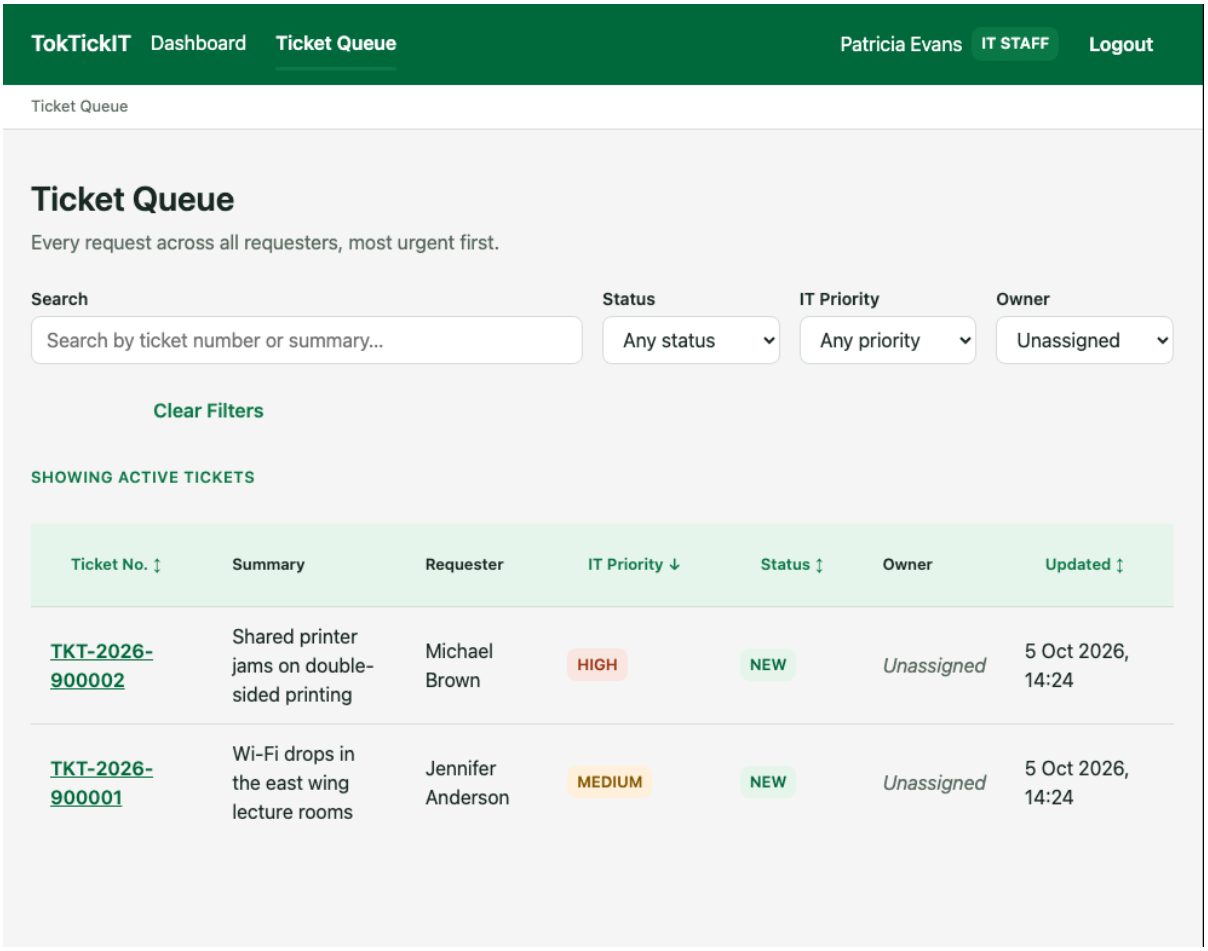


Figure 5.4 — Positive Unassigned drill-down in Ticket Queue (992 px). Crop shows the selected filter and both matching rows; full capture.

Additional state captures — Part 5

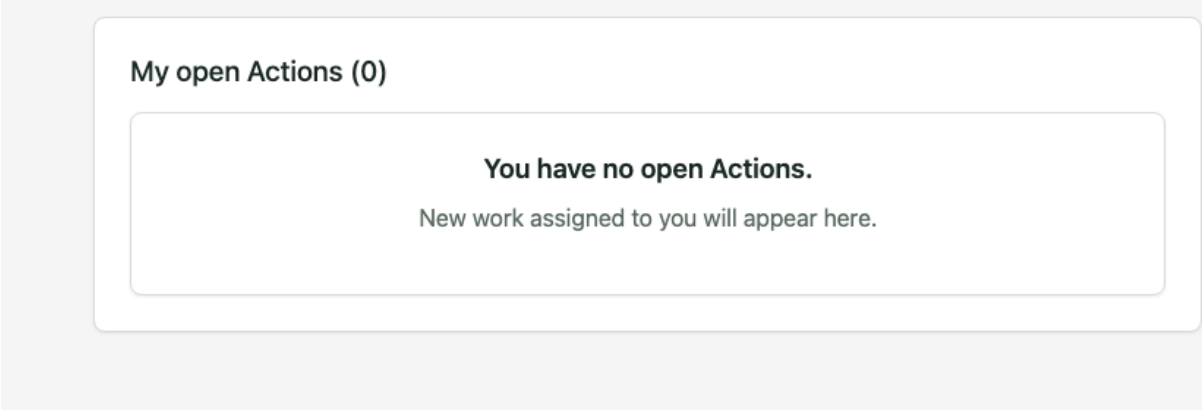


Figure 5.5 — Administrator Dashboard with no open Actions. Crop isolates the complete My open Actions (0) state; full capture.

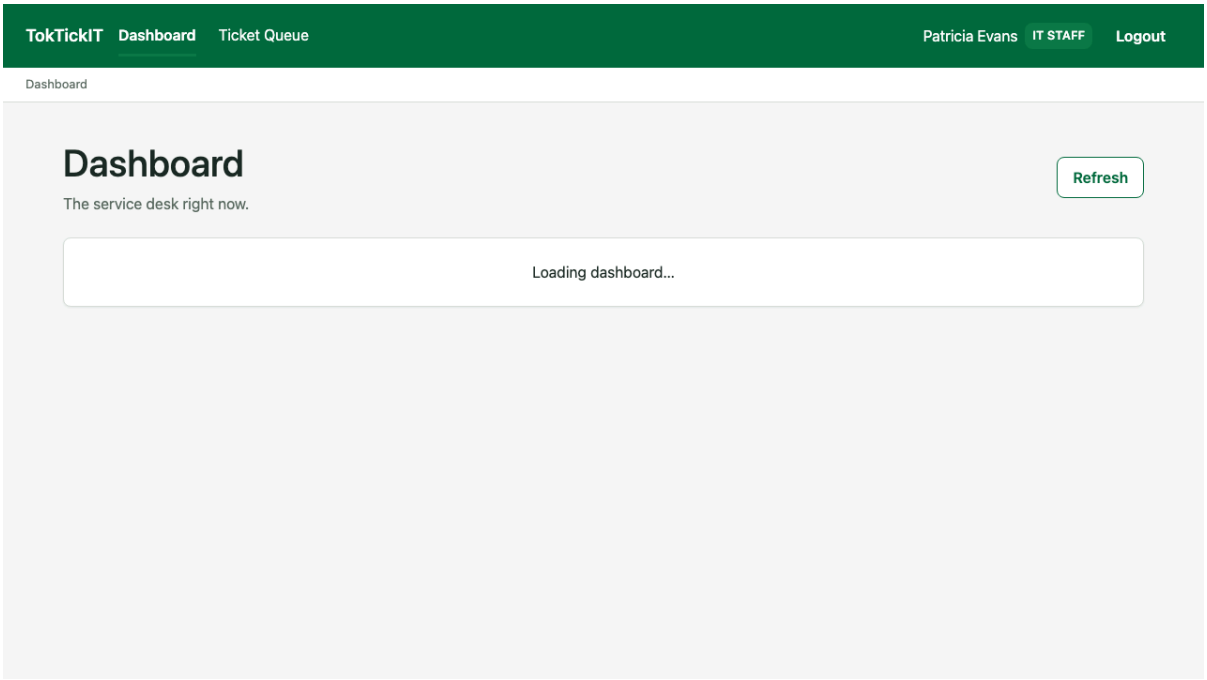


Figure 5.6 — Loading Dashboard state (fixture-backed); full capture.

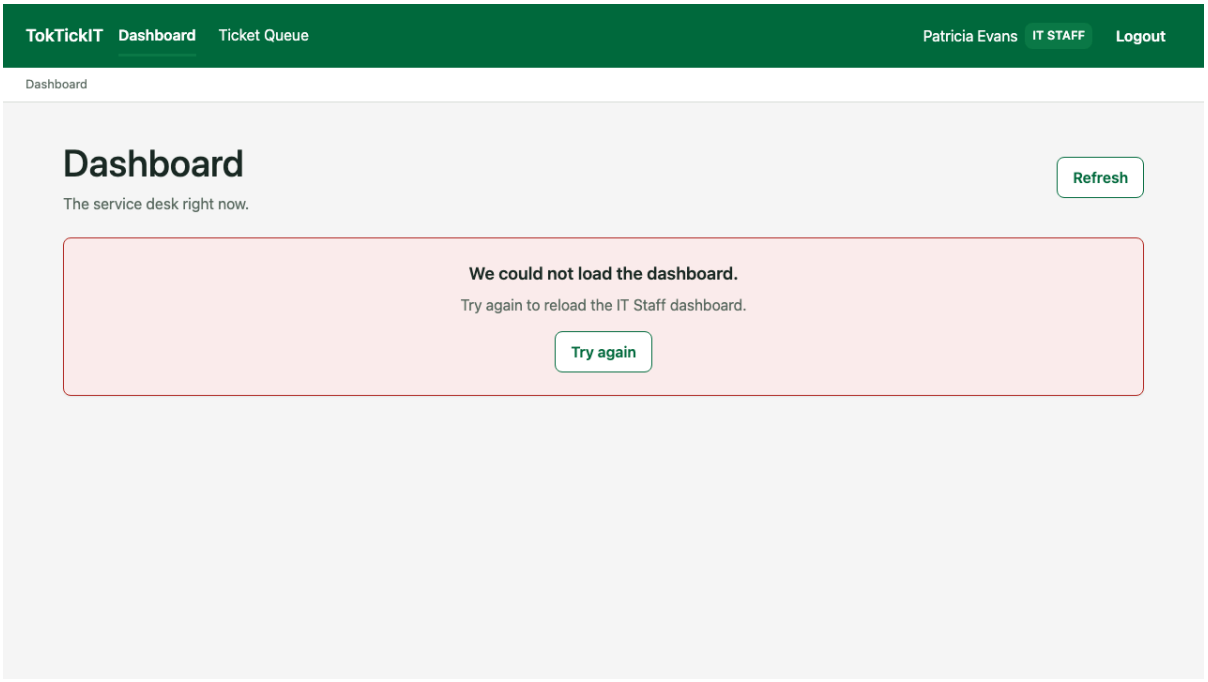


Figure 5.7 — Safe Dashboard load failure with retry (fixture-backed); full capture.

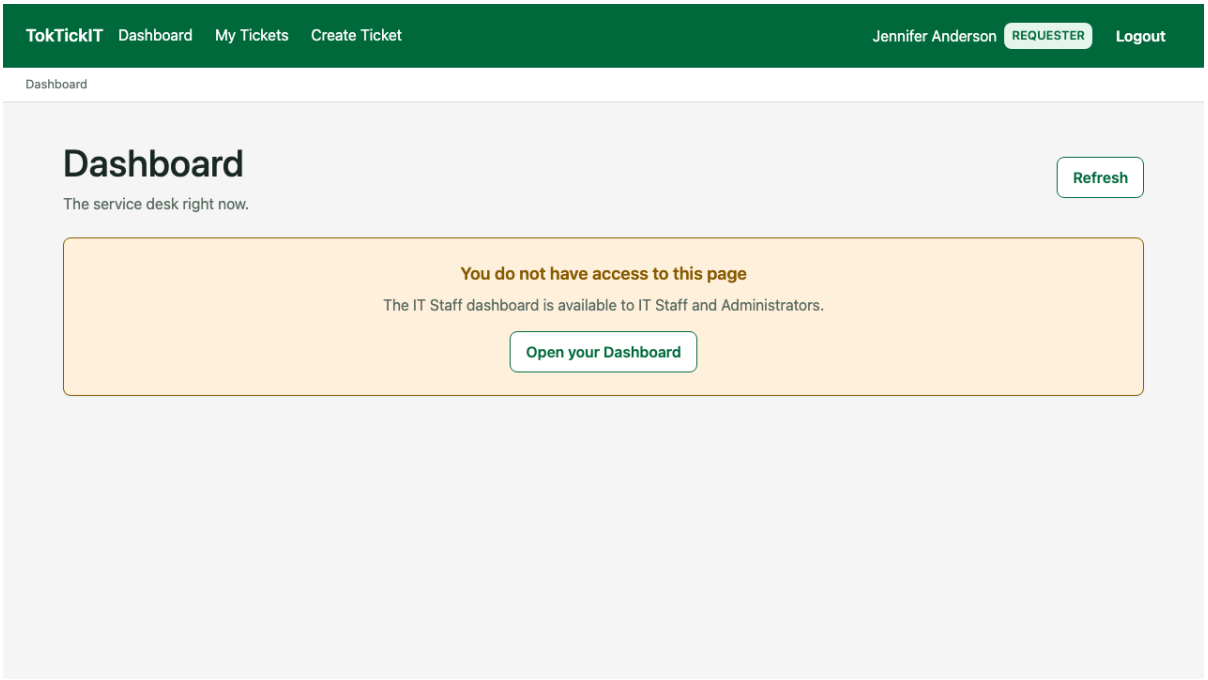


Figure 5.8 — Requester refused access to the staff Dashboard; full capture.

Answer Part 6: Working Actions Taken UI

Question — Can authorized users record and manage ordered Actions Taken while invalid operations are rejected?

Actions Taken records several separately identified pieces of work on a Ticket. Description, Result, Follow-up Required, Follow-up Note, Attachment Notes, Performed by and Recorded at have distinct meanings. Assignee and lifecycle status extend the labsheet fields. System identities and timestamps are readable text, never editable inputs.

Authorized staff create, assign, start, edit a workable non-terminal Action, complete it with a Result, or cancel it with a reason. Follow-up Note is required only when selected and clears when unselected. The API rejects inactive assignees, unauthorized roles, stale writes and unworkable Tickets. Requesters can read Actions on their own Tickets but cannot mutate them.

Behaviour	Evidence
Create and assign	Action list and creation captures at desktop, tablet and phone (Figures 6.1–6.6).
Edit and reassign	Editable form at desktop, tablet and phone (Figures 6.7–6.8, 6.11).
Requester read-only	Owned Action list and detail at three widths (Figures 6.9–6.10, 6.12).
Validate, transition and refuse	Required fields, lifecycle, inactive assignee, unworkable Ticket, stale write and safe failure (Figures 6.13–6.23).

The refreshed captures were recorded on 8 October in separate test journeys, so Action IDs and actors vary. Each caption links the complete source capture. Failure injection is labelled as fixture-backed. The mobile edit capture shows the responsive editable form; its full source is linked.

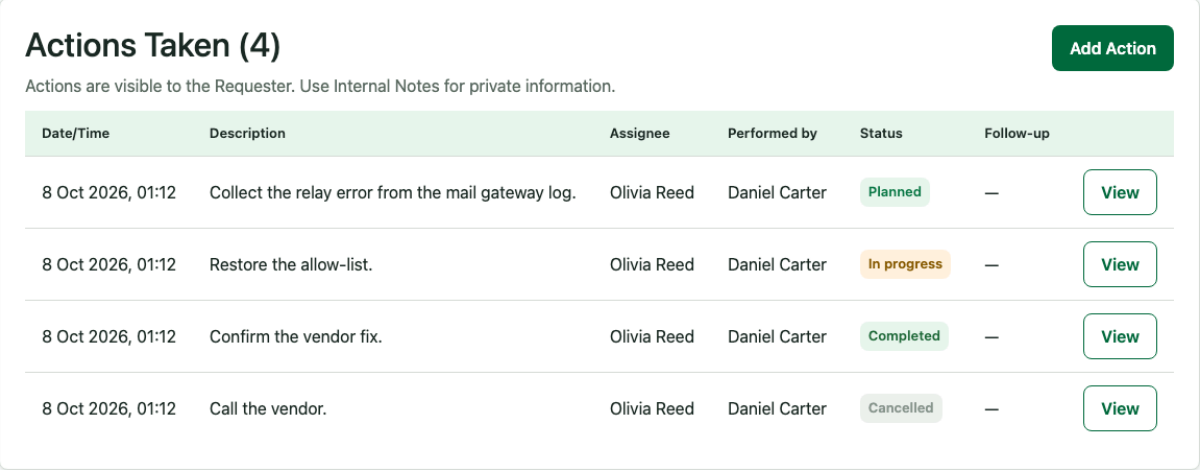


Figure 6.1 — Actions Taken list at desktop width (1,280 px), showing four lifecycle statuses; full capture.

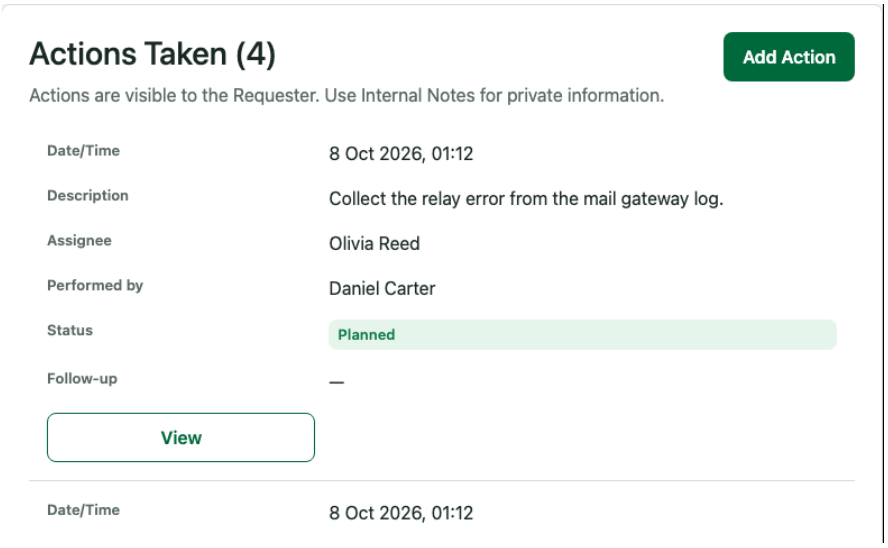


Figure 6.2 — Tablet Actions list; crop shows one complete tablet Action card and its status; full capture.

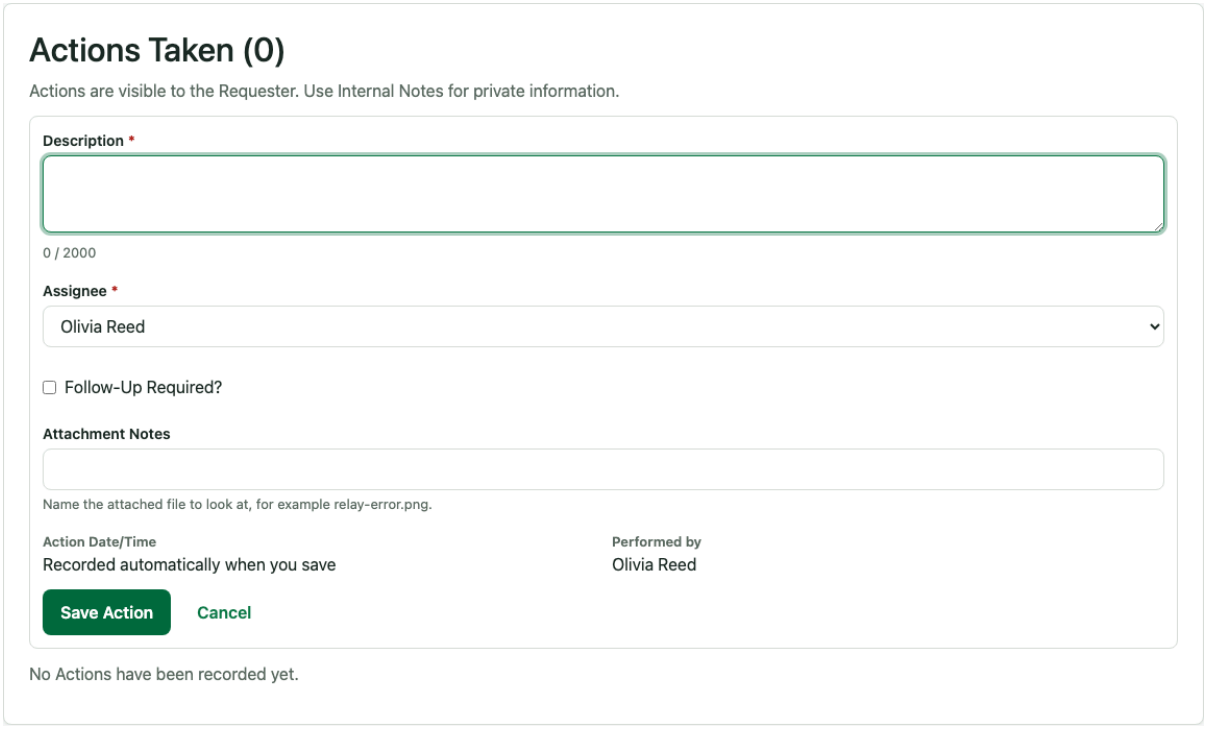


Figure 6.3 — Create Action form at desktop width, including the empty state; full capture.

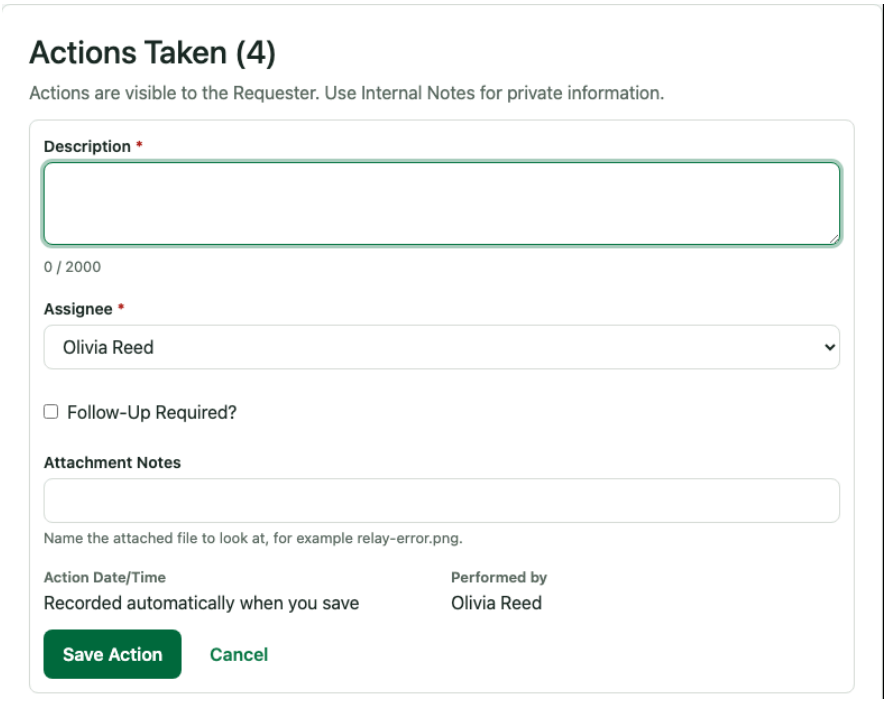
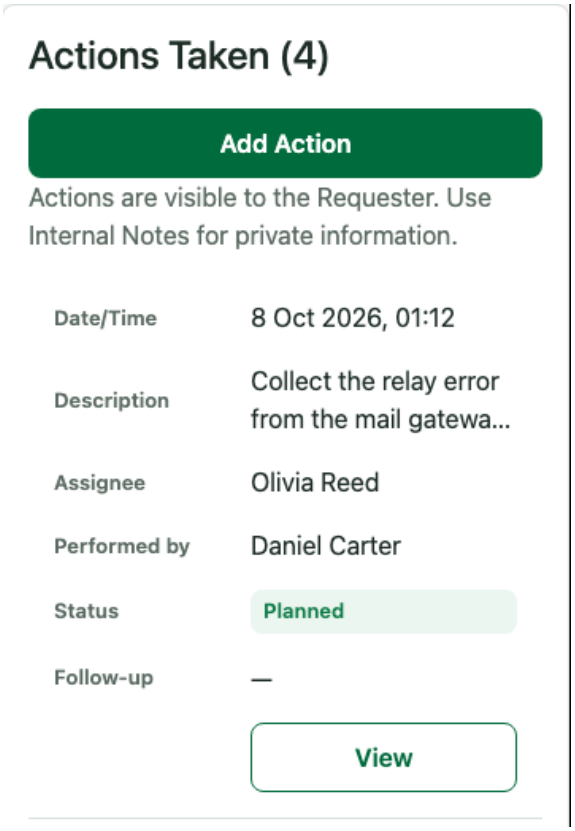


Figure 6.4 — Tablet create form; crop shows required fields, the Follow-Up option and Save controls; full capture.



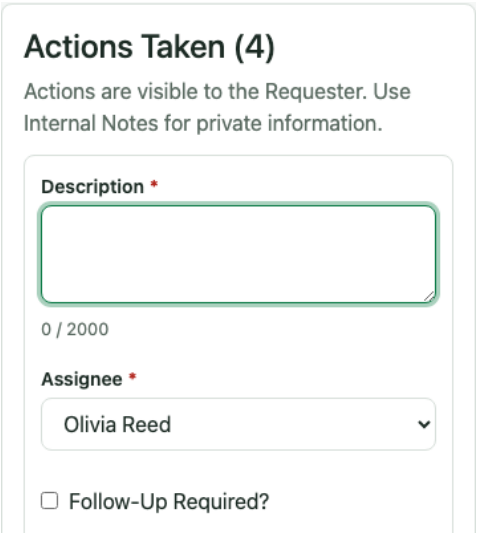


Figure 6.5 (left) — Mobile Actions list; crop shows one complete responsive card; full capture.

Figure 6.6 (right) — Mobile create form; crop shows Description, Assignee and Follow-Up controls; full capture.

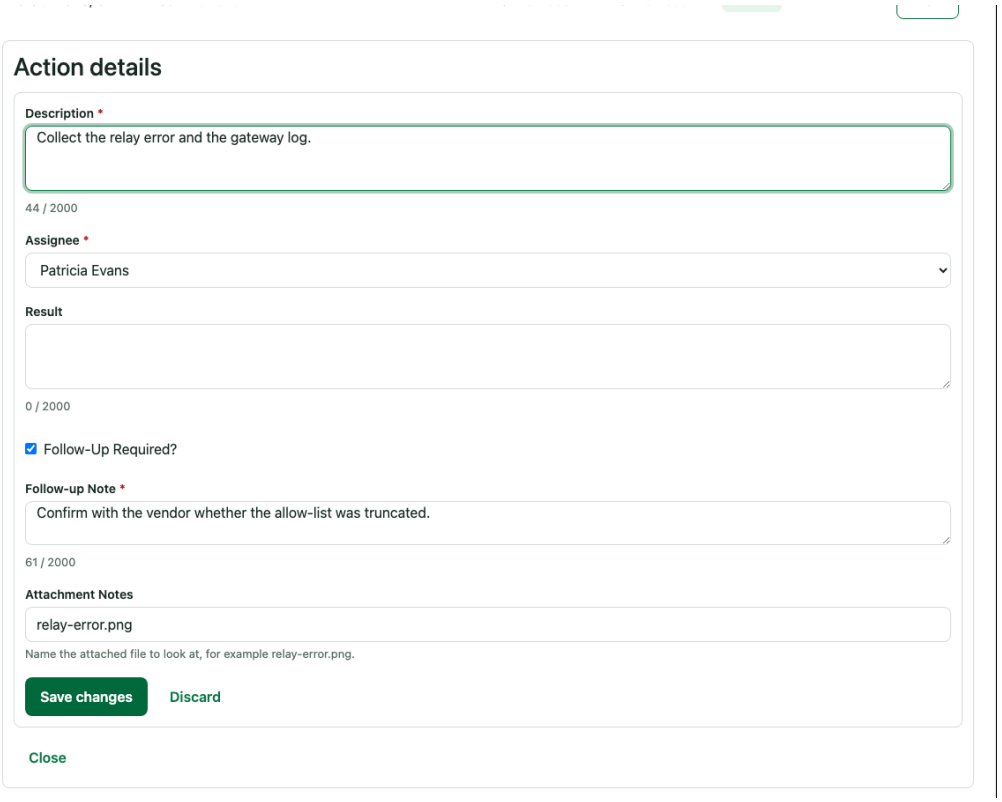


Figure 6.7 — Desktop edit/reassign form; crop shows editable fields and Save changes; full capture.

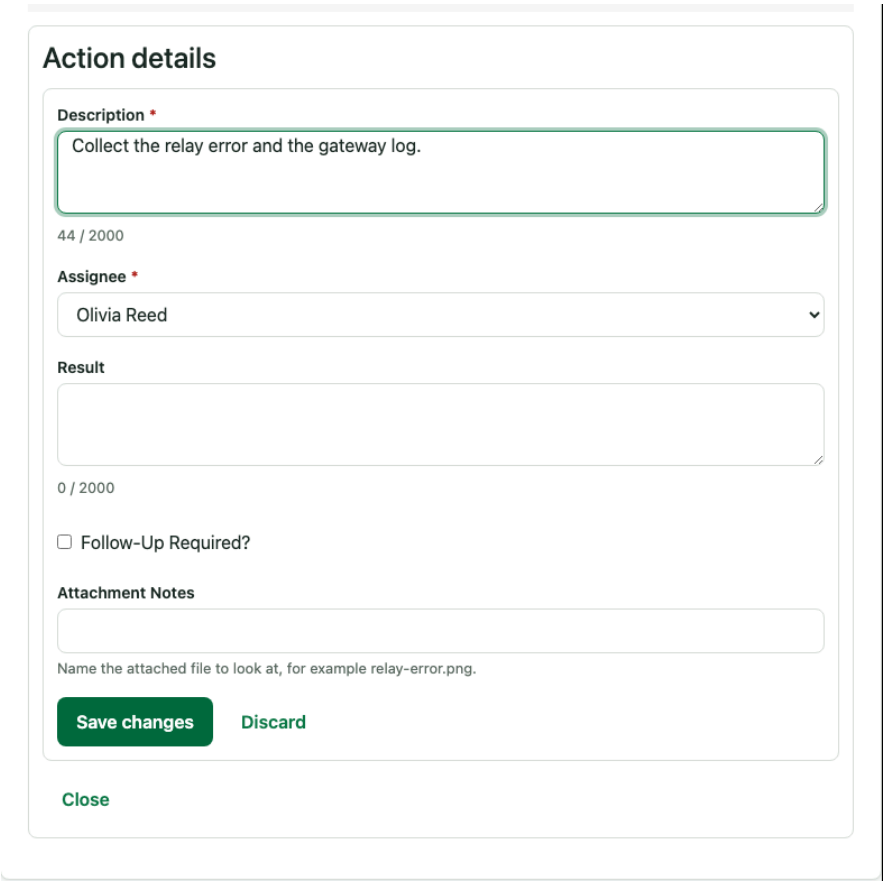


Figure 6.8 — Tablet edit/reassign form; crop shows fields through Save changes; full capture.

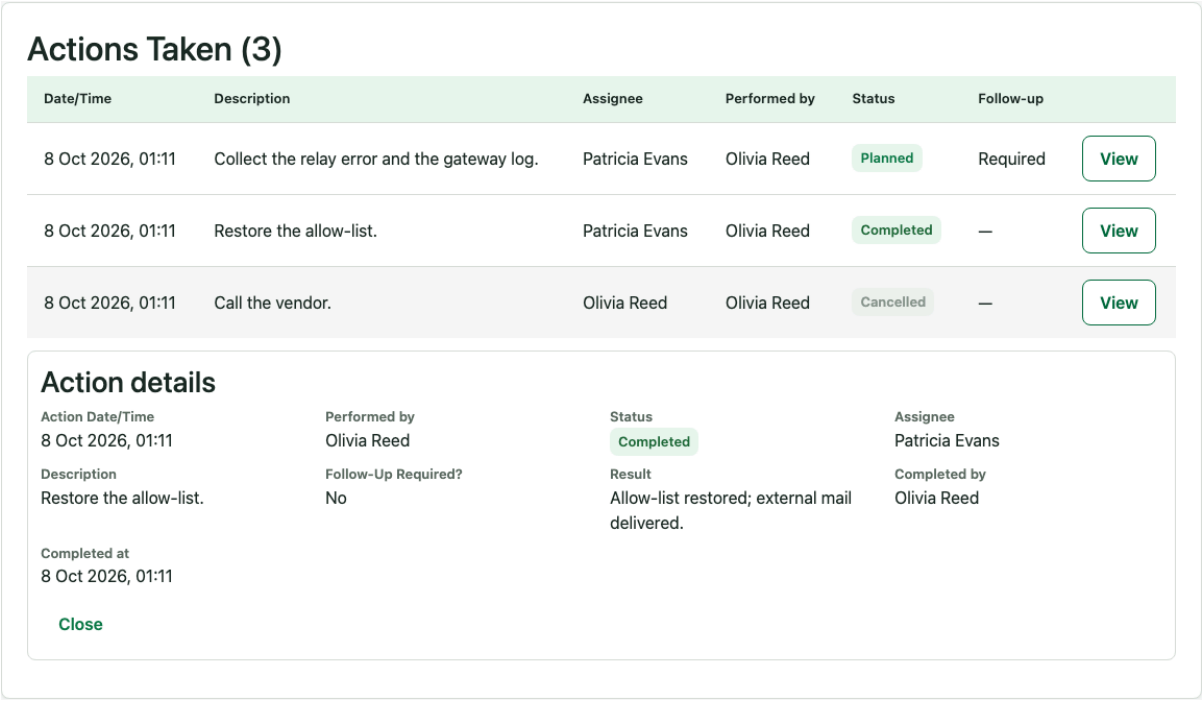


Figure 6.9 — Requester read-only Actions list and selected Action detail at desktop width; full capture.

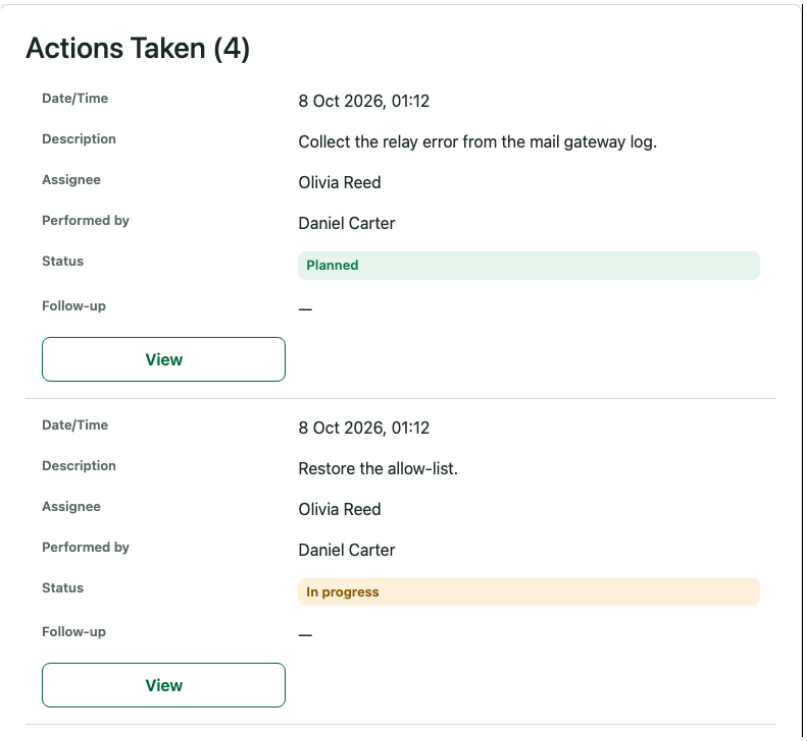
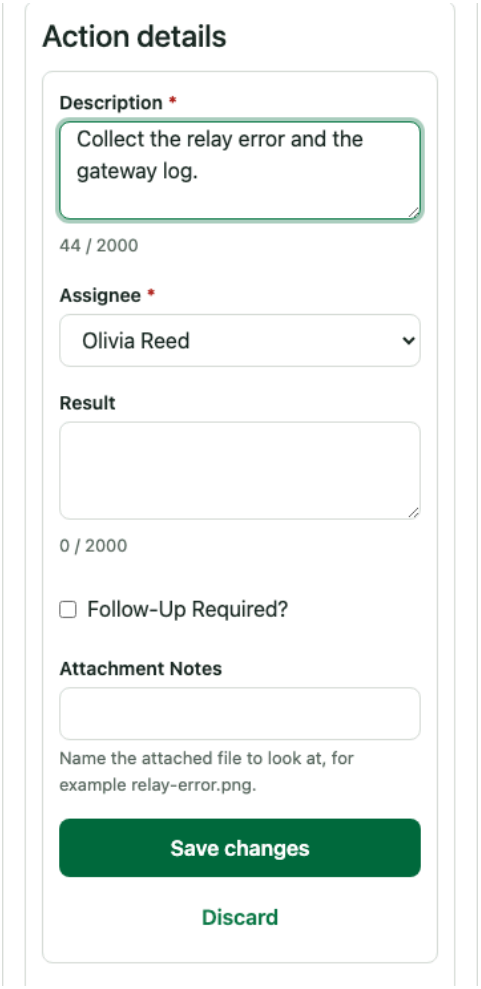


Figure 6.10 — Requester read-only Action cards at tablet width; crop shows two complete tablet Action cards; full capture.



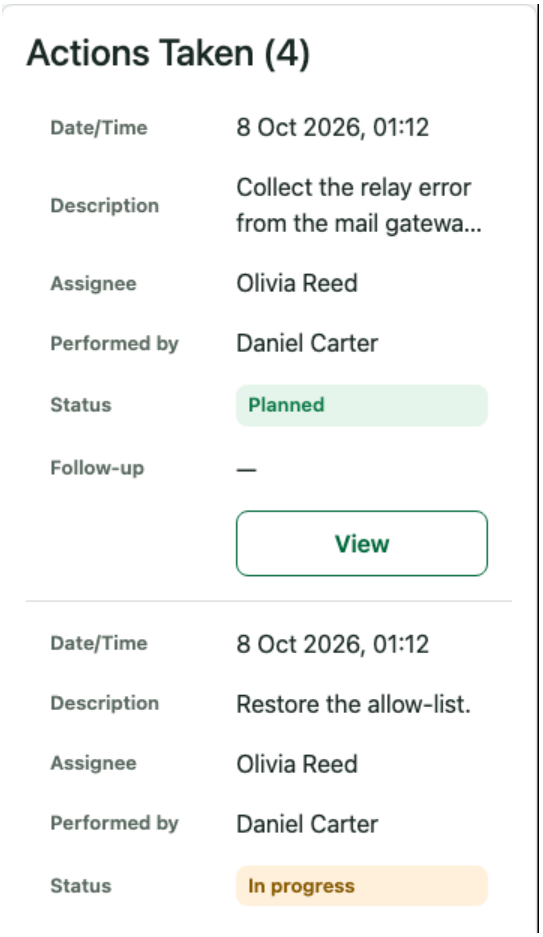


Figure 6.11 (left) — Mobile editable Action form; crop shows Description, Assignee, Result and Save changes; full capture.

Figure 6.12 (right) — Requester read-only Actions list on mobile; crop shows responsive Action cards and statuses; full capture.

Additional state captures — Part 6

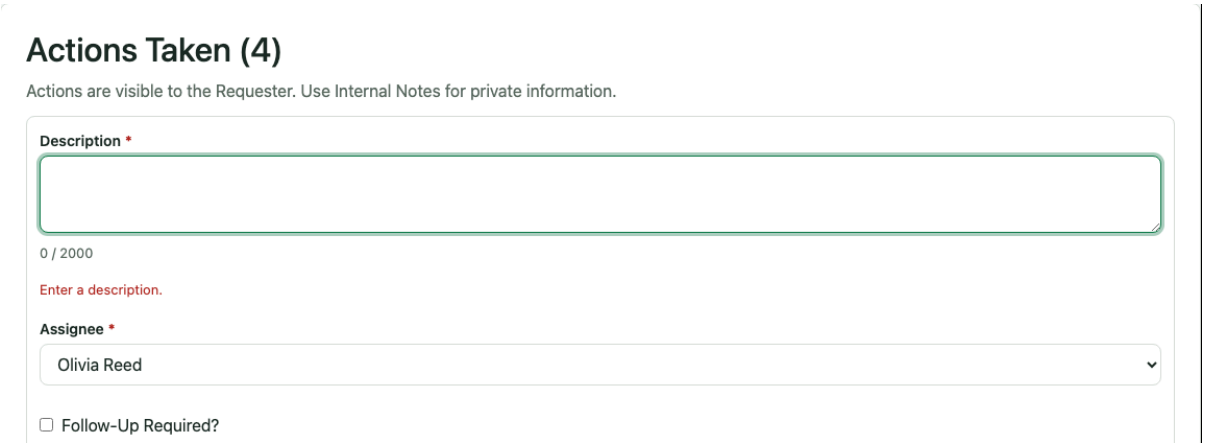


Figure 6.13 — Description validation appears below the required field; crop; full capture.

Figure 6.14 — Required Follow-up Note validation; crop shows the field, error message and Save control; full capture.

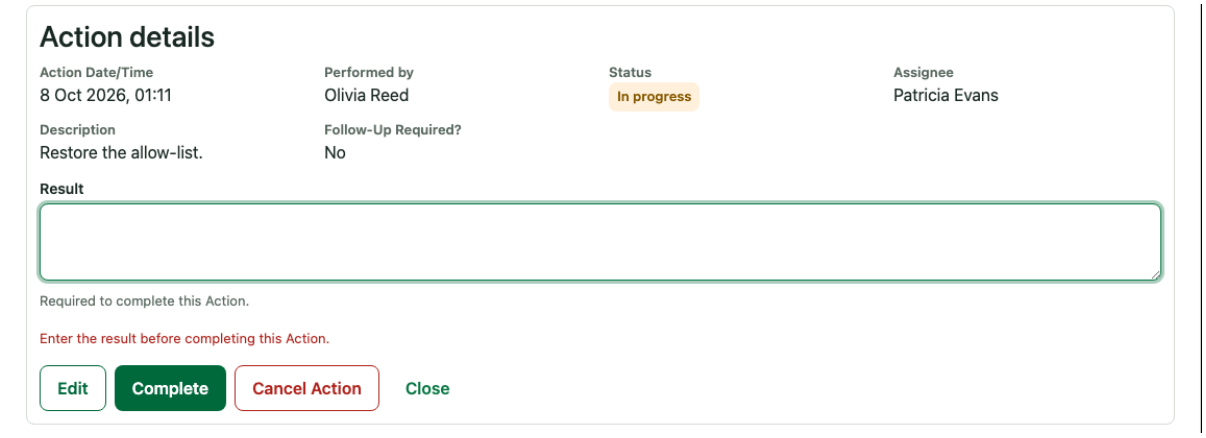


Figure 6.15 — Completion is refused until a Result is entered; crop; full capture.

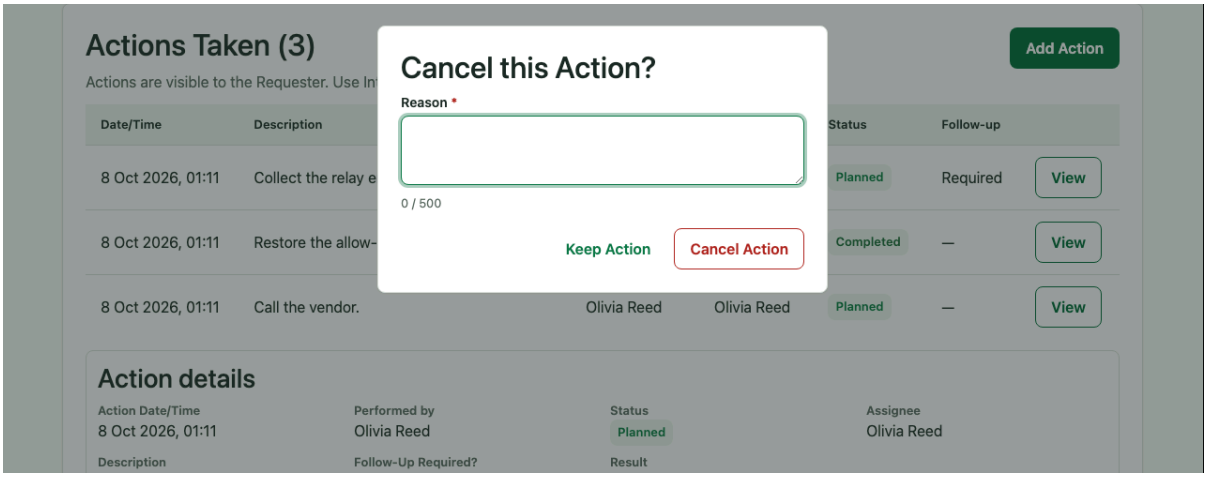


Figure 6.16 — Cancellation dialog requires a reason; crop; full capture.

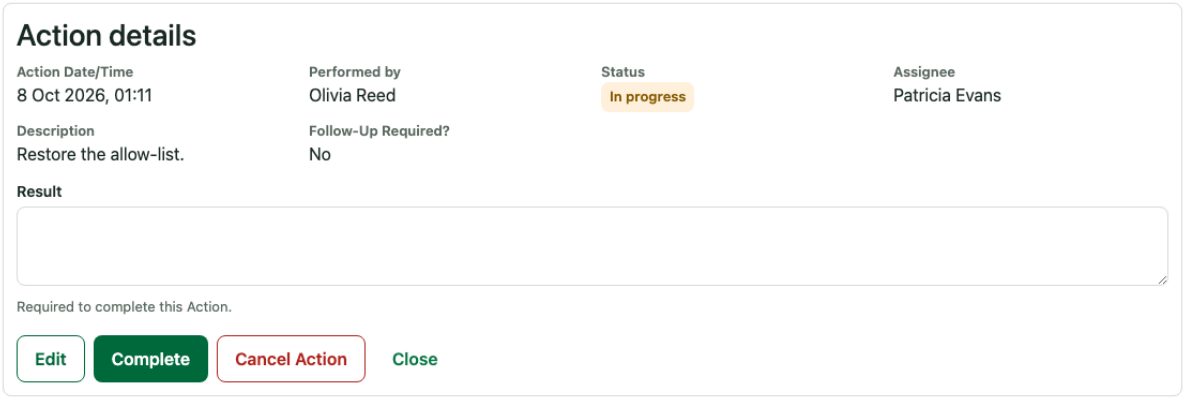


Figure 6.17 — In-progress Action detail for "Restore the allow-list"; crop shows status, performer, assignee and workflow controls; full capture.

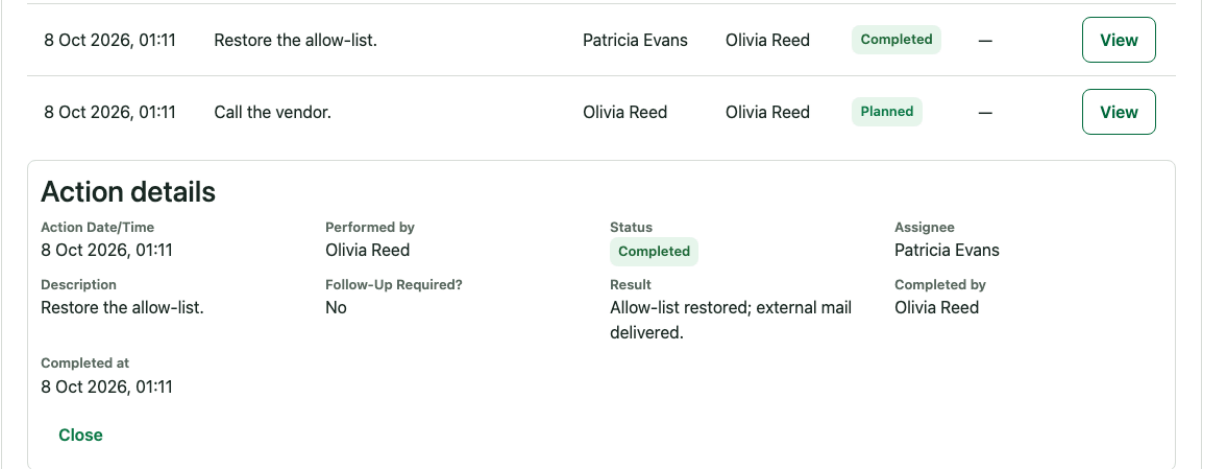


Figure 6.18 — Completed Action detail shows Result, Performed by, Completed by and Completed at; crop; full capture.

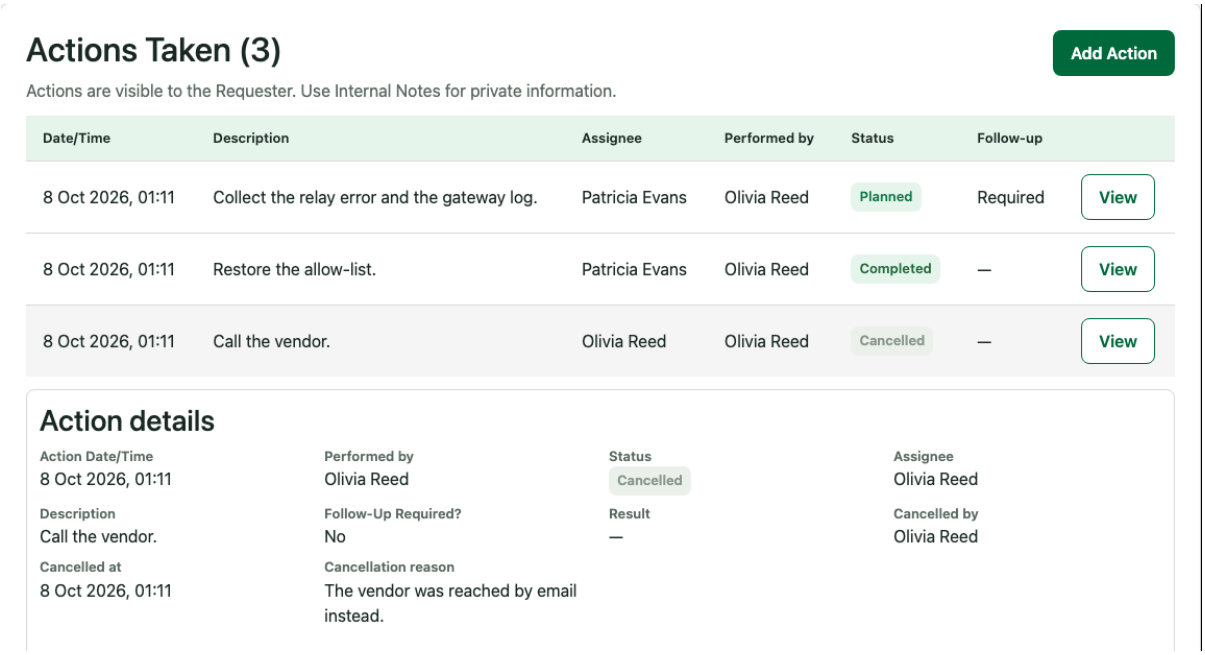


Figure 6.19 — Cancelled Action with reason and actor; crop; full capture.

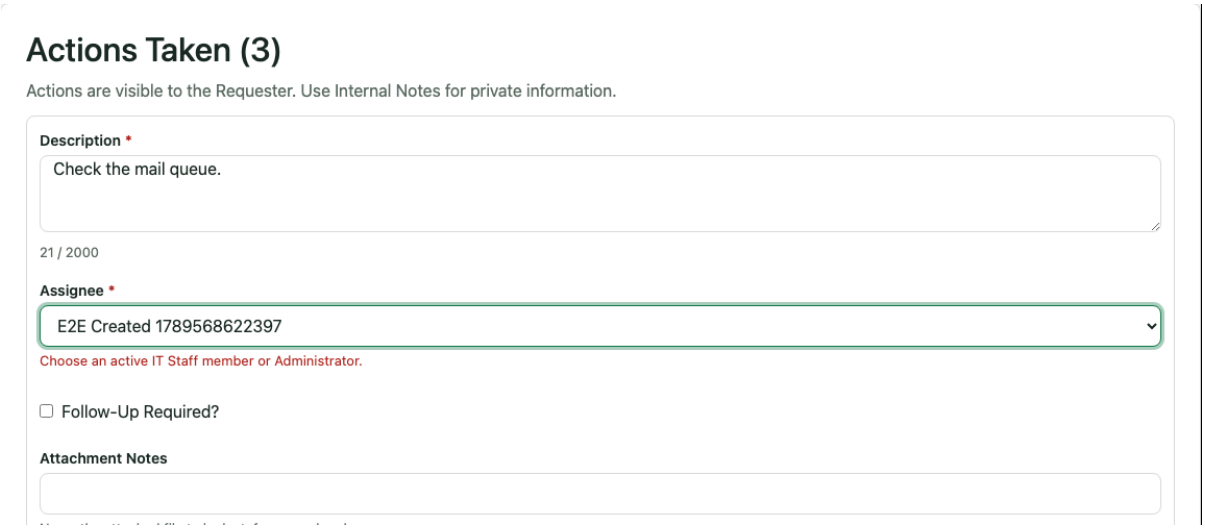


Figure 6.20 — Inactive assignee is rejected with field validation; crop; full capture.

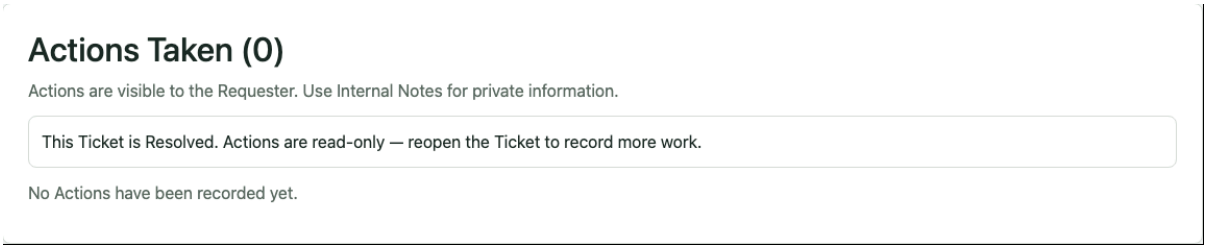


Figure 6.21 — Resolved Ticket is read-only and has no Actions; full capture.

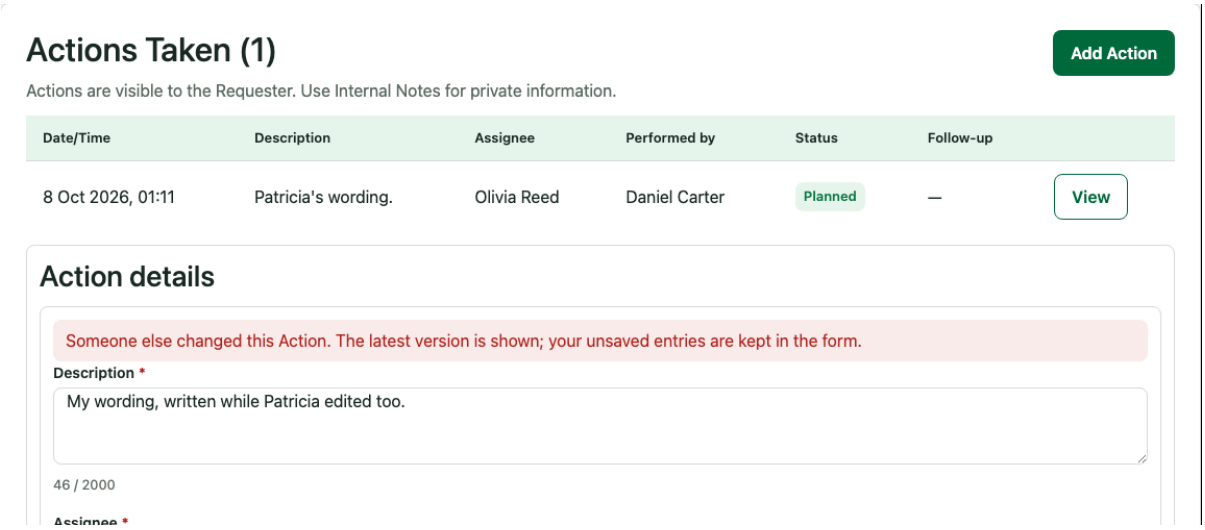


Figure 6.22 — Stale-write conflict preserves the user's unsaved text; crop; full capture.

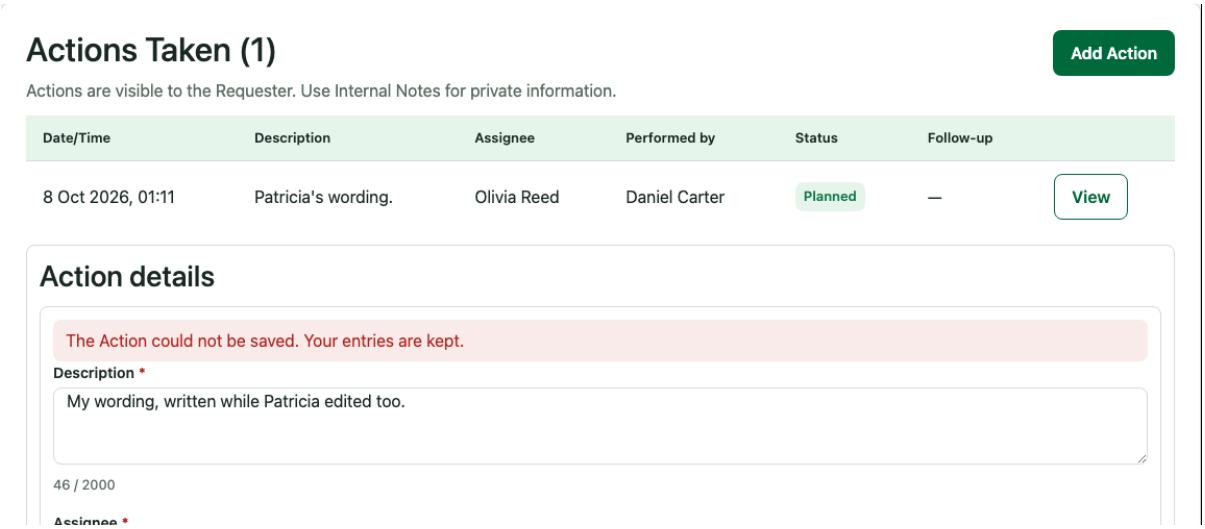


Figure 6.23 — Fixture-backed save failure keeps entered values; crop; full capture.

Cancellation-reason validation is asserted by UI-05/E2E-01 but has no separately captured error-state image. Part 3 names the direct negative checks; successful forms alone do not prove input validation.

Answer Part 7: Working Ticket Workflow

Question — Are ticket transitions, resolution conditions, cancellation effects, and history enforced consistently?

The status control offers only role-permitted transitions and explains a blocked resolution when PLANNED or IN_PROGRESS Actions remain. The backend repeats the gate inside the transaction. After work completes, the permitted transition refreshes the real Ticket status and announces it.

Ticket cancellation cancels open Actions in the same transaction and appends History. Versioned changes reject stale expectedVersion values. Staff History includes permitted material work events; Requester History is status-only and cannot expose staff-only payloads. Stable ordering does not depend on an editable client narrative.

Behaviour	**Evidence**
Resolution gate	Open Actions block resolution at desktop and tablet; the mobile view is paired with staff History (Figures 7.1–7.2, 7.8).
After work completes	Permitted resolution updates the Ticket status (Figure 7.3).
History visibility	Staff sees work events; Requesters see status-only events across widths (Figures 7.4–7.7, 7.9–7.10).
Cancellation and stale version	Ticket cancellation cascades to open Actions; stale expectedVersion is refused (Figures 7.11–7.12).

The workflow captures remain the accepted-staging screenshots at 876511f and are unchanged in main. The release gate at 1141738 passed on tree 91a8f1a, identical to main @ d07fefe. A known cascade limit remains: each Action's updatedAt is stamped with the request instant without the never-backwards rule applied to the Ticket. Nothing uses an Action's updatedAt for ordering.

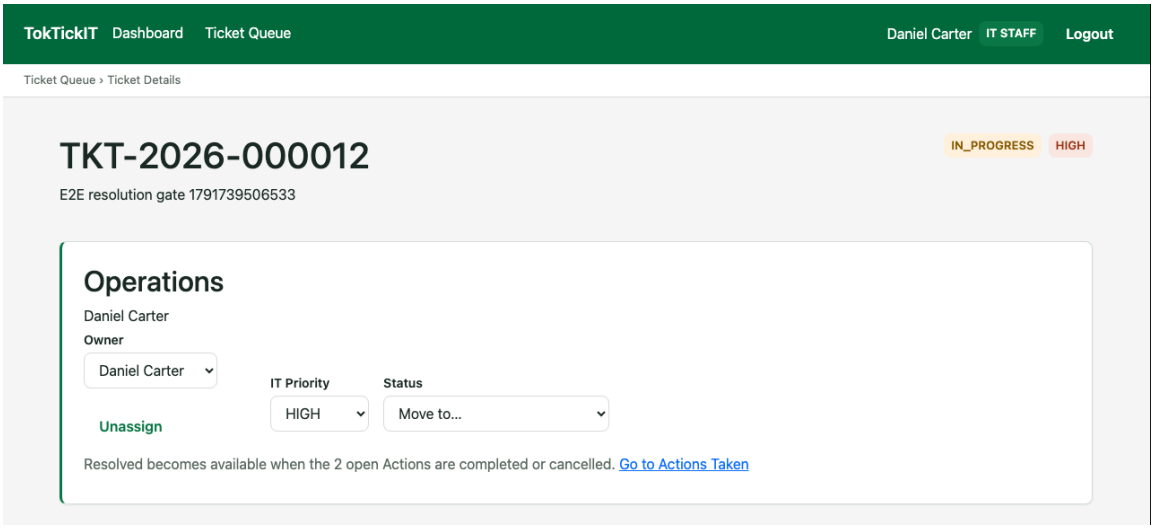


Figure 7.1 — Desktop resolution gate; crop shows the Operations panel and the message that two open Actions block RESOLVED; full capture.

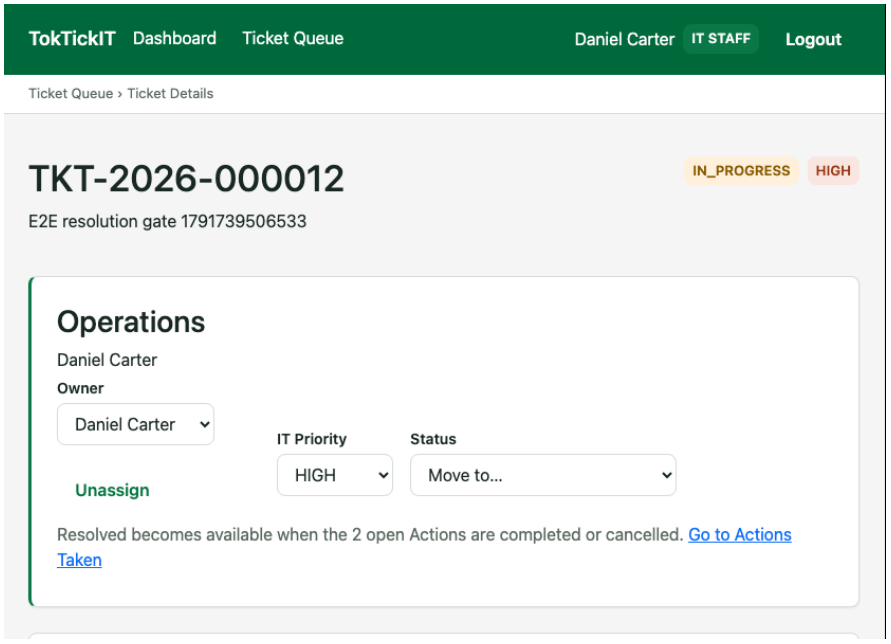


Figure 7.2 — Tablet resolution gate; crop shows the blocked status transition and Go to Actions link; full capture.

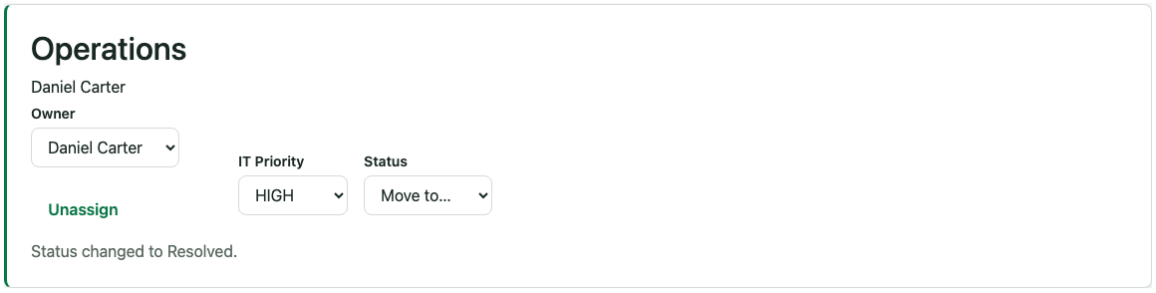


Figure 7.3 — Desktop Ticket after work completes; status changes to Resolved; full capture.

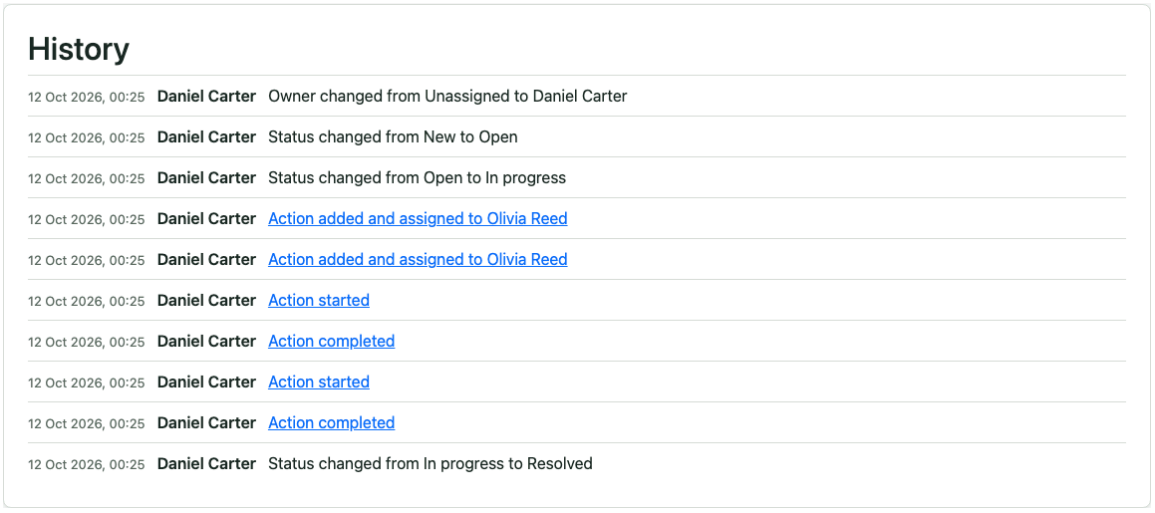


Figure 7.4 — Staff History at desktop width, including ownership, status and Action events; full capture.

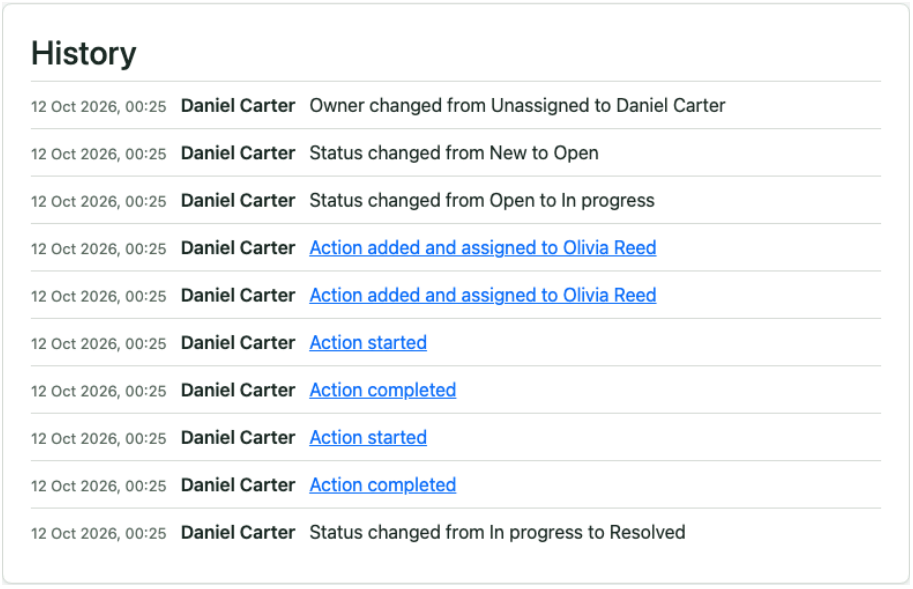


Figure 7.5 — Staff History at tablet width; full capture.

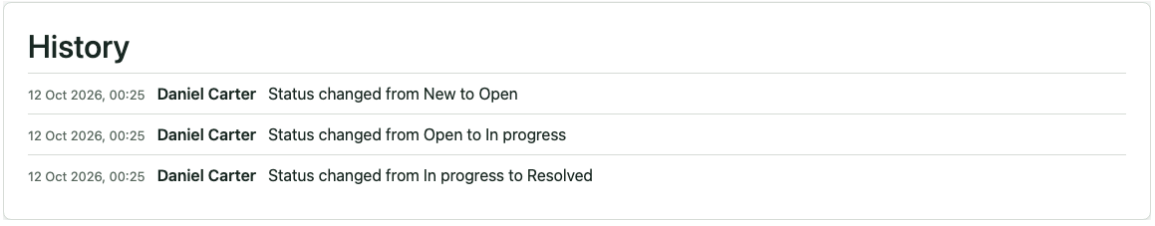


Figure 7.6 — Requester History at desktop width, showing only status changes; full capture.

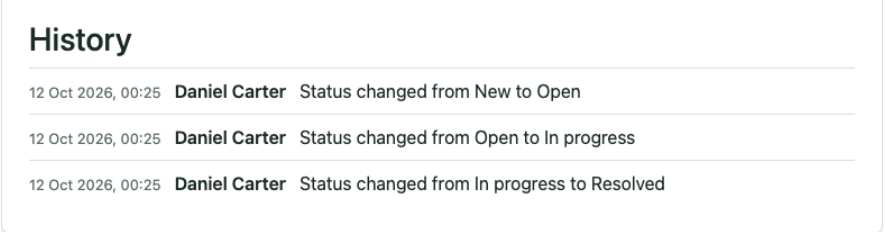


Figure 7.7 — Requester History at tablet width, showing only status changes; full capture.

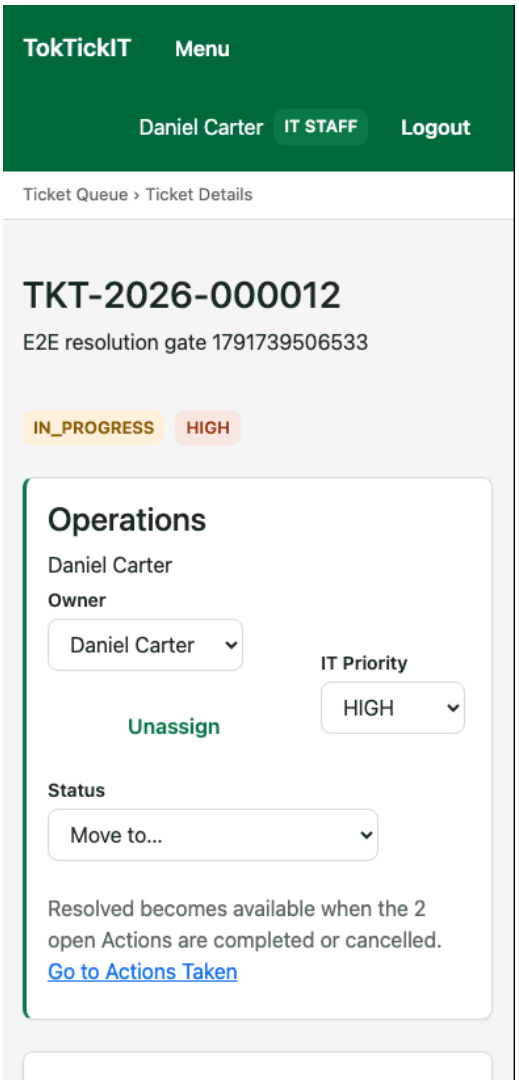


Figure 7.8 (left) — Mobile resolution gate; crop shows open Actions block resolution and links to Actions; full capture.

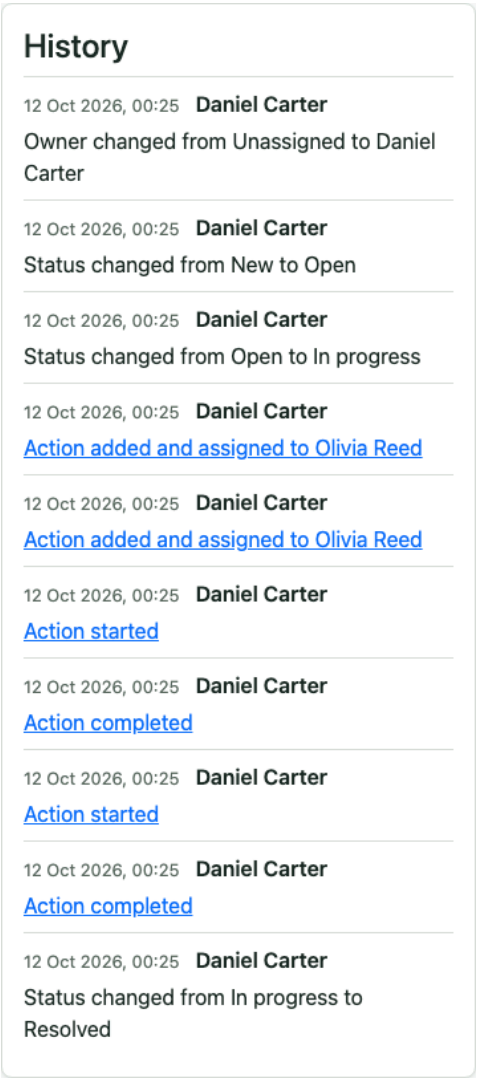


Figure 7.9 (right) — Staff History on mobile, with ordered status and Action events; full capture.

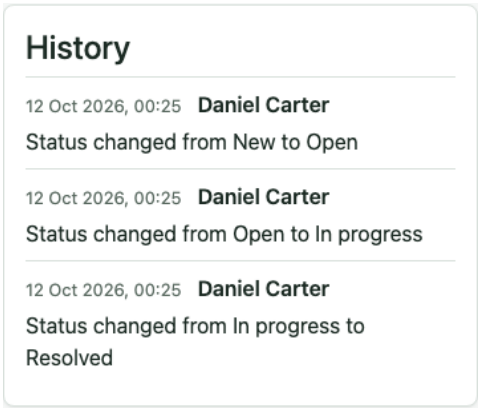


Figure 7.10 — Requester History on mobile, showing status-only events; full capture.

Additional state captures — Part 7

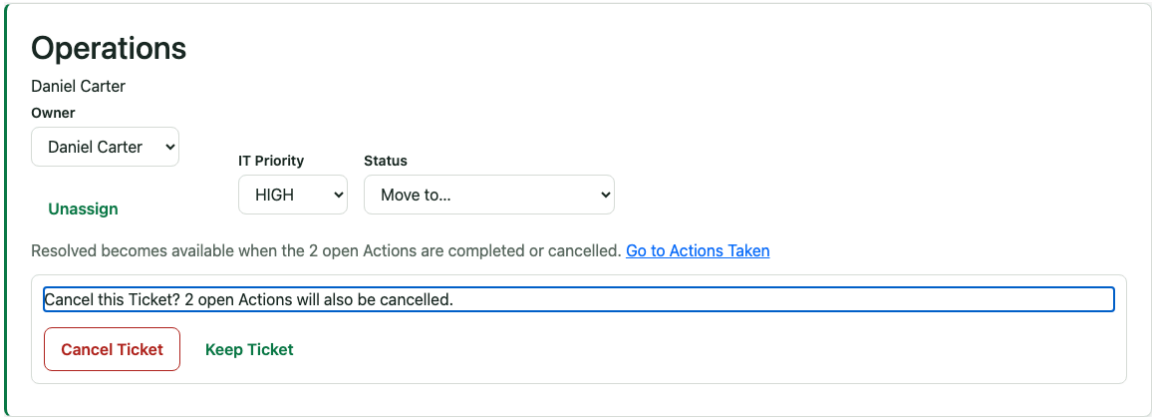


Figure 7.11 — Ticket cancellation warns that its two open Actions will also be cancelled; full capture.

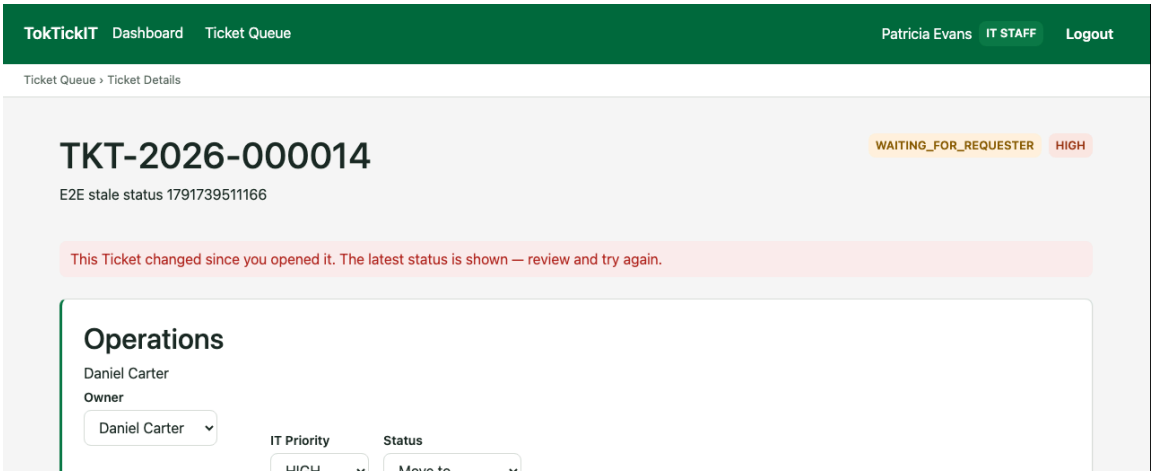


Figure 7.12 — Stale Ticket version is refused while the latest status is shown; crop; full capture.

Answer Part 8: Working Requester Dashboard and Final Regression UI

Question — Does the Requester dashboard protect ownership and preserve the earlier user journeys after Lab 4 changes?

The Requester Dashboard is the post-login landing page. Each query is scoped to the signed-in requester; counts and recent Tickets match independently computed owned-data results.

Behaviour	Ownership rule / destination
Open	requesterId = me and ACTIVE → My Tickets statusGroup=active
Needs my attention	requesterId = me and WAITING_FOR_REQUESTER → matching My Tickets status
Resolved	requesterId = me and RESOLVED → matching My Tickets status
Closed	requesterId = me and CLOSED → matching My Tickets status
Recently updated	Five newest owned Tickets; updatedAt DESC, id DESC → Ticket Detail / sorted My Tickets

DASH-07–DASH-10 and SEC-05 cover ownership, status predicates, and the newest-five limit. E2E-04 follows Open and Needs my attention to My Tickets and verifies the routes and API totals (Figures 8.4–8.5).

Requester query and independent SQL-comparison tests

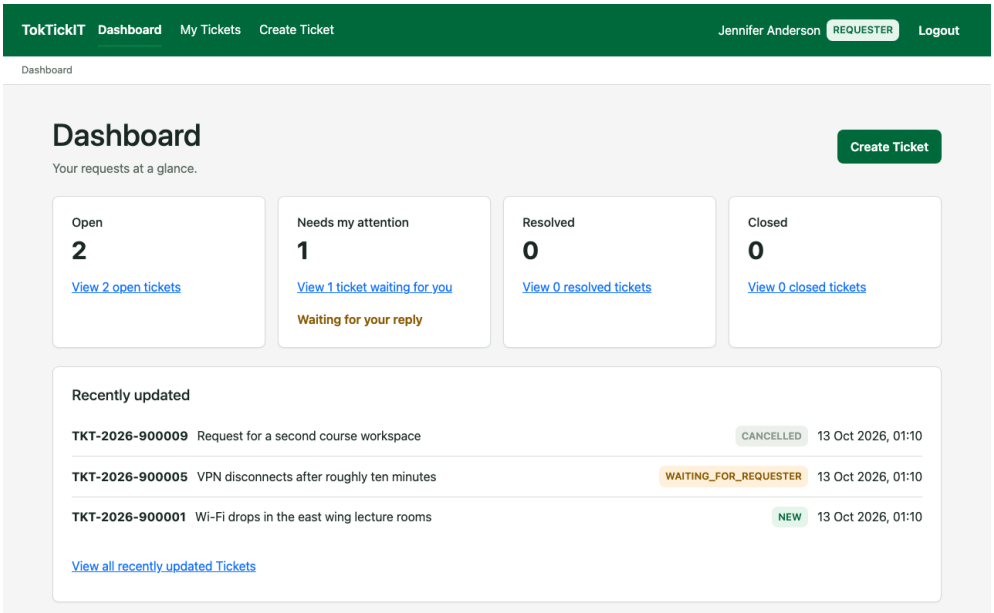


Figure 8.1 — Requester Dashboard at desktop width, showing the owned-ticket summary cards and recent-ticket list; full tracked capture.

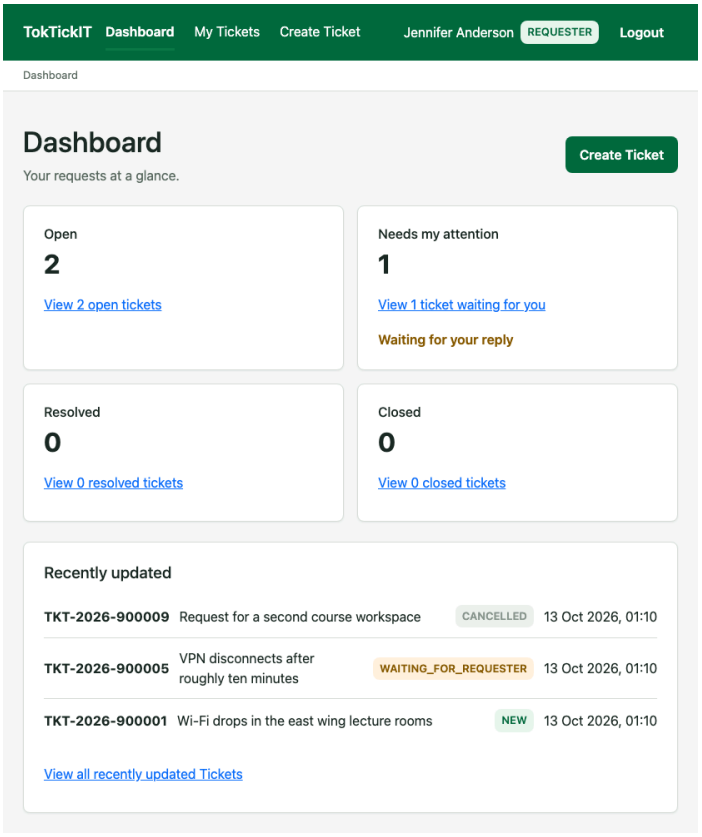


Figure 8.2 — Tablet layout of the Requester Dashboard, with the summary cards and recent-ticket section; full tracked capture.

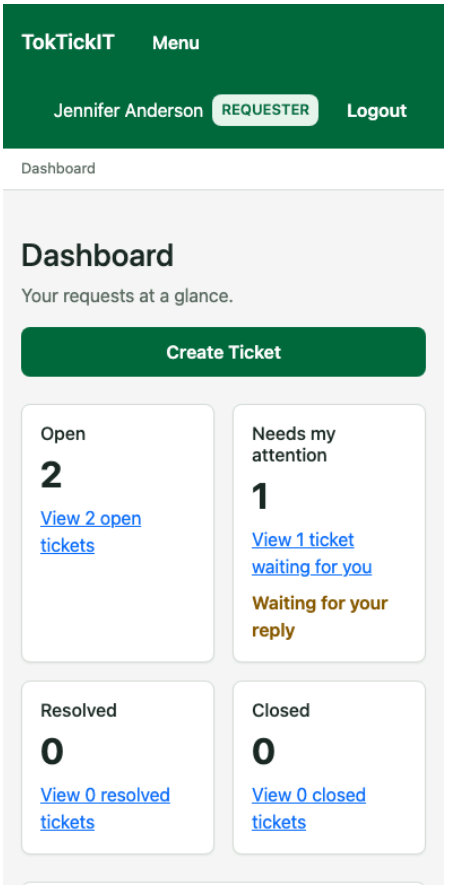


Figure 8.3 — Mobile layout of the Requester Dashboard, with all four summary cards arranged for a narrow viewport; full tracked capture.

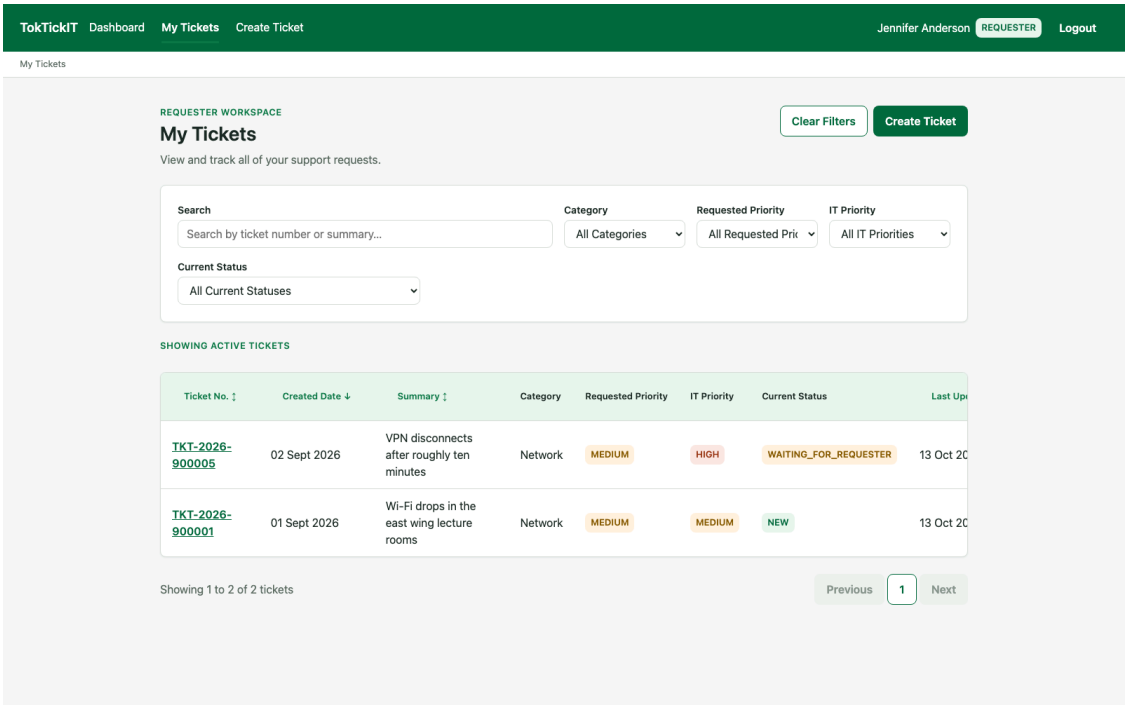


Figure 8.4 — Open-ticket drilldown, left-hand viewport. Both active-ticket rows appear through Current Status; Last Updated lies beyond this viewport. Source: full tracked capture.

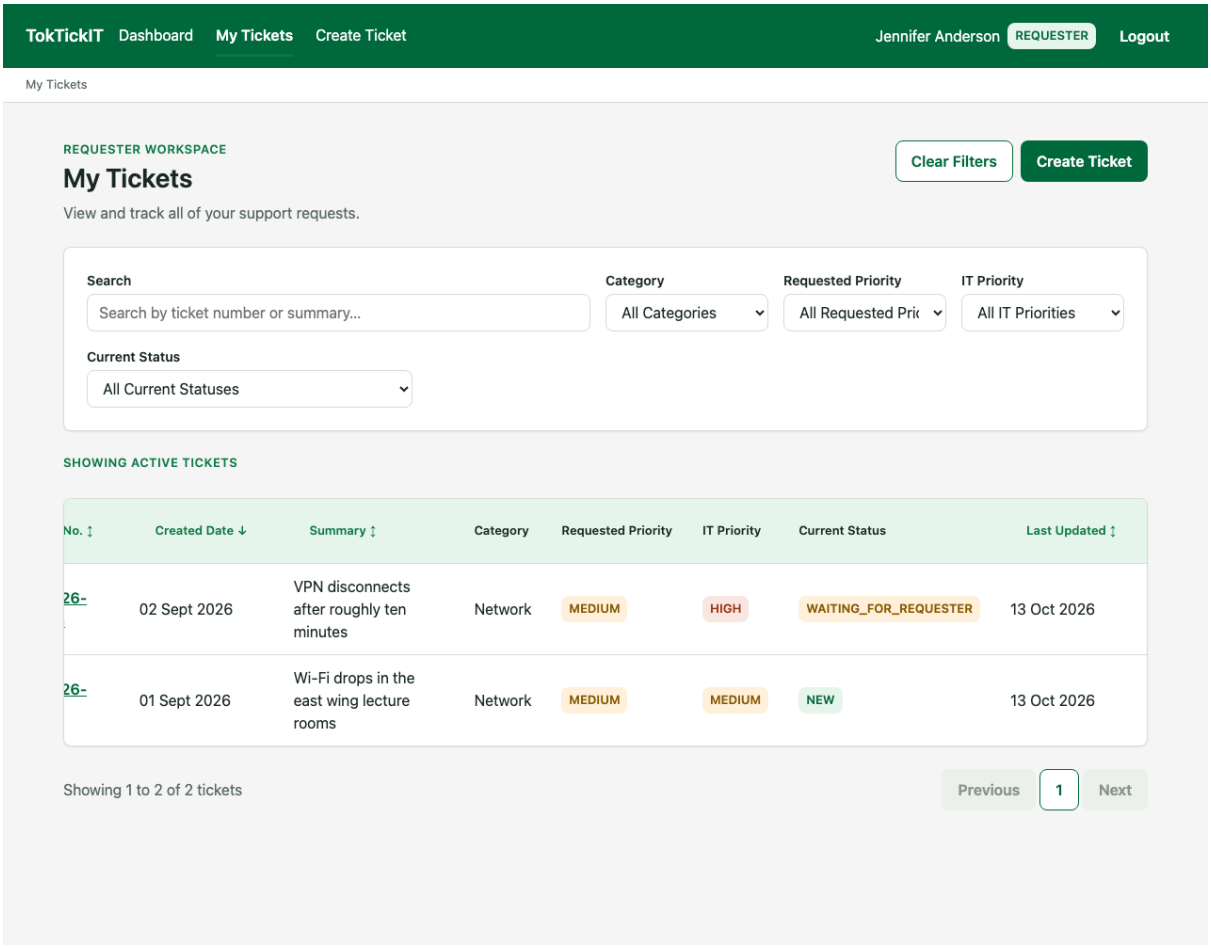


Figure 8.4a — Open-ticket drilldown, right-edge view. The captured interface uses the same two displayed rows as Figure 8.4, with the table horizontally scrolled to Last Updated; Ticket No. is partly outside this viewport. Visual probe on main @ `d07fefe` using fixture-backed API responses; not a database-backed test capture or tracked product screenshot.

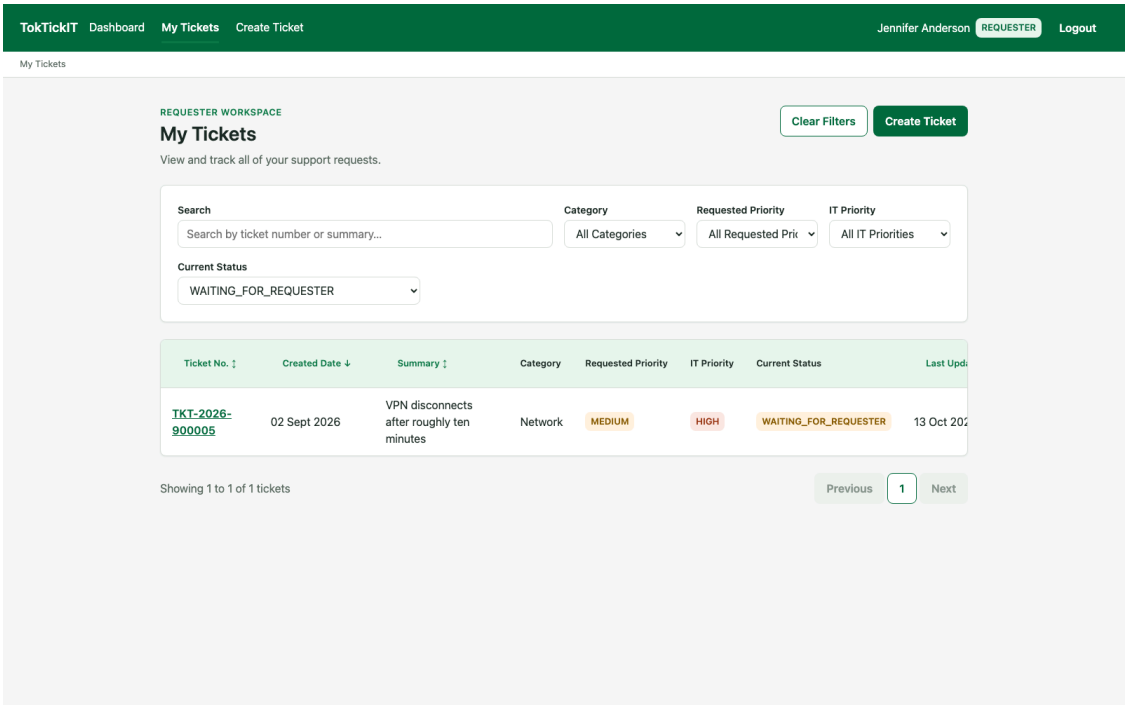


Figure 8.5 — Needs-my-attention drilldown, left-hand viewport. The filtered row appears through Current Status; Last Updated lies beyond this viewport. Source: full tracked capture.

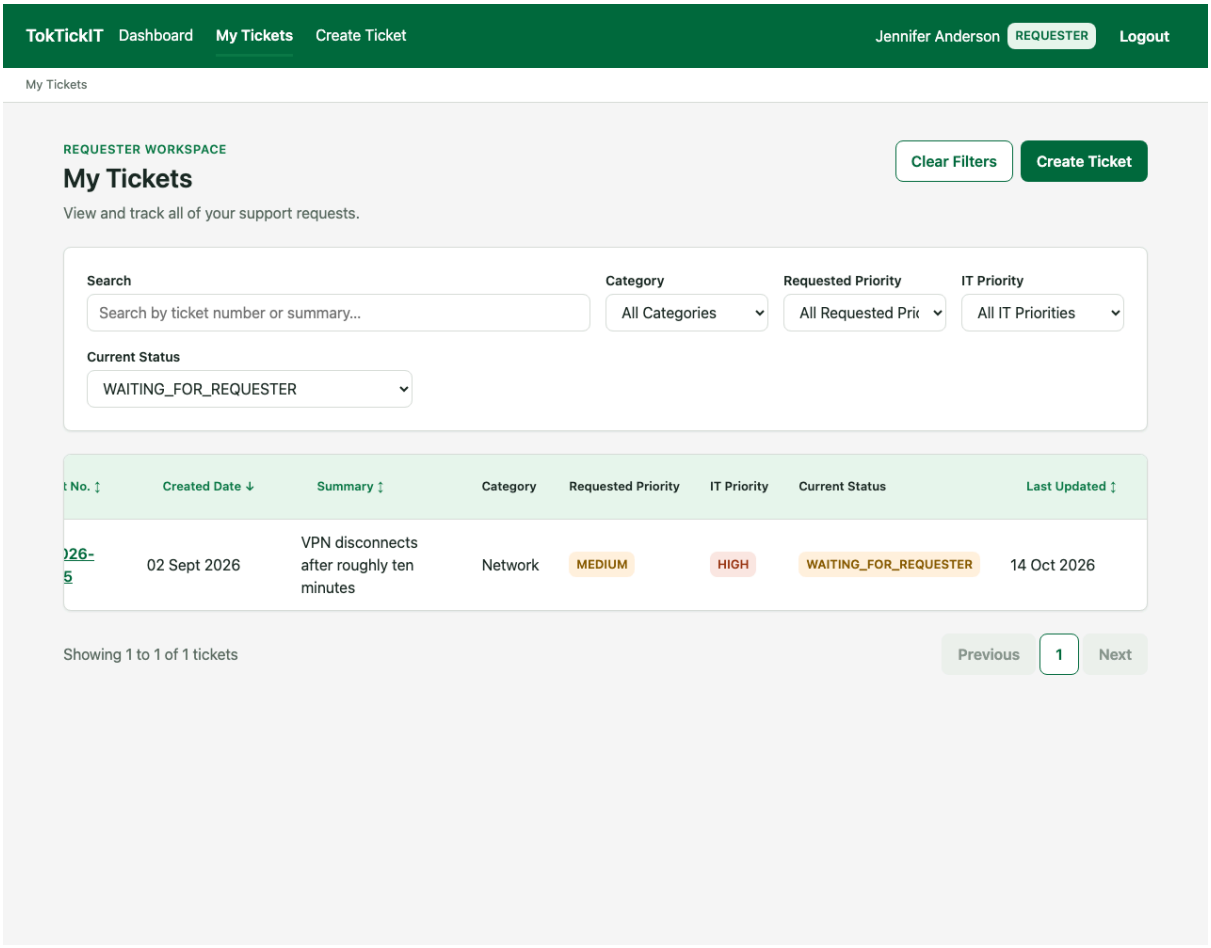


Figure 8.5a — Complementary right-edge view of the same Needs my attention result shown in Figure 8.5. Captured on main @ d07fefe at 1,280 px with the table horizontally scrolled to Last Updated; the Ticket No. column is partly outside this viewport. This is a direct probe capture, not a tracked product-repository screenshot.

Additional state captures — Part 8

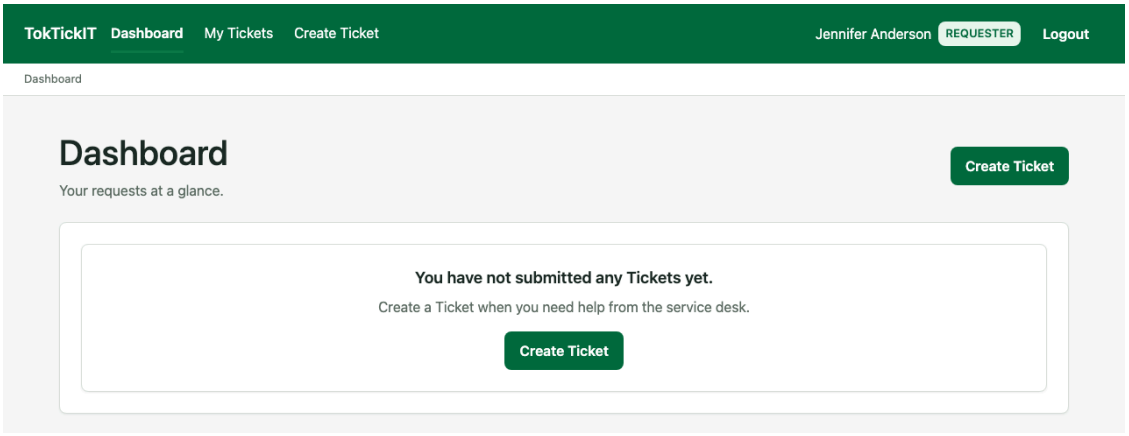


Figure 8.6 — Fixture-backed first-use empty state for the Requester Dashboard; full tracked capture.

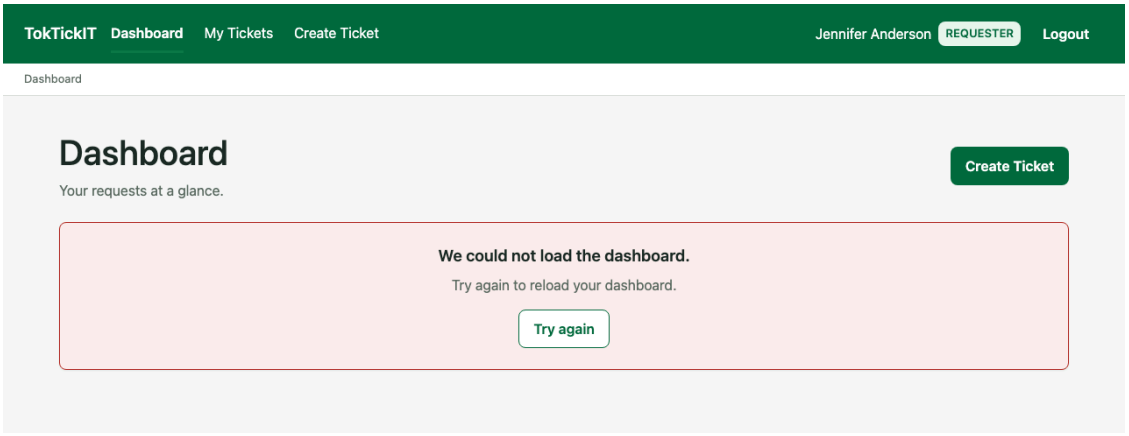


Figure 8.7 — Fixture-backed safe failure state when dashboard data cannot be loaded; full tracked capture.

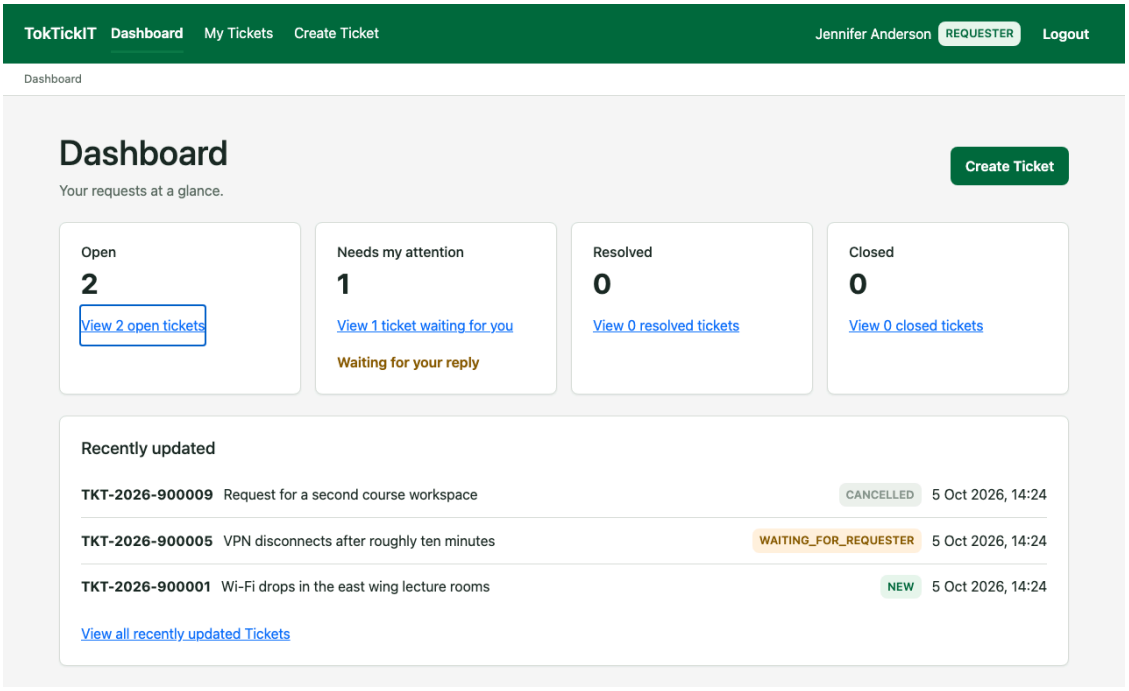


Figure 8.8 — Visible keyboard focus on a Requester Dashboard drilldown control; full tracked capture.

Inherited regression journeys

REG-01–REG-04 and inherited journeys passed in the full Chromium E2E run at the PR #103 release gate on tree 91a8f1a, identical to main @ d07fefe (64/64; 0 failed, 0 skipped). Screenshots below are accepted-staging captures at 876511f, unchanged in main. Lab 4 changes post-login landing assertions to Dashboard; /tickets remains My Tickets. No screen-reader run was performed.

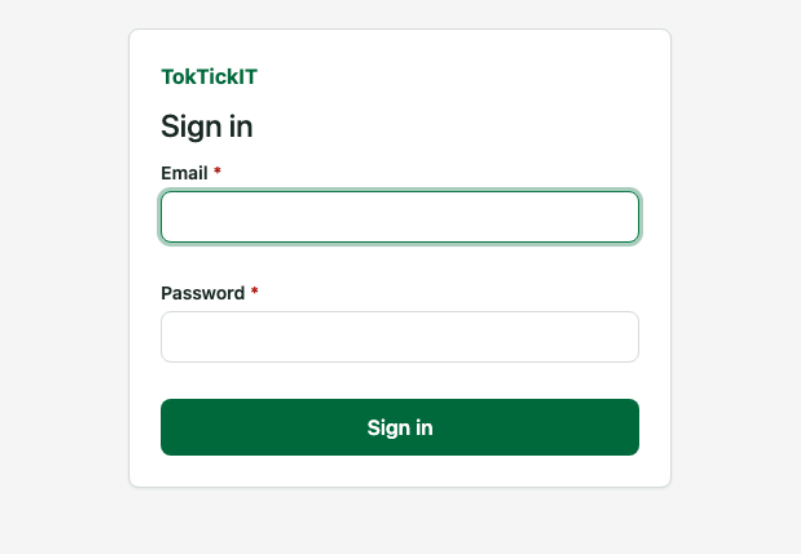


Figure 8.9 — Inherited login-screen regression capture; full tracked capture.

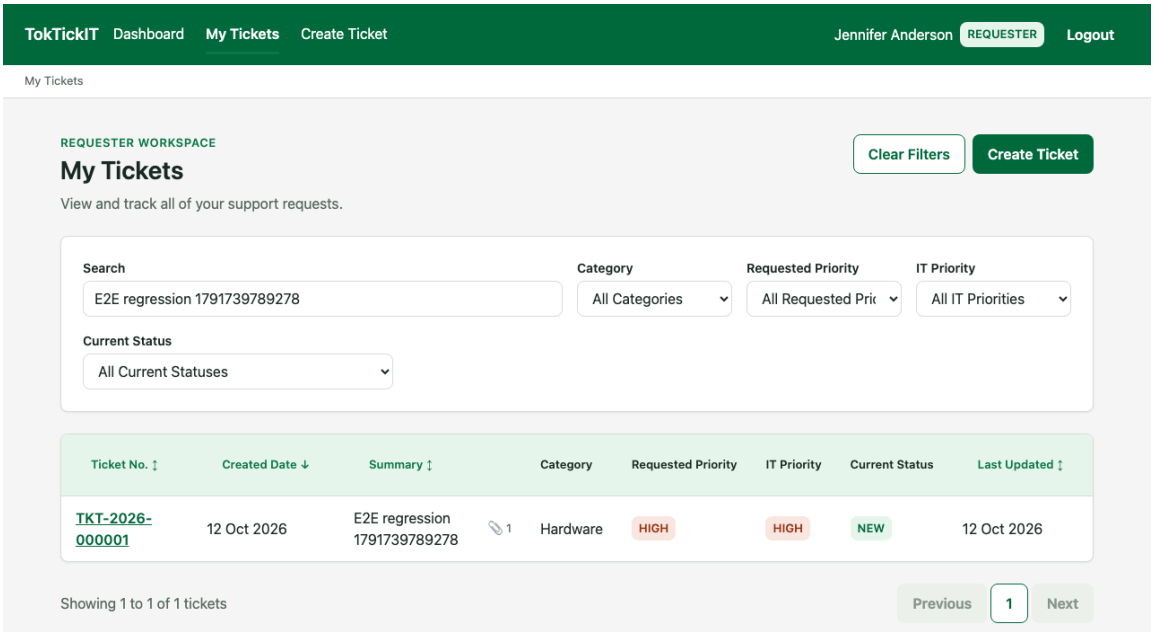


Figure 8.10 — Inherited Requester My Tickets list regression capture; full tracked capture.

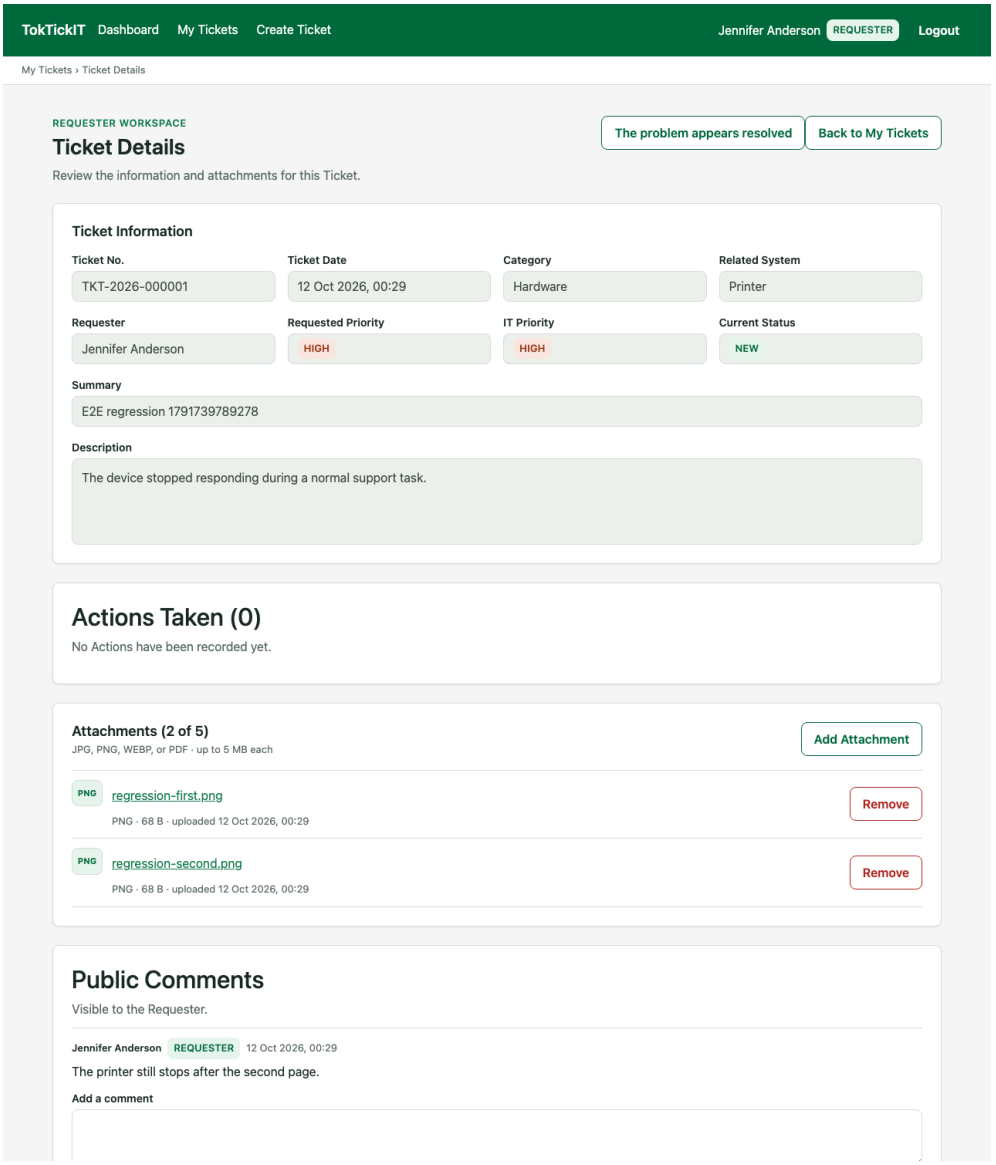


Figure 8.11 — Inherited Requester Ticket Detail capture with attachments and public comments; full tracked capture.

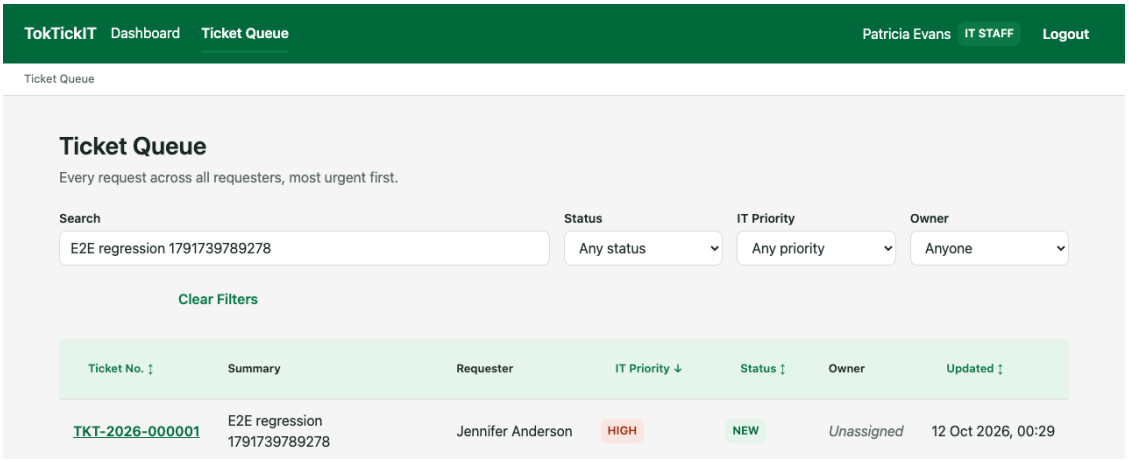


Figure 8.12 — Inherited Staff Queue regression capture; full tracked capture.

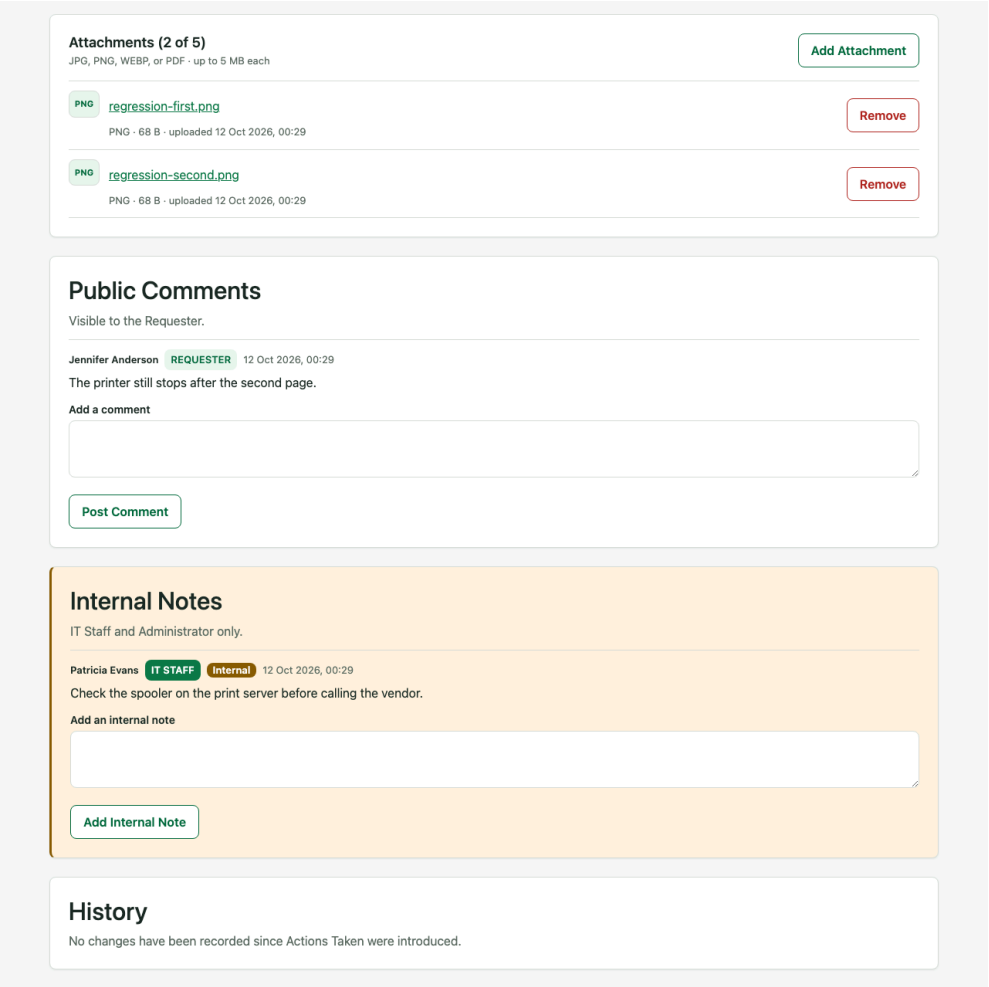


Figure 8.13 — Inherited Staff Ticket Detail capture showing attachments, public comments, Internal Notes, and History; full tracked capture.

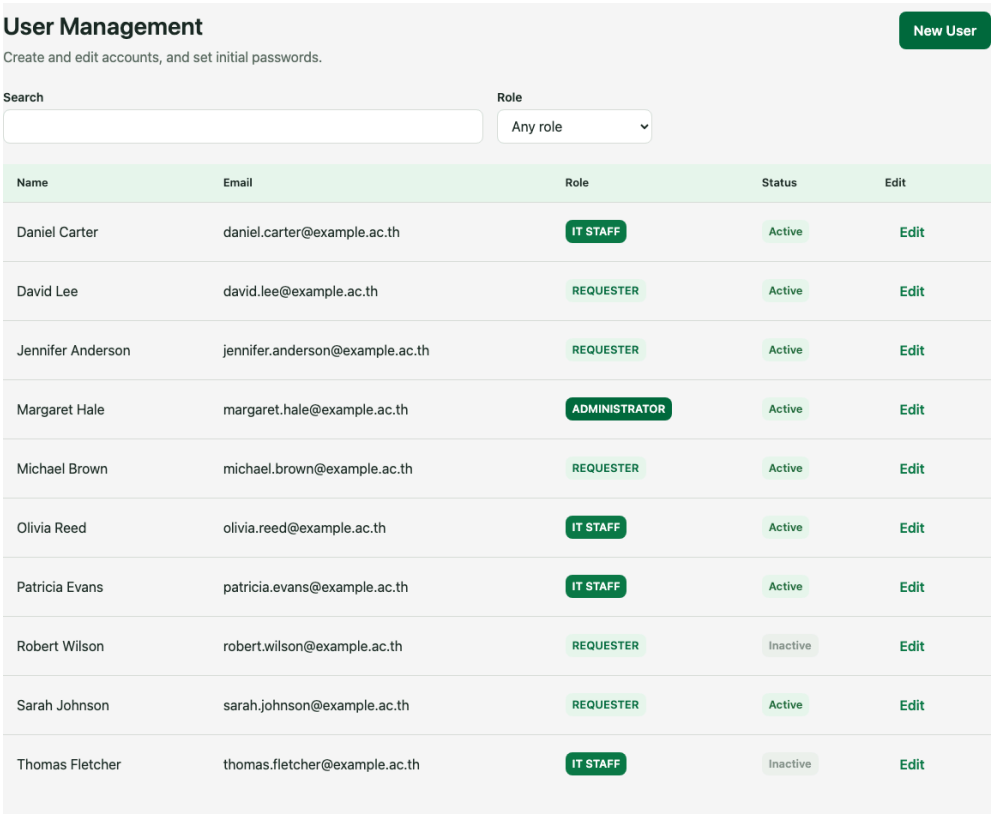


Figure 8.14 — Inherited Administrator User Management regression capture; full tracked capture.

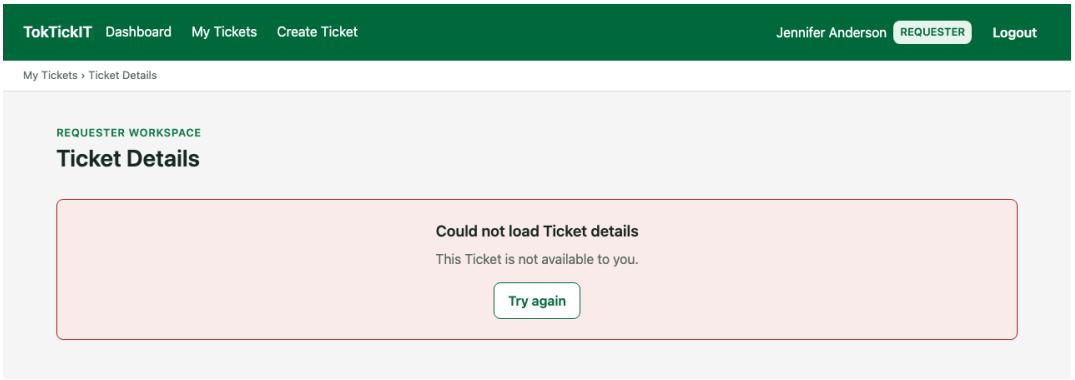


Figure 8.15 — Inherited Requester-facing unavailable-Ticket state; full tracked capture.

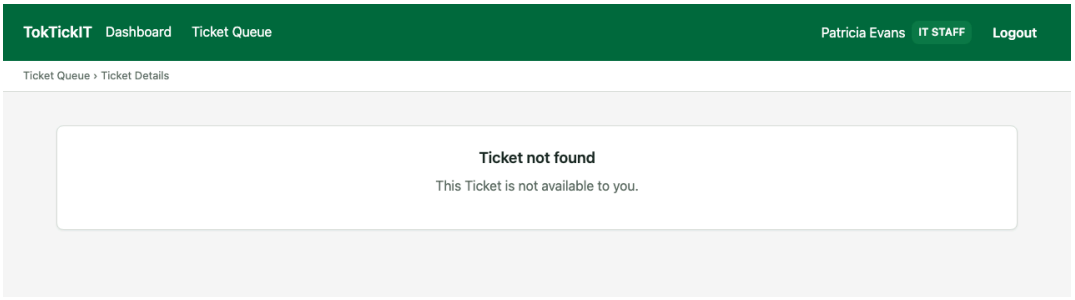


Figure 8.16 — Inherited Staff not-found Ticket state; full tracked capture.

Answer Part 9: Zen Green UI, Responsive, Accessibility, and Final Polish

Question — Does the finished UI retain the Zen Green design and meet the required responsive and accessibility checks?

The UI retains Zen Green tokens, text-labelled badges, field-level validation and distinct loading/empty/error/forbidden states. Dashboard is first in the shell. Desktop tables begin at 992 px; below 992 px Actions become cards. Tested widths include 1280, 834 and 390 px and boundaries 767, 768, 991 and 992 px.

Accessibility evidence comprises component assertions, accessible roles/names/states, keyboard journeys, visible focus, modal containment, Escape/return-focus and live announcements. No screen reader walk was completed. WebKit's default Tab behavior causes two A11Y-01 failures; Chromium, Firefox and Edge pass those walks. The audit detects declared console/network failures, empty/broken links and placeholders, not arbitrary no-op controls unless a journey uses a control and asserts its effect.

Long field content is checked for horizontal page scroll. The shell header remains a known limit for a user's own 120-character name. At 390 px, E2E-10 measures every button and link in the Actions list, Add form and edit panel. PR #104 added E2E-04/RESP-01 checks that measure every visible link on both dashboards against the 44 px minimum; the 64/64 release-gate E2E run at d07fefe exercised them on tree 91a8f1a, identical to main at d07fefe. Dashboard non-link controls are outside that link-specific check; RESP-01 separately checks the View button. Visual inspection sampled each kind of phone screen rather than individually inspecting all 86 original captures.

Figures 9.1a–9.1i reproduce the tracked ui-spec.md at main 6d38208. Figures 9.2a–9.4b are six selected responsive long-content views from accepted staging at 876511f, not a continuous capture; each caption links to the complete tracked source. Responsive workflow evidence is also shown in Parts 5–8.

Visual and accessibility checklist

#	Check	Candidate evidence / limits
1	New screens use only --zen-* tokens; no Bootstrap colour utilities	PASS 12 Oct 2026 — client/tests/lab-03/StyleContract.test.tsx (STYLE-01) scans every file of client/src for colour literals outside the :root token block and for Bootstrap colour utilities; it passes in the client suite on the release candidate (tests.md §6). The reviewer also read the CSS diff of #101 for literals (7 Oct review)

#	Check	Candidate evidence / limits
2	Action and Ticket status badges always show text	PASS 12 Oct 2026 — STYLE-02 (one badge mapping, a text label for each of the four Action statuses); badges carry text in every capture of actions-taken/ and both dashboards
3	Behaviour cards show label + value + descriptive link name	PASS 12 Oct 2026 — UI-14 and UI-16 query each card's link by its accessible name ("View 2 unassigned tickets") beside its value; staff-dashboard/desktop and requester-dashboard/mobile show label, value and link
4	Every drill-down opens a list whose total equals the card	PASS — API/query tests and E2E-04 verify the Open and Needs my attention filters and their matching list totals. The Open result has two rows (Figures 8.4 and 8.4a); the Needs my attention result has one (Figures 8.5 and 8.5a). Each pair presents the tracked left-hand viewport and a right-edge companion after horizontal scrolling. Together the views expose the displayed row values, including Last Updated, but no single viewport shows every column at once; the table’s horizontal overflow remains a disclosed interface limit.
5	Editable vs read-only fields are distinct; system fields never inputs	PASS 12 Oct 2026 — STYLE-05 (system fields are text, never inputs); create-form-{desktop,tablet,mobile} and requester-read-only-{desktop,tablet,mobile}
6	Validation messages sit below their fields; input text does not turn red	PASS 12 Oct 2026, Chromium, 1280 / 834 / 390 px — STYLE-04 (message below its field, error token, input text not red); captured for the Description (description-required-*), the follow-up note (follow-up-required) and the result (complete-result-required). The cancellation reason's message is asserted (UI-05, E2E-01) but not captured
7	Follow-up Note appears, is required, and clears when unticked	PASS 12 Oct 2026 — UI-02 and E2E-01; follow-up-required
8	Focus is visible on every new control; tab order matches the visual order	PASS on Chromium, Firefox and Microsoft Edge, 12 Oct 2026 — A11Y-01 keyboard walks (Action form, cancel dialog, status control, dashboard drill-down) and A11Y-02 focus rules, with the focus ring visible in create-form-*. WebKit: not passed — its default Tab order skips buttons and checkboxes (tests.md §7)

#	Check	Candidate evidence / limits
9	Cancel dialog traps focus, closes on Escape, returns focus	PASS 12 Oct 2026 — UI-05, A11Y-01 and A11Y-02 (the dialog keeps the focus, closes on Escape, returns the focus; Tab and Shift+Tab never leave it while it is busy)
10	Status change and refresh are announced (live region)	PASS 12 Oct 2026 — UI-12 ("Status changed to Resolved." in a polite live region) and UI-28 (the dashboard refresh is announced and cleared), each failing for four injected faults
11	No clipping or overlap at 1280, 834 and 390 px	PASS for the sampled visual review, 12 Oct 2026 — E2E-10, RESP-01, and the Ticket-workflow/regression specs assert no horizontal page scroll at 1280, 834 and 390 px. Claude Code visually inspected representative phone-width images for each screen type: the Actions list, form, validation and edit panel; Requester view and cancel dialog; Ticket workflow and both History views; authentication, My Tickets, Queue, Ticket Detail, attachments, comments, Internal Notes, User Management, not-found and unavailable states; and both dashboards. No clipping or overlap remained after the View-button fix (01c58de). The inspection sampled screen types rather than opening all 86 images individually; automated assertions cover the remaining cases.
12	No horizontal page scroll at 767, 768, 991 and 992 px	PASS 12 Oct 2026 — RESP-01 at 767, 768, 991 and 992 px (Ticket Detail with Actions) and the dashboards' viewport and boundary checks (E2E-04)
13	2000-character text and long names wrap without widening the page	PASS 12 Oct 2026 — RESP-01 (2000-character Description, Follow-up Note and Result, 500-character Attachment Notes, a 120-character name); list-long-content-*. The header of a person whose own name is 120 characters long is a known limit (tests.md §7)
14	Actions list becomes cards below 992 px	PASS 12 Oct 2026 — RESP-01 asserts a table from 992 px and cards below it; list-tablet and list-mobile
15	Touch targets ≥ 44 px on mobile	PASS, 13 Oct 2026, 390 px — E2E-10 measures every button and link in the Actions list, Add form and edit panel. PR #104's E2E-04/RESP-01 measures every visible link on both dashboards against the 44 px minimum; the 64/64 release-gate E2E run at d07fefe exercised these checks on tree 91a8f1a, identical to main at d07fefe. Dashboard non-link controls are outside that dashboard-link measurement; RESP-01 separately checks the View button.

#	Check	Candidate evidence / limits
16	Forbidden, not-found and failure stay visually distinct	PASS 12 Oct 2026 — forbidden (staff-dashboard/forbidden-requester), not found (regression/staff-ticket-not-found-*), unavailable (regression/requester-ticket-unavailable-*) and failure (failure-fixture, save-failure-fixture) are four different messages and styles. A "not found" message has no heading of its own (specification §11.19, item 4)
17	Conflict and save failure keep entered values	PASS 12 Oct 2026 — E2E-03 and UI-10; conflict, save-failure-fixture
18	Dashboard is first and active in navigation; it is the landing page for each role	PASS 12 Oct 2026 — UI-18 and E2E-04 (each role lands on its first Dashboard item, marked current); the captures of both dashboards show the active item
19	No console error, broken link, placeholder text or no-op control in the journeys	PASS for asserted coverage, 12 Oct 2026 — E2E-06 runs the audit in every Lab 4 journey, and audit.spec.ts follows every link on each role's screens. It fails on undeclared console/page errors, failed requests or error responses, empty or # links, missing anchors and placeholder text. Each exception specifies its expected status or failure, method and exact path, and allows only the corresponding console line (UNIT-09); UI-27 fixed the broken anchor found by the audit. The audit does not detect arbitrary no-op controls: those are caught only where a journey exercises the control and asserts its effect (tests.md §7). The signed-out session check's expected 401 is explicitly declared (specification §11.19, item 2).
20	The Lab 4 journeys also pass on Microsoft Edge, Firefox and WebKit, not only Chromium	PASS with one limit, 12 Oct 2026 — e2e/lab-04 on Firefox, WebKit and Microsoft Edge 154 with the throwaway cross-browser config: every test passes except A11Y-01's Tab walks on WebKit (two of them); run output in tests.md §6

#	Check	Candidate evidence / limits
21	Names, roles and states are exposed to assistive technology: every input has a label; a validation message is linked to its field and the field is marked invalid; the cancel dialog is a named modal dialog; a refusal is an alert and a status change is announced in a polite live region	PASS for the assertions, 12 Oct 2026, Chromium (A11Y-01 and A11Y-02 also on Firefox and Edge). Not run: a screen reader; a pass on these assertions says the names, roles and states are exposed, not that the speech is understandable

Lab 4 UI Specification

Version: 1.0 — approved for implementation, 3 October 2026 **Extends:** `docs/lab-03/ui-spec.md` and `docs/lab-02/ui-spec.md`, both still in force. This document covers only what is **new or changed**.

Lab 2's tokens, typography, control states, button hierarchy, badge rules, responsive rules and accessibility requirements remain binding, and so do Lab 3's forbidden state and shell. `style-contract.md` STY-001…030 applies unchanged.

The test of every new screen is the labsheet's: one coherent application, not a second visual system. A new screen that needs a new component shape or colour is a finding to justify, not a default.

1. Tokens and badges

No new colour tokens are planned. Action status badges reuse the existing Ticket status badge palettes through the single badge mapping (STY-018). The text label carries the meaning (STY-019).

Action status	Label	Palette reused from
`PLANNED`	Planned	Ticket `OPEN`
`IN_PROGRESS`	In progress	Ticket `IN_PROGRESS`
`COMPLETED`	Completed	Ticket `RESOLVED`
`CANCELLED`	Cancelled	Ticket `CANCELLED`

If implementation shows that a token is genuinely needed, it is added to the global `:root` block only, recorded here and checked by the existing style-contract test.

2. Application shell — changed

Dashboard becomes the first destination for every role. Because the post-login redirect uses the first navigation item (`routes.ts homeFor`), it also becomes the landing page (FR-21, AC-32).

Role	Navigation	Dashboard route
Requester	**Dashboard** · My Tickets · Create Ticket	`/dashboard`
IT Staff	**Dashboard** · Ticket Queue	`/staff/dashboard`
Administrator	**Dashboard** · Ticket Queue · User Management	`/staff/dashboard`

- Active-page indication (`--zen-secondary`, STY-007), the mobile toggle, Escape-to-close and focus restoration are unchanged.
- An unauthorised dashboard route shows the Lab 3 forbidden state, with a link to the user's own dashboard.
- **Times:** every time and date on every screen is Asia/Bangkok whatever the device's zone (api-spec §1, BR-38), including the Lab 2 and Lab 3 screens and the date-only column of My Tickets.
- **Removed (FR-26):** any leftover development-only route reference. The E2E readiness probe moves off `/select-requester`.

3. Screen: IT Staff Dashboard (`/staff/dashboard`)

Also used by Administrators. Every value comes from `GET /api/staff/dashboard`.

Structure, top to bottom:

1. **Header:** "Dashboard" (h1), the subtitle "The service desk right now", and a secondary Refresh button that shows busy

Figure 9.1a — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 1/9).

Structure, top to bottom:

1. **Header:** "Dashboard" (h1), the subtitle "The service desk right now.", and a secondary **Refresh** button that shows busy while it reloads.
2. **Four metric cards:** Unassigned · Assigned to me · Urgent · Waiting for Requester.
   - Each card shows a label, a large value and a link.
   - Each link names its result, for example **"View 2 unassigned tickets"**, and opens the Queue with the metric's returned `query` (BR-37).
3. **My open Actions (N):**
   - Up to 10 items, oldest first.
   - Each shows the Ticket number and summary, the Action description on one line, the Action status badge and the Action Date/Time.
   - Each item opens the Ticket Detail at its Actions section.
   - Footer when `total > 10`: "Showing the 10 oldest of N open Actions."
4. **Tickets by status:** eight rows, one per status. Each row has a status badge, a count and a link to the Queue filtered by that status. Zero rows are shown, not hidden.
5. **Recently updated:**
   - Five Tickets, each with number, summary, status and IT Priority badges, owner (or *Unassigned*) and updated time.
   - **View all** opens the Queue sorted by `updatedAt:desc`.

Layout:

- **≥ 992 px:** the four cards in one row. Then *My open Actions* takes two thirds of the width beside *Tickets by status*. *Recently updated* is full width below.
- **768–991 px:** cards 2 × 2; every panel full width.
- **< 768 px:** cards 2 per row with 44 px touch targets; panels stacked; lists as stacked items.

States:

State	Presentation
Loading	Header visible; Lab 2 loading state in the content area
Safe failure	Error state, "We could not load the dashboard.", with **Try again** (AC-31)
Forbidden	Lab 3 forbidden state, with a link to the user's own dashboard
Zero metric	`0` shown; the link still opens the Queue's empty or no-results state (AC-29)
No open Actions	Inside the panel: "You have no open Actions."
Refresh in flight	Refresh disabled, labelled "Refreshing..."; current values stay visible

4. Screen: Requester Dashboard (`/dashboard`)

Only the authenticated Requester's Tickets (BR-39). Values come from `GET /api/requester/dashboard`.

Structure:

1. **Header:** "Dashboard" (h1), "Your requests at a glance.", and the primary **Create Ticket** button.
2. **Four metric cards:** Open · Needs my attention · Resolved · Closed.
   - Each has a link with a descriptive name, for example "View 1 ticket waiting for you".
   - *Needs my attention* adds the text "Waiting for your reply" when its count is above zero, so the cue is never colour alone.
3. **Recently updated:**
   - Up to five Tickets, each with number, summary, status badge and updated time.
   - Each opens its Ticket Detail

Figure 9.1b — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 2/9).

3. **Recently updated:**
   - Up to five Tickets, each with number, summary, status badge and updated time.
   - Each opens its Ticket Detail.
   - **View all** opens My Tickets sorted by `updatedAt:desc`.

First use (`totalTickets = 0`): the cards and list are replaced by the empty state "You have not submitted any Tickets yet." with **Create Ticket**. It is not four zeros.

Layout: cards 4 in a row at ≥ 992 px, 2 × 2 from 768 to 991 px, and 2 per row below 768 px. The list stacks.

States: loading, safe failure with Try again, and forbidden (IT Staff or Administrator opening `/dashboard` are offered their own dashboard), as in §3.

5. Ticket Detail — Actions Taken

Placement

Screen	Section order
IT Staff Ticket Detail	Operations · Ticket Information · **Actions Taken** · Attachments · Public Comments · Internal Notes · **History**
Requester Ticket Detail	Ticket Information · **Actions Taken** (read-only) · Attachments · Public Comments · **History** (status changes only)

List

- **Heading:** "Actions Taken (N)". IT Staff on a workable Ticket also see an **Add Action** button.
- **Hint under the heading (staff only):** "Actions are visible to the Requester. Use Internal Notes for private information." (BR-17)
- **≥ 992 px:** a table with the columns Date/Time · Description (two lines, then truncated) · Assignee · Performed by · Status · Follow-up.
  - Each row ends with a **View** button whose accessible name includes the Date/Time.
  - A deactivated assignee shows "(inactive)" after the name (BR-08).
- **< 992 px:** labelled cards with the same fields, matching the Lab 3 Queue's card breakpoint.
- **Order:** oldest first (BR-29).
- **Empty:** "No Actions have been recorded yet."

Create mode (IT Staff, workable Ticket)

Add Action opens an inline form at the top of the section, and focus moves to its first field.

Field	Control	Rules
Description *	Textarea	Required, 1–2000, with a character count
Assignee *	Select from `assignableOwners`	Defaults to the current user
Follow-Up Required?	Checkbox	Ticking reveals Follow-up Note; unticking hides and clears it (BR-14)
Follow-up Note *	Textarea	Required while Follow-Up Required is ticked, 1–2000
Attachment Notes	Text input	Optional, ≤ 500. Helper: "Name the attached file to look at, for example relay-error.png."
Action Date/Time ·	Read-only text	"Recorded automatically when you save" — never inputs (STY-

Figure 9.1c — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 3/9).

		example relay-error.png."
Action Date/Time · Performed by	Read-only text	"Recorded automatically when you save" — never inputs (STY-010)

- **Save Action** (primary) and **Cancel** (tertiary).
- While saving, the button is disabled and shows progress (STY-022). One submission carries one `requestId`, and a retry reuses it (BR-34).
- **A lost answer never costs what was typed (AC-36).** When a save fails without an answer (network failure or a 5xx), **Try again** resends exactly what was sent, under the same `requestId`. If the entries were changed meanwhile, the first entries are resent first, to find out whether they were saved, and the changes are then saved onto that Action as an update. Nothing is dropped and no second Action is created. If that update fails, **Try again** repeats only the update.
- **A form open when the Ticket stops taking Actions becomes read-only (BR-23).** The entries stay visible but cannot be changed, **Save Action** and **Try again** are withdrawn, and the form says "This Ticket no longer takes Actions, so these entries cannot be saved." **Cancel** stays. The edit form does the same.
- Times in the list and the panel are Asia/Bangkok whatever the device's zone (api-spec §1).

View and edit mode

View opens an inline detail panel, and focus moves to its heading.

- **System fields are always read-only** (STY-009, STY-010): Action Date/Time, Performed by, Status, and Completed by / at or Cancelled by / at with the reason.
- **Non-terminal Action, IT Staff:**
  - **Edit** switches Description, Result, Assignee, Follow-up and Attachment Notes into inputs, with **Save changes** and **Discard**.
  - Status buttons depend on the status:
    - **Start** (from Planned);
    - **Complete** (from In progress);
    - **Cancel Action** (destructive; never primary, STY-015).
- **Complete without a Result:** the Result field shows "Enter the result before completing this Action." below it and receives focus. The server enforces the same rule (BR-11).
- **Cancel Action:** opens a dialog with a required **Reason** field (1–500), **Cancel Action** (destructive) and **Keep Action**.
  - The dialog follows Lab 3's modal rules: labelled, focus trapped, Escape closes, and focus returns to the trigger.
- **Terminal Action:** everything is read-only. The badge and completion or cancellation details are shown, with no Edit or status buttons.
- **Requester:** the same panel, entirely read-only, with no buttons. People appear by display name only.

Feedback states

Situation	Presentation
Validation	Field-level messages below each field (STY-012); the form stays open
Inactive or ineligible assignee (`ASSIGNEE_NOT_ELIGIBLE`)	Under Assignee: "Choose an active IT Staff member or Administrator." The assignee list is refetched
Ticket not workable	Banner replacing Add Action: "This Ticket is Resolved. Actions are read-only — reopen the Ticket to record more work."
Conflict (`STALE_VERSION`)	"Someone else changed this Action. The latest version is shown; your unsaved entries are kept in the form." The user can review and save again (AC-36)
Terminal (`ACTION_TERMINAL`)	"This Action is already completed or cancelled." The panel refreshes to read-only
Save failure (network or 500)	"The Action could not be saved. Your entries are kept." with **Try again**, which resends the same `requestId` and the same entries; changes made since are then

Figure 9.1d — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 4/9).

Save failure (network or 500)	"The Action could not be saved. Your entries are kept." with **Try again**, which resends the same `requestId` and the same entries; changes made since are then saved as an update (see Create mode)
Ticket stops taking Actions while a form is open	The form turns read-only with its entries kept, and says why; Save is withdrawn (BR-23)
Ticket status changes	The Actions are loaded again, so a cancellation's cascade shows at once (BR-22)
Load failure	Section-level error state with Try again; the rest of the Ticket stays usable

6. Ticket status control and resolution feedback

- The **Move to…** control lists only `permittedTransitions` (AC-18).
- When `blockedTransitions` is present, a helper line explains the omission, for example "Resolved and Closed become available when the 2 open Actions are completed or cancelled." It links to the Actions section.
- Cancelling a Ticket asks for confirmation, and states the cascade when one applies: "Cancel this Ticket? 2 open Actions will also be cancelled."
- **On success:**
  - the header status badge, Operations and permitted transitions refresh from the response;
  - a polite live region announces "Status changed to Resolved." (AC-21).
- **On `OPEN_ACTIONS_BLOCK_RESOLUTION`:** an inline message shows the open-Action count, the Ticket refetches, and the previous status remains shown.
- **On `STALE_VERSION`:** "This Ticket changed since you opened it. The latest status is shown — review and try again." The Ticket refetches.

7. History

- **IT Staff and Administrators:** every event as one line: time · actor · readable description.
  - Examples: "Status changed from In progress to Resolved", "Action assigned from Daniel Carter to Olivia Reed".
  - Action events link to their Action.
- **Requester:** status changes only (BR-28).
- **Order:** oldest first, which is also the order the changes were committed in (BR-29, specification §11.17).
- **Time:** Asia/Bangkok whatever the device's zone (api-spec §1).
- **Empty:** "No changes have been recorded since Actions Taken were introduced." This is true for every seeded or legacy Ticket until its first change.
- Event text is generated from the event type and identifiers only. No free text from users appears in history (BR-30).

8. Responsive

Lab 2's three viewports are unchanged: desktop ≥ 992, tablet 768–991, mobile < 768.

- **Representative captures:** 1280 × 900, 834 × 1112 and 390 × 844.
- **Boundary checks:** automated no-horizontal-overflow assertions at 767, 768, 991 and 992 px for both dashboards and for Ticket Detail with Actions (AC-34).
- **Long content:**
  - A 2000-character Description, Result or Follow-up Note wraps inside its cell or card, never widening the page.
  - Long display names and Ticket summaries wrap.
  - Dashboard descriptions are truncated to one line, with the full text one click away.
- **Touch targets:** every control keeps the 44 px minimum on mobile (`lab-02 §11.26`).

9. Accessibility

Figure 9.1e — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 5/9).

9. Accessibility

- Every input has a programmatic label (STY-026).
- Errors are linked through `aria-describedby`; the first invalid field, in the order shown, receives focus on submit. When the error comes from the server, the focus moves once the controls are enabled again. The cancel dialog's Reason is such a field.
- Metric values and badges are text, so colour never carries meaning alone (STY-019).
- Card links have descriptive accessible names, not "View all" alone.
- Panels open with focus moved to their heading, and closing a panel, form or dialog gives the focus back to the control that opened it. If that control is gone (an Action that was just cancelled offers none), the focus goes to the details heading or the section heading, never to the page.
- A dialog keeps the focus while its save is in flight: with every control disabled, the dialog itself holds the focus and Tab and Shift+Tab stay inside (AC-35).
- Status changes and dashboard refreshes are announced through polite live regions.
- Lists use list semantics. Tables use `<th scope>`.
- Keyboard order follows the visual order, and focus is visibly indicated with `--zen-focus-ring` (STY-027) (AC-35).

10. Screenshot paths

Captures are written only when `CAPTURE_EVIDENCE=1` is set (specification §11.13). Actions Taken captures are cropped to the Actions Taken card (the cancel dialog to the visible viewport) so that they stay readable; `list-long-content-*` is the 2000-character and 120-character-name stress case of RESP-01, not the ordinary list. A normal test run writes nothing tracked. A failure, empty or loading state produced by intercepting the API is labelled `fixture` in its filename and caption.

`artifacts/lab-04/screenshots/`

```
staff-dashboard/      desktop · tablet · mobile · loading-fixture · failure-fixture ·
                      admin-no-open-actions · forbidden-requester · drilldown-unassigned
requester-dashboard/  desktop · tablet · mobile · first-use-empty-fixture · failure-fixture ·
                      drilldown-open · drilldown-needs-attention
actions-taken/        list-{desktop,tablet,mobile} · list-long-content-{desktop,tablet,mobile} ·
                      create-form-{desktop,tablet,mobile} · description-required-{desktop,tablet,mobile} ·
                      edit-reassign-{desktop,tablet,mobile} · requester-read-only-{desktop,tablet,mobile}
                      follow-up-required · inactive-assignee-refused · started · complete-result-required
                      completed · cancel-dialog · cancelled · ticket-not-workable · conflict ·
                      save-failure-fixture
ticket-workflow/      {resolve-blocked, resolved-after-work-complete, cancel-cascade, stale-status,
                      history-staff, history-requester}-{desktop,tablet,mobile}
regression/           {login, my-tickets, requester-detail-attachments-comments, staff-queue,
                      staff-detail-internal-notes, staff-ticket-not-found,
                      requester-ticket-unavailable, user-management}-{desktop,tablet,mobile}
```

Capture dates and runs: `staff-dashboard/` and `requester-dashboard/` on 5 October 2026 (Issue #93), `actions-taken/` on 8 October 2026 (Issue #91), `ticket-workflow/` and `regression/` on 12 October 2026 on a freshly created E2E database (Issue #94). The 12 October run also regenerated the dashboards and Actions Taken; its staff dashboard (desktop) was compared with the 5 October capture and differs only in the dates the seed writes, so the earlier, reviewed captures were kept. The Ticket workflow captures are cropped to the Operations card, the History section, or the top of the page, and each states its viewport in its name.

Lab 2 and Lab 3 capture folders are never written by a Lab 4 run.

11. Visual and accessibility checklist

Run against the integrated staging candidate before the release, and again on final `main` for evidence. Lab 2's and Lab 3's checklists still apply. Every row was NOT RUN until the evidence run of 12 October 2026. A row is marked only with the date

Figure 9.1f — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 6/9).

Run against the integrated staging candidate before the release, and again on final `main` for evidence. Lab 2's and Lab 3's checklists still apply. Every row was NOT RUN until the evidence run of 12 October 2026. A row is marked only with the date, viewport or width, role and evidence that prove it.

#	Check	Evidence required	Status
1	New screens use only `--zen-*` tokens; no Bootstrap colour utilities	Style-contract test output; grep audit	PASS 12 Oct 2026 — `client/tests/lab-03/StyleContract.test.tsx` (STYLE-01) scans every file of `client/src` for colour literals outside the `:root` token block and for Bootstrap colour utilities; it passes in the client suite on the release candidate (`tests.md` §6). The reviewer also read the CSS diff of #101 for literals (7 Oct review)
2	Action and Ticket status badges always show text	Component test; screenshot	PASS 12 Oct 2026 — STYLE-02 (one badge mapping, a text label for each of the four Action statuses); badges carry text in every capture of `actions-taken/` and both dashboards
3	Metric cards show label + value + descriptive link name	Component test; accessibility tree snippet	PASS 12 Oct 2026 — UI-14 and UI-16 query each card's link by its accessible name ("View 2 unassigned tickets") beside its value; `staff-dashboard/desktop` and `requester-dashboard/mobile` show label, value and link
4	Every drill-down opens a list whose total equals the card	E2E assertion; paired screenshots	PASS 12 Oct 2026 — DASH-02 and DASH-08 call each metric's returned query and compare the Tickets with SQL (API, every metric); E2E-04 follows the unassigned (staff) and both Requester links; `drilldown-unassigned` , `drilldown-open` and `drilldown-needs-attention` are the paired captures. `drilldown-needs-attention` shows My Tickets filtered to `WAITING_FOR_REQUESTER` ("Showing 1 to 1 of 1 tickets"). In both Requester captures the table is wider than its card and the last column, Last Updated, is cut at the card's edge (`tests.md` §7)
5	Editable vs read-only fields are distinct; system fields never inputs	Screenshot of create and view modes	PASS 12 Oct 2026 — STYLE-05 (system fields are text, never inputs); `create-form-{desktop,tablet,mobile}` and `requester-read-only-{desktop,tablet,mobile}`
6	Validation messages sit below their fields; input text does not turn red	Screenshot (follow-up, result, reason)	PASS 12 Oct 2026, Chromium, 1280 / 834 / 390 px — STYLE-04 (message below its field, error token, input text not red); captured for the Description (`description-required-*`), the follow-up note (`follow-up-required`) and the result (`complete-result-required`). The cancellation reason's message is asserted (UI-05, E2E-01) but not captured
7	Follow-up Note appears, is required, and clears when unticked	Component test; screenshot	PASS 12 Oct 2026 — UI-02 and E2E-01; `follow-up-required`
8	Focus is visible on every new control; tab order matches the visual order	Keyboard walk notes; screenshot with focus ring	PASS on Chromium, Firefox and Microsoft Edge, 12 Oct 2026 — A11Y-01 keyboard walks (Action form, cancel dialog, status control, dashboard drill-down) and A11Y-02 focus rules, with the focus ring visible in `create-form-*` . **WebKit: not passed** — its default Tab order skips buttons and checkboxes (`tests.md` §7)
9	Cancel dialog traps focus, closes	Component test; E2E	PASS 12 Oct 2026 — UI-05, A11Y-01 and A11Y-02 (the dialog keeps the focus, closes on Escape, returns the focus; Tab and

Figure 9.1g — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 7/9).

9	Cancel dialog traps focus, closes on Escape, returns focus	Component test; E2E	PASS 12 Oct 2026 — UI-05, A11Y-01 and A11Y-02 (the dialog keeps the focus, closes on Escape, returns the focus; Tab and Shift+Tab never leave it while it is busy)
10	Status change and refresh are announced (live region)	Component test	PASS 12 Oct 2026 — UI-12 ("Status changed to Resolved." in a polite live region) and UI-28 (the dashboard refresh is announced and cleared), each failing for four injected faults
11	No clipping or overlap at 1280, 834 and 390 px	Screenshots	PASS for what was inspected, 12 Oct 2026, with one exception found on 14 Oct 2026 (last sentence) — E2E-10, RESP-01 and the Ticket workflow and regression specs assert no horizontal page scroll at 1280, 834 and 390 px. Claude Code opened an image of every kind of screen at phone width (Actions list, Add form, validation, edit panel, Requester view, cancel dialog; Ticket workflow states and both Histories; sign-in, My Tickets, Queue, Ticket Detail with attachments, comments and an Internal Note, User Management, the not-found and unavailable states; both dashboards) and found no clipping or overlap after the fix of the View button (`01c58de`). It did not open every one of the 86 images; the assertions cover the rest. **Exception, measured on `d07fefe` on 14 Oct 2026 with a throwaway probe that is not a committed test:** the Requester's My Tickets table is 1219 to 1231 px wide (by list) in a card 1150 px wide at 1600 and 1280 px, 974 px at 1024 px and 784 px at 834 px, because the status badge `WAITING_FOR_REQUESTER` does not wrap, so the last column, Last Updated, is reached by scrolling the table sideways inside its card (the container has scrolled since Lab 2; `tests.md` §7). The page itself did not scroll sideways at any width measured (1600, 1280, 1024, 834 and 390 px) on My Tickets, both dashboards, the Staff Queue and User Management; the Staff Queue table is 2 px wider than its card at 1024 px only. At 390 px no container scrolls
12	No horizontal page scroll at 767, 768, 991 and 992 px	E2E width assertions	PASS 12 Oct 2026 — RESP-01 at 767, 768, 991 and 992 px (Ticket Detail with Actions) and the dashboards' viewport and boundary checks (E2E-04)
13	2000-character text and long names wrap without widening the page	E2E fixture; screenshot	PASS 12 Oct 2026 — RESP-01 (2000-character Description, Follow-up Note and Result, 500-character Attachment Notes, a 120-character name); `list-long-content-*` . The header of a person whose own name is 120 characters long is a known limit (`tests.md` §7)
14	Actions list becomes cards below 992 px	Tablet and mobile screenshots	PASS 12 Oct 2026 — RESP-01 asserts a table from 992 px and cards below it; `list-tablet` and `list-mobile`
15	Touch targets ≥ 44 px on mobile	E2E measurement	PASS for the Actions Taken screens, 12 Oct 2026, 390 px — E2E-10 measures every button and link of the list, Add form and edit panel; RESP-01 measures the View button. The 13 Oct review-readiness follow-up (PR #104, merged into `lab4-staging` and released to `main` by PR #103) measures every visible link on both dashboards at 390 px (E2E-04); all must be at least 44 px, and the two footer links that measured 18 px were fixed. Dashboard buttons and other non-link controls are not covered by that measurement. Runs: `tests.md` §6.3 and §6.4.
			PASS 12 Oct 2026 — forbidden (`staff-dashboard/forbidden-`

Figure 9.1h — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 8/9).

			`tests.md` §6.3 and §6.4.
16	Forbidden, not-found and failure stay visually distinct	Screenshots	PASS 12 Oct 2026 — forbidden (`staff-dashboard/forbidden-requester`), not found (`regression/staff-ticket-not-found-*`), unavailable (`regression/requester-ticket-unavailable-*`) and failure (`failure-fixture` , `save-failure-fixture`) are four different messages and styles. A "not found" message has no heading of its own (specification §11.19, item 4)
17	Conflict and save failure keep entered values	E2E; screenshot	PASS 12 Oct 2026 — E2E-03 and UI-10; `conflict` , `save-failure-fixture`
18	Dashboard is first and active in navigation; it is the landing page for each role	E2E; screenshots	PASS 12 Oct 2026 — UI-18 and E2E-04 (each role lands on its first Dashboard item, marked current); the captures of both dashboards show the active item
19	No console error, broken link, placeholder text or no-op control in the journeys	E2E console and link audit output	PASS for what is asserted, 12 Oct 2026 — E2E-06: every Lab 4 journey runs under the audit (a console error, page error, failed request or error response it did not declare fails it), and `audit.spec.ts` follows every link on every screen of each role: no empty or `#` link, no anchor without its element, no placeholder text. Each declared exception names its status or failure, its method and its exact address, and licenses only the one console line it causes (UNIT-09); the audit found one broken anchor, fixed under UI-27. **Not asserted: a no-op control.** The audit does not click every control; a control that does nothing is caught only where a journey uses it and asserts its effect (`tests.md` §7). One declared line is common: the session check of a signed-out visit is answered 401 and Chromium logs it (specification §11.19, item 2)
20	The Lab 4 journeys also pass on Microsoft Edge, Firefox and WebKit, not only Chromium	Run output of the cross-browser command in `tests.md` §5, on the staging candidate	PASS with one limit, 12 Oct 2026 — `e2e/lab-04` on Firefox, WebKit and Microsoft Edge 154 with the throwaway cross-browser config: every test passes except A11Y-01's Tab walks on WebKit (two of them); run output in `tests.md` §6
21	Names, roles and states are exposed to assistive technology: every input has a label; a validation message is linked to its field and the field is marked invalid; the cancel dialog is a named modal dialog; a refusal is an `alert` and a status change is announced in a polite live region	Component assertions in STYLE-03, STYLE-04, UI-05 and UI-12 (accessible name, `aria-modal` , `aria-invalid` , `aria-describedby` , `role=alert` , `aria-live`), and Playwright finding the controls by role and name in A11Y-01 and A11Y-02. No screen reader was used (`tests.md` §7)	PASS for the assertions, 12 Oct 2026, Chromium (A11Y-01 and A11Y-02 also on Firefox and Edge). **Not run:** a screen reader; a pass on these assertions says the names, roles and states are exposed, not that the speech is understandable

Figure 9.1i — Rendered ui-spec.md in GitHub Preview; source at main 6d38208 (L2, overlapping segment 9/9).

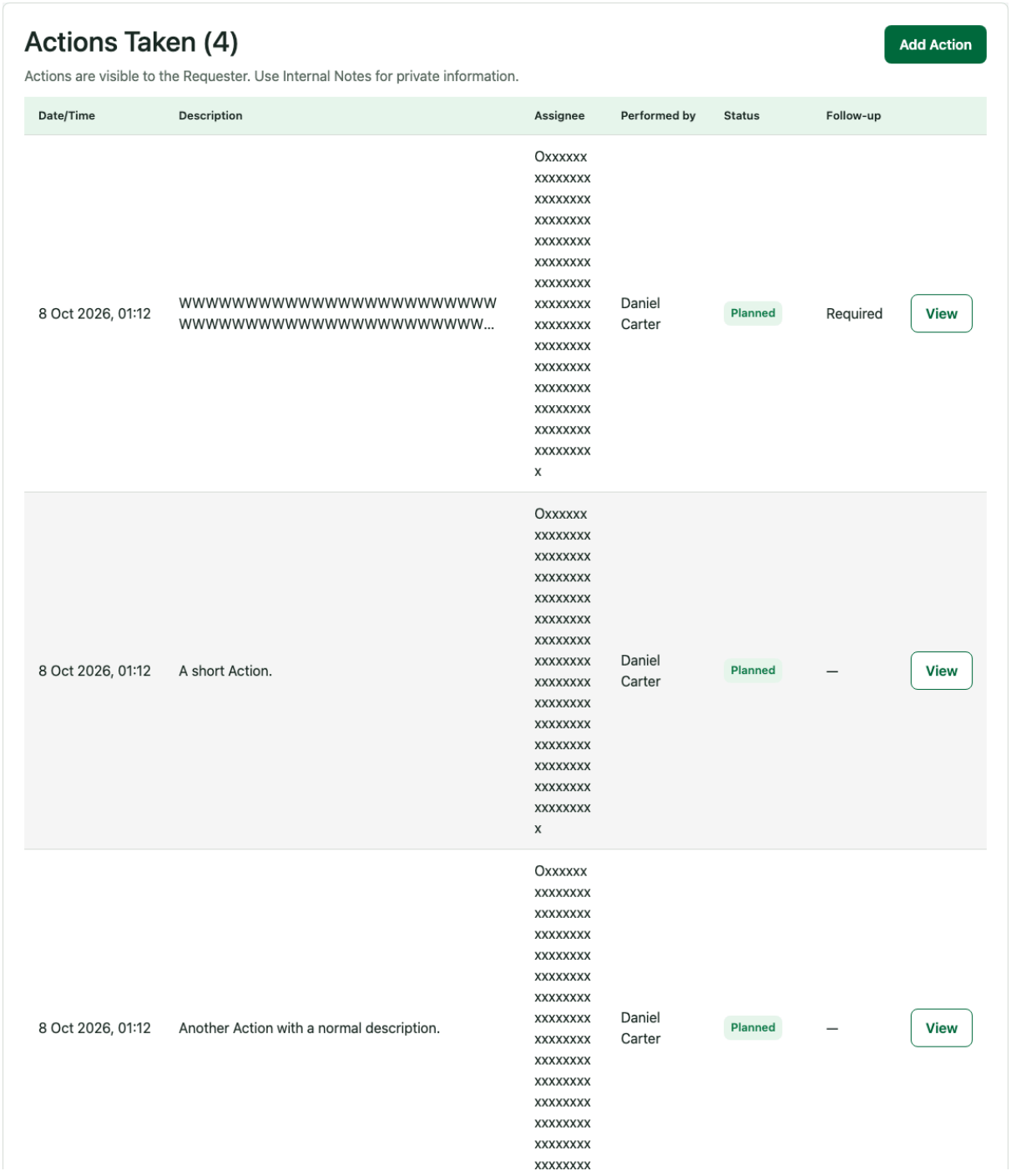


Figure 9.2a — Long-content layout, desktop; Chromium, 1280 px (selected portions; source at lab4-staging 876511f).

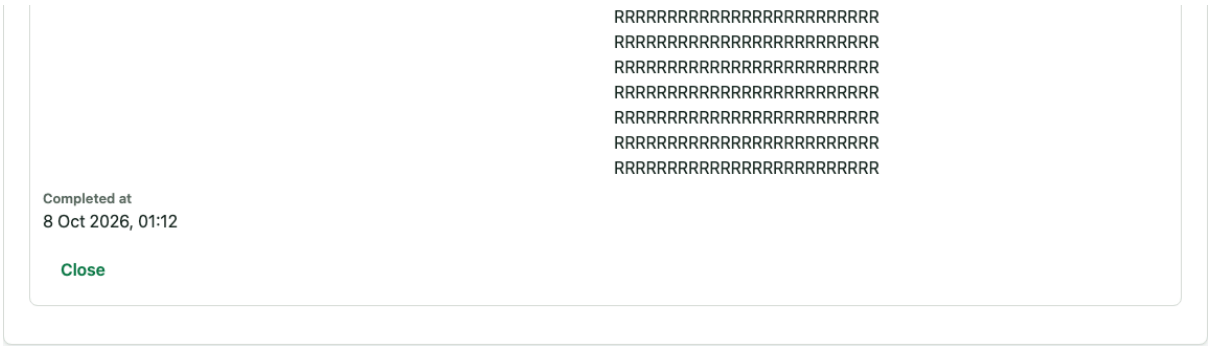
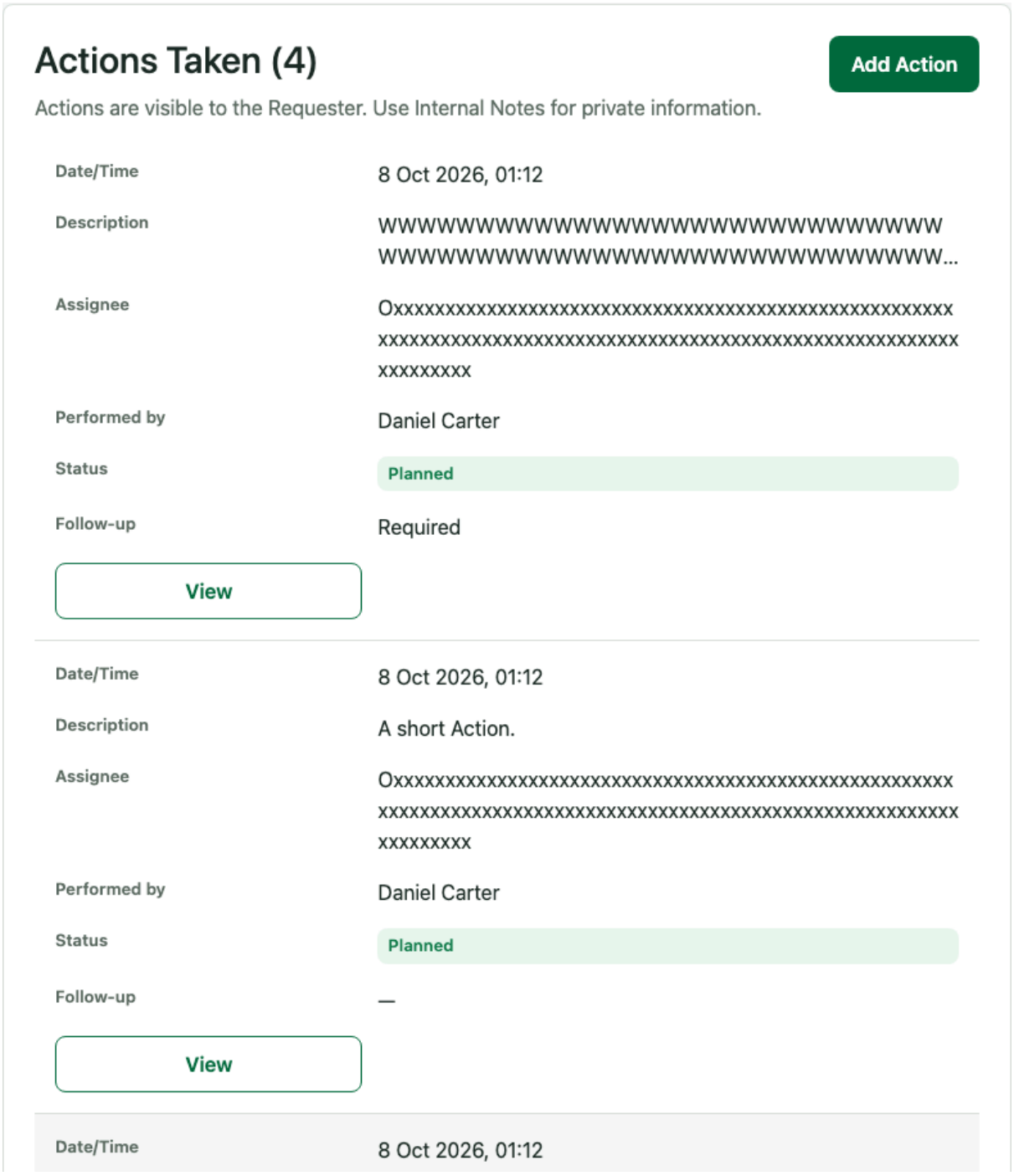
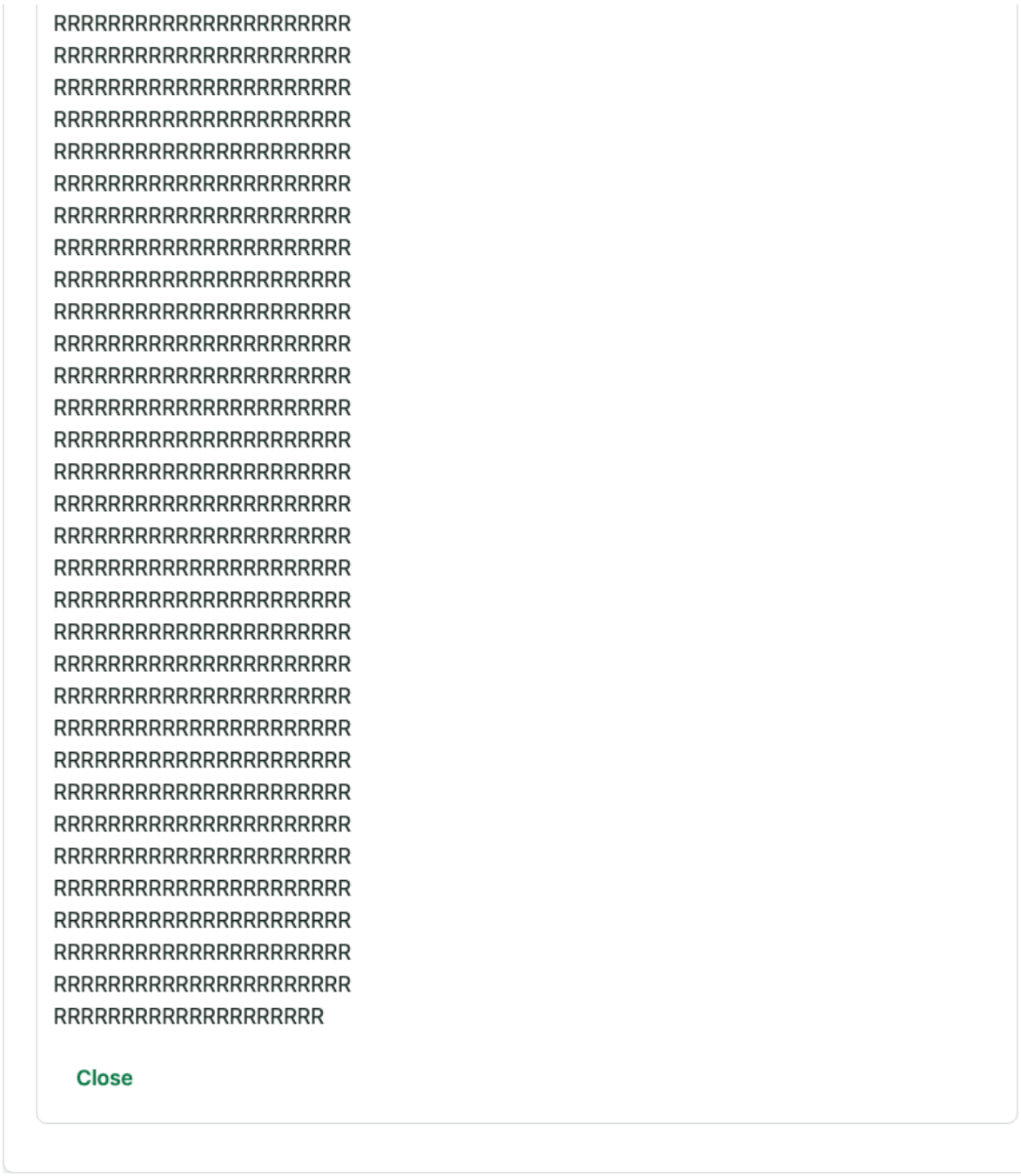
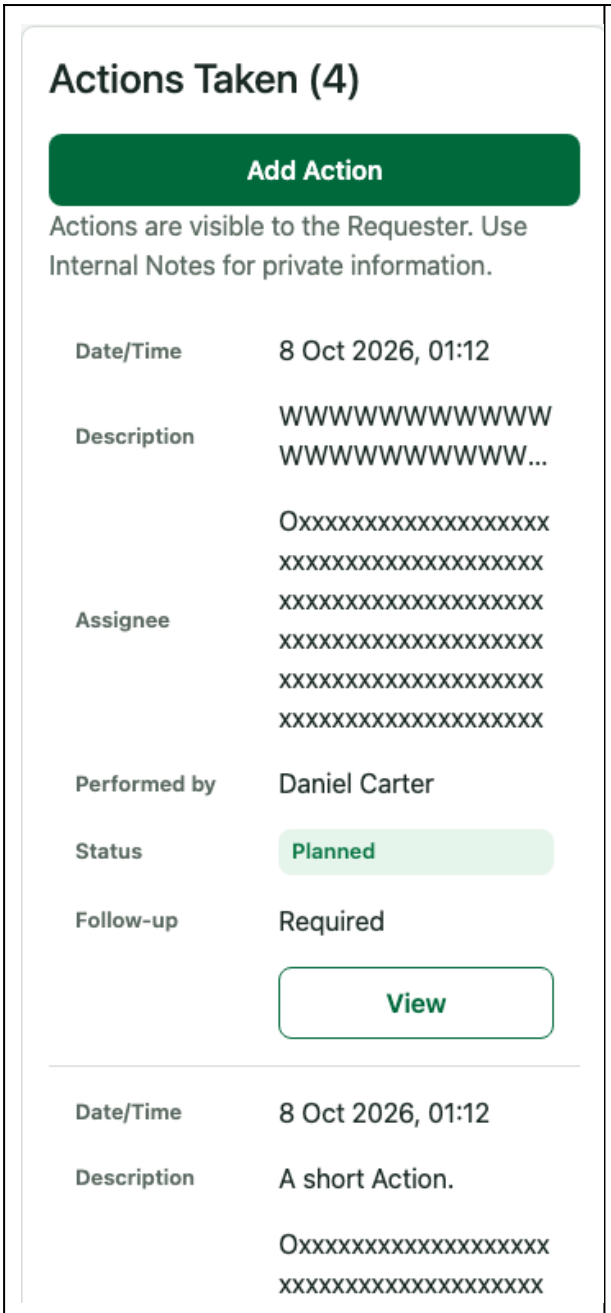


Figure 9.2b — Long-content layout, desktop; Chromium, 1280 px (selected portions; source at lab4-staging 876511f).


Actions Taken (4)
Add Action
Actions are visible to the Requester. Use Internal Notes for private information.
Date/Time
8 Oct 2026, 01:12
Description
WWWWWWWWWWWWWWWWWWWWWWWWWWWWW
WWWWWWWWWWWWWWWWWWWWWWWWWWWWW...
Assignee
Oxxx


RRRRRRRRRRRRRRRRRRRR
RRRRRRRRRRRRRRRRRRRR
RRRRRRRRRRRRRRRRRRRR



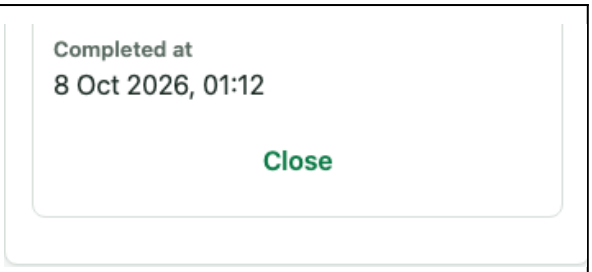


Figure 9.4a — Long-content layout, mobile; Chromium, 390 px (selected portions; source at lab4-staging 876511f).

Figure 9.4b — Long-content layout, mobile; Chromium, 390 px (selected portions; source at lab4-staging 876511f).

Release and submission gate

Lab 4's product code was released to main at d07fefe (tree 91a8f1a). The 14 October release-gate results were run on that exact tree. PR #105 later merged the documentation-only final-main record at 6d38208 on 17 October. Main 6d38208 differs from d07fefe only in ai-use.md, reviewer.md, tests.md and ui-spec.md; no test rerun on 6d38208 is claimed. The current tracked reviewer.md, tests.md and ui-spec.md are rendered in Parts 1, 3 and 9; Part 4's AI-use record is checked against the merged ai-use.md.

Figure 1.2 and the Project API show all seven Lab 4 Issues in Done. The 64/64 release-gate E2E run at d07fefe includes PR #104's E2E-04/RESP-01 checks for visible dashboard links at the 44 px minimum at 390 px; this coverage is recorded against the tested code tree. Appendix A under Part 3 contains compact test output from the d07fefe run, with shuffled-server and three-browser outputs labeled as excerpts.

Accessibility limits: two WebKit A11Y-01 Tab-walk failures; no screen-reader evaluation; and no general detection of arbitrary no-op controls unless a tested journey asserts the control's effect.

Responsive limits: a 120-character user name can reach the header limit. At 390 px, the Requester's My Tickets table also exceeds its 1,150 px card (1,219–1,231 px for the measured row), as recorded in tests.md §7 and ui-spec.md row 11.

Evidence limits: the My Tickets table is wider than its card for these desktop results, so Last Updated lies beyond the initial viewport. Figures 8.4 and 8.5 are full tracked left-hand captures; Figures 8.4a and 8.5a show the same respective results after scrolling to the right edge. Together, each pair shows the row values, but not in one viewport. Figure 8.4a is a fixture-backed UI probe replaying the two Open rows from the tracked capture on main at `d07fefe`; it is not a database-backed test capture or a tracked repository screenshot. Figure 8.5a is the direct seeded-data probe.